Supplementary tables

Every table is written by `scripts/analyze_revision.py` from the campaign results in `results/revision`. Cells give the mean and the half-width of the 95% t-interval over seeds; differences are paired by seed. Tables in the main text select the loss-threshold attack and the primary non-members from these.

1 Membership inference per attack and non-member set

Table S1: Membership inference before and after centralized unlearning (Loss thr., primary non-members). Gap is the post-unlearning AUC minus the retrained model's (same seed); advantage is $|\mathrm{AUC} - 0.5|$; outliers by base-model difficulty, stratum-matched.

		MIA AUC				Post-unlearning advantage		
Model	**Method**	**Pre**	**Post**	**Gap**	**TPR@1%**	**All**	**Typical**	**Outlier**
DistilGPT-2	Retrain	0.60 ± 0.02	0.45 ± 0.02	0.00 ± 0.00	0.01 ± 0.01	0.05 ± 0.02	0.03 ± 0.02	0.07 ± 0.05
DistilGPT-2	GA†	0.60 ± 0.02	0.47 ± 0.03	0.03 ± 0.03	0.01 ± 0.00	0.03 ± 0.02	0.04 ± 0.01	0.03 ± 0.04
DistilGPT-2	GD	0.60 ± 0.02	0.40 ± 0.02	-0.04 ± 0.02	0.01 ± 0.00	0.10 ± 0.02	0.09 ± 0.03	0.10 ± 0.05
DistilGPT-2	NPO	0.60 ± 0.02	0.39 ± 0.02	-0.06 ± 0.01	0.00 ± 0.00	0.11 ± 0.02	0.11 ± 0.03	0.07 ± 0.08
GPT-2	Retrain	0.63 ± 0.02	0.44 ± 0.02	0.00 ± 0.00	0.02 ± 0.01	0.06 ± 0.02	0.03 ± 0.03	0.08 ± 0.04
GPT-2	GA†	0.63 ± 0.02	0.46 ± 0.04	0.02 ± 0.05	0.01 ± 0.00	0.04 ± 0.04	0.04 ± 0.03	0.07 ± 0.04
GPT-2	GD	0.63 ± 0.02	0.39 ± 0.04	-0.05 ± 0.04	0.01 ± 0.00	0.11 ± 0.04	0.09 ± 0.03	0.13 ± 0.10
GPT-2	NPO	0.63 ± 0.02	0.36 ± 0.10	-0.09 ± 0.09	0.01 ± 0.01	0.14 ± 0.10	0.14 ± 0.09	0.13 ± 0.14
GPT-2-medium	Retrain	0.73 ± 0.02	0.43 ± 0.02	0.00 ± 0.00	0.01 ± 0.00	0.07 ± 0.02	0.04 ± 0.02	0.05 ± 0.03
GPT-2-medium	GA†	0.73 ± 0.02	0.50 ± 0.04	0.07 ± 0.04	0.01 ± 0.01	0.02 ± 0.02	0.02 ± 0.02	0.05 ± 0.06
GPT-2-medium	GD	0.73 ± 0.02	0.03 ± 0.04	-0.40 ± 0.05	0.00 ± 0.00	0.47 ± 0.04	0.47 ± 0.04	0.47 ± 0.03
GPT-2-medium	NPO	0.73 ± 0.02	0.09 ± 0.10	-0.34 ± 0.12	0.00 ± 0.00	0.41 ± 0.10	0.41 ± 0.10	0.41 ± 0.12
Pythia-1B (LoRA)	Retrain	0.58 ± 0.02	0.44 ± 0.02	0.00 ± 0.00	0.01 ± 0.01	0.06 ± 0.02	0.03 ± 0.02	0.09 ± 0.07
Pythia-1B (LoRA)	GA†	0.58 ± 0.02	0.43 ± 0.17	-0.01 ± 0.17	0.01 ± 0.00	0.07 ± 0.17	0.09 ± 0.17	0.11 ± 0.13
Pythia-1B (LoRA)	GD†	0.58 ± 0.02	0.06 ± 0.04	-0.38 ± 0.05	0.00 ± 0.01	0.44 ± 0.04	0.43 ± 0.06	0.47 ± 0.02
Pythia-1B (LoRA)	NPO	0.58 ± 0.02	0.04 ± 0.03	-0.40 ± 0.04	0.00 ± 0.00	0.46 ± 0.03	0.46 ± 0.03	0.47 ± 0.05

Mean $\pm$ half-width of the 95% t-interval over seeds (up to 5); a single value means one seed. †Model collapsed: retain loss rose more than threefold in 5/5 seeds. GA: all four central models; GD: Pythia-1B (LoRA).

Table S2: Membership inference before and after centralized unlearning (Loss thr., holdout10 non-members). Gap is the post-unlearning AUC minus the retrained model's (same seed); advantage is $|\text{AUC} - 0.5|$; outliers by base-model difficulty, stratum-matched.

		MIA AUC				Post-unlearning advantage		
Model	**Method**	**Pre**	**Post**	**Gap**	**TPR@1%**	**All**	**Typical**	**Outlier**
DistilGPT-2	Retrain	0.60 ± 0.00	0.43 ± 0.00	0.00 ± 0.00	0.02 ± 0.00	0.07 ± 0.00	0.07 ± 0.01	0.08 ± 0.01
DistilGPT-2	GA†	0.60 ± 0.00	0.55 ± 0.11	0.11 ± 0.11	0.04 ± 0.02	0.08 ± 0.04	0.13 ± 0.07	0.08 ± 0.05
DistilGPT-2	GD	0.60 ± 0.00	0.49 ± 0.01	0.06 ± 0.01	0.01 ± 0.00	0.01 ± 0.01	0.09 ± 0.01	0.02 ± 0.01
DistilGPT-2	NPO	0.60 ± 0.00	0.43 ± 0.04	0.00 ± 0.04	0.00 ± 0.00	0.07 ± 0.04	0.03 ± 0.02	0.04 ± 0.04
GPT-2	Retrain	0.63 ± 0.00	0.43 ± 0.00	0.00 ± 0.00	0.02 ± 0.00	0.07 ± 0.00	0.07 ± 0.00	0.08 ± 0.01
GPT-2	GA†	0.63 ± 0.00	0.55 ± 0.10	0.12 ± 0.10	0.02 ± 0.03	0.08 ± 0.06	0.11 ± 0.09	0.10 ± 0.07
GPT-2	GD	0.63 ± 0.00	0.47 ± 0.06	0.05 ± 0.06	0.01 ± 0.00	0.03 ± 0.05	0.08 ± 0.05	0.05 ± 0.07
GPT-2	NPO	0.63 ± 0.00	0.40 ± 0.10	-0.03 ± 0.10	0.01 ± 0.01	0.10 ± 0.10	0.05 ± 0.09	0.09 ± 0.12
GPT-2-medium	Retrain	0.75 ± 0.00	0.41 ± 0.00	0.00 ± 0.00	0.02 ± 0.00	0.09 ± 0.00	0.05 ± 0.00	0.11 ± 0.01
GPT-2-medium	GA†	0.75 ± 0.00	0.55 ± 0.11	0.14 ± 0.12	0.02 ± 0.02	0.09 ± 0.06	0.13 ± 0.09	0.08 ± 0.07
GPT-2-medium	GD	0.75 ± 0.00	0.05 ± 0.05	-0.36 ± 0.05	0.00 ± 0.00	0.45 ± 0.05	0.44 ± 0.06	0.46 ± 0.04
GPT-2-medium	NPO	0.75 ± 0.00	0.13 ± 0.12	-0.28 ± 0.12	0.00 ± 0.00	0.37 ± 0.12	0.36 ± 0.13	0.38 ± 0.13
Pythia-1B (LoRA)	Retrain	0.56 ± 0.00	0.41 ± 0.00	0.00 ± 0.00	0.02 ± 0.00	0.09 ± 0.00	0.04 ± 0.01	0.09 ± 0.01
Pythia-1B (LoRA)	GA†	0.56 ± 0.00	0.49 ± 0.24	0.08 ± 0.24	0.03 ± 0.03	0.14 ± 0.14	0.18 ± 0.11	0.13 ± 0.15
Pythia-1B (LoRA)	GD†	0.56 ± 0.00	0.09 ± 0.06	-0.32 ± 0.06	0.00 ± 0.01	0.41 ± 0.06	0.39 ± 0.07	0.43 ± 0.05
Pythia-1B (LoRA)	NPO	0.56 ± 0.00	0.07 ± 0.04	-0.35 ± 0.05	0.00 ± 0.00	0.43 ± 0.04	0.43 ± 0.04	0.43 ± 0.09

Mean $\pm$ half-width of the 95% t-interval over seeds (up to 5); a single value means one seed. †Model collapsed: retain loss rose more than threefold in 5/5 seeds. GA: all four central models; GD: Pythia-1B (LoRA).

Table S3: Membership inference before and after centralized unlearning (Min-K%, primary non-members). Gap is the post-unlearning AUC minus the retrained model's (same seed); advantage is $|\text{AUC} - 0.5|$; outliers by base-model difficulty, stratum-matched.

		MIA AUC				Post-unlearning advantage		
Model	**Method**	**Pre**	**Post**	**Gap**	**TPR@1%**	**All**	**Typical**	**Outlier**
DistilGPT-2	Retrain	0.63 ± 0.01	0.44 ± 0.00	0.00 ± 0.00	0.01 ± 0.01	0.06 ± 0.00	0.05 ± 0.01	0.04 ± 0.04
DistilGPT-2	GA†	0.63 ± 0.01	0.45 ± 0.04	0.01 ± 0.03	0.01 ± 0.00	0.05 ± 0.04	0.05 ± 0.03	0.05 ± 0.02
DistilGPT-2	GD	0.63 ± 0.01	0.40 ± 0.02	-0.04 ± 0.02	0.00 ± 0.00	0.10 ± 0.02	0.11 ± 0.02	0.03 ± 0.03
DistilGPT-2	NPO	0.63 ± 0.01	0.41 ± 0.03	-0.03 ± 0.03	0.00 ± 0.00	0.09 ± 0.03	0.11 ± 0.03	0.04 ± 0.03
GPT-2	Retrain	0.66 ± 0.01	0.44 ± 0.01	0.00 ± 0.00	0.01 ± 0.00	0.06 ± 0.01	0.06 ± 0.02	0.04 ± 0.02
GPT-2	GA†	0.66 ± 0.01	0.46 ± 0.04	0.02 ± 0.04	0.01 ± 0.01	0.04 ± 0.04	0.05 ± 0.03	0.08 ± 0.03
GPT-2	GD	0.66 ± 0.01	0.38 ± 0.05	-0.05 ± 0.05	0.01 ± 0.01	0.12 ± 0.05	0.12 ± 0.03	0.07 ± 0.09
GPT-2	NPO	0.66 ± 0.01	0.35 ± 0.11	-0.08 ± 0.11	0.01 ± 0.01	0.15 ± 0.11	0.16 ± 0.11	0.09 ± 0.14
GPT-2-medium	Retrain	0.75 ± 0.02	0.41 ± 0.01	0.00 ± 0.00	0.01 ± 0.01	0.09 ± 0.01	0.07 ± 0.02	0.07 ± 0.02
GPT-2-medium	GA†	0.75 ± 0.02	0.47 ± 0.02	0.06 ± 0.03	0.01 ± 0.01	0.03 ± 0.02	0.04 ± 0.03	0.05 ± 0.06
GPT-2-medium	GD	0.75 ± 0.02	0.03 ± 0.04	-0.38 ± 0.04	0.00 ± 0.00	0.47 ± 0.04	0.46 ± 0.04	0.47 ± 0.04
GPT-2-medium	NPO	0.75 ± 0.02	0.09 ± 0.10	-0.32 ± 0.11	0.00 ± 0.00	0.41 ± 0.10	0.41 ± 0.10	0.40 ± 0.13

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Table S3 (continued)

Model	Method	MIA AUC		Gap	TPR@1%	Post-unlearning advantage		
		Pre	Post			All	Typical	Outlier
Pythia-1B (LoRA)	Retrain	0.58 ± 0.01	0.42 ± 0.01	0.00 ± 0.00	0.01 ± 0.01	0.08 ± 0.01	0.07 ± 0.02	0.06 ± 0.05
Pythia-1B (LoRA)	GA†	0.58 ± 0.01	0.44 ± 0.18	0.02 ± 0.17	0.01 ± 0.01	0.09 ± 0.16	0.09 ± 0.16	0.09 ± 0.15
Pythia-1B (LoRA)	GD†	0.58 ± 0.01	0.05 ± 0.04	-0.37 ± 0.05	0.00 ± 0.00	0.45 ± 0.04	0.44 ± 0.06	0.47 ± 0.02
Pythia-1B (LoRA)	NPO	0.58 ± 0.01	0.03 ± 0.03	-0.38 ± 0.03	0.00 ± 0.00	0.47 ± 0.03	0.46 ± 0.03	0.48 ± 0.04

Mean ± half-width of the 95% t-interval over seeds (up to 5); a single value means one seed. †Model collapsed: retain loss rose more than threefold in 5/5 seeds. GA: all four central models; GD: Pythia-1B (LoRA).

Table S4: Membership inference before and after centralized unlearning (Min-K%, holdout10 non-members). Gap is the post-unlearning AUC minus the retrained model's (same seed); advantage is $|\text{AUC} - 0.5|$; outliers by base-model difficulty, stratum-matched.

Model	Method	MIA AUC		Gap	TPR@1%	Post-unlearning advantage		
		Pre	Post			All	Typical	Outlier
DistilGPT-2	Retrain	0.63 ± 0.00	0.42 ± 0.01	0.00 ± 0.00	0.02 ± 0.01	0.08 ± 0.01	0.02 ± 0.01	0.08 ± 0.01
DistilGPT-2	GA†	0.63 ± 0.00	0.52 ± 0.06	0.10 ± 0.06	0.01 ± 0.01	0.04 ± 0.02	0.08 ± 0.05	0.05 ± 0.04
DistilGPT-2	GD	0.63 ± 0.00	0.51 ± 0.02	0.09 ± 0.02	0.00 ± 0.00	0.01 ± 0.01	0.06 ± 0.02	0.04 ± 0.02
DistilGPT-2	NPO	0.63 ± 0.00	0.48 ± 0.03	0.06 ± 0.03	0.00 ± 0.00	0.02 ± 0.03	0.03 ± 0.02	0.04 ± 0.03
GPT-2	Retrain	0.65 ± 0.00	0.41 ± 0.00	0.00 ± 0.00	0.02 ± 0.00	0.09 ± 0.00	0.01 ± 0.01	0.08 ± 0.01
GPT-2	GA†	0.65 ± 0.00	0.54 ± 0.03	0.14 ± 0.03	0.02 ± 0.02	0.04 ± 0.03	0.09 ± 0.04	0.07 ± 0.05
GPT-2	GD	0.65 ± 0.00	0.48 ± 0.06	0.07 ± 0.06	0.01 ± 0.00	0.03 ± 0.04	0.05 ± 0.04	0.05 ± 0.07
GPT-2	NPO	0.65 ± 0.00	0.40 ± 0.11	-0.01 ± 0.12	0.00 ± 0.00	0.10 ± 0.11	0.07 ± 0.11	0.09 ± 0.10
GPT-2-medium	Retrain	0.76 ± 0.00	0.38 ± 0.00	0.00 ± 0.00	0.02 ± 0.00	0.12 ± 0.00	0.01 ± 0.01	0.13 ± 0.01
GPT-2-medium	GA†	0.76 ± 0.00	0.52 ± 0.12	0.13 ± 0.12	0.01 ± 0.01	0.08 ± 0.03	0.14 ± 0.04	0.06 ± 0.04
GPT-2-medium	GD	0.76 ± 0.00	0.06 ± 0.06	-0.32 ± 0.06	0.00 ± 0.00	0.44 ± 0.06	0.43 ± 0.07	0.46 ± 0.04
GPT-2-medium	NPO	0.76 ± 0.00	0.13 ± 0.12	-0.25 ± 0.12	0.00 ± 0.00	0.37 ± 0.12	0.36 ± 0.12	0.37 ± 0.14
Pythia-1B (LoRA)	Retrain	0.54 ± 0.00	0.38 ± 0.01	0.00 ± 0.00	0.02 ± 0.00	0.12 ± 0.01	0.03 ± 0.01	0.10 ± 0.01
Pythia-1B (LoRA)	GA†	0.54 ± 0.00	0.42 ± 0.20	0.04 ± 0.20	0.02 ± 0.02	0.11 ± 0.17	0.12 ± 0.17	0.10 ± 0.18
Pythia-1B (LoRA)	GD†	0.54 ± 0.00	0.09 ± 0.06	-0.29 ± 0.06	0.00 ± 0.01	0.41 ± 0.06	0.40 ± 0.07	0.43 ± 0.05
Pythia-1B (LoRA)	NPO	0.54 ± 0.00	0.06 ± 0.04	-0.32 ± 0.05	0.00 ± 0.00	0.44 ± 0.04	0.43 ± 0.03	0.43 ± 0.08

Mean ± half-width of the 95% t-interval over seeds (up to 5); a single value means one seed. †Model collapsed: retain loss rose more than threefold in 5/5 seeds. GA: all four central models; GD: Pythia-1B (LoRA).

Table S5: Membership inference before and after centralized unlearning (Ref.-calib., primary non-members). Gap is the post-unlearning AUC minus the retrained model's (same seed); advantage is $|\text{AUC} - 0.5|$; outliers by base-model difficulty, stratum-matched.

		MIA AUC				Post-unlearning advantage		
Model	**Method**	**Pre**	**Post**	**Gap**	**TPR@1%**	**All**	**Typical**	**Outlier**
DistilGPT-2	Retrain	0.75 ± 0.02	0.46 ± 0.03	0.00 ± 0.00	0.01 ± 0.00	0.04 ± 0.03	0.04 ± 0.02	0.06 ± 0.04
DistilGPT-2	GA†	0.75 ± 0.02	0.48 ± 0.03	0.03 ± 0.05	0.01 ± 0.01	0.02 ± 0.01	0.04 ± 0.01	0.03 ± 0.04
DistilGPT-2	GD	0.75 ± 0.02	0.40 ± 0.02	-0.05 ± 0.04	0.01 ± 0.01	0.10 ± 0.02	0.10 ± 0.02	0.09 ± 0.06
DistilGPT-2	NPO	0.75 ± 0.02	0.40 ± 0.03	-0.05 ± 0.01	0.00 ± 0.00	0.10 ± 0.03	0.13 ± 0.02	0.06 ± 0.06
GPT-2	Retrain	0.80 ± 0.02	0.45 ± 0.02	0.00 ± 0.00	0.01 ± 0.00	0.05 ± 0.02	0.05 ± 0.02	0.07 ± 0.02
GPT-2	GA†	0.80 ± 0.02	0.47 ± 0.04	0.03 ± 0.05	0.01 ± 0.01	0.03 ± 0.03	0.04 ± 0.03	0.07 ± 0.04
GPT-2	GD	0.80 ± 0.02	0.39 ± 0.06	-0.06 ± 0.05	0.00 ± 0.00	0.11 ± 0.06	0.11 ± 0.04	0.12 ± 0.10
GPT-2	NPO	0.80 ± 0.02	0.35 ± 0.12	-0.10 ± 0.11	0.00 ± 0.00	0.15 ± 0.12	0.17 ± 0.10	0.13 ± 0.14
GPT-2-medium	Retrain	0.89 ± 0.02	0.45 ± 0.01	0.00 ± 0.00	0.01 ± 0.00	0.05 ± 0.01	0.06 ± 0.02	0.04 ± 0.04
GPT-2-medium	GA†	0.89 ± 0.02	0.51 ± 0.04	0.06 ± 0.04	0.01 ± 0.01	0.02 ± 0.02	0.02 ± 0.02	0.05 ± 0.06
GPT-2-medium	GD	0.89 ± 0.02	0.03 ± 0.04	-0.42 ± 0.05	0.00 ± 0.00	0.47 ± 0.04	0.47 ± 0.04	0.47 ± 0.03
GPT-2-medium	NPO	0.89 ± 0.02	0.09 ± 0.11	-0.36 ± 0.12	0.00 ± 0.00	0.41 ± 0.11	0.41 ± 0.10	0.41 ± 0.12
Pythia-1B (LoRA)	Retrain	0.74 ± 0.03	0.46 ± 0.02	0.00 ± 0.00	0.00 ± 0.00	0.04 ± 0.02	0.05 ± 0.01	0.04 ± 0.04
Pythia-1B (LoRA)	GA†	0.74 ± 0.03	0.44 ± 0.18	-0.02 ± 0.19	0.01 ± 0.00	0.08 ± 0.17	0.09 ± 0.17	0.11 ± 0.13
Pythia-1B (LoRA)	GD†	0.74 ± 0.03	0.05 ± 0.04	-0.40 ± 0.05	0.00 ± 0.00	0.45 ± 0.04	0.44 ± 0.06	0.47 ± 0.02
Pythia-1B (LoRA)	NPO	0.74 ± 0.03	0.04 ± 0.04	-0.42 ± 0.06	0.00 ± 0.00	0.46 ± 0.04	0.46 ± 0.03	0.47 ± 0.05

Mean $\pm$ half-width of the 95% t-interval over seeds (up to 5); a single value means one seed. †Model collapsed: retain loss rose more than threefold in 5/5 seeds. GA: all four central models; GD: Pythia-1B (LoRA).

Table S6: Membership inference before and after centralized unlearning (Ref.-calib., holdout10 non-members). Gap is the post-unlearning AUC minus the retrained model's (same seed); advantage is $|\text{AUC} - 0.5|$; outliers by base-model difficulty, stratum-matched.

		MIA AUC				Post-unlearning advantage		
Model	**Method**	**Pre**	**Post**	**Gap**	**TPR@1%**	**All**	**Typical**	**Outlier**
DistilGPT-2	Retrain	0.86 ± 0.00	0.61 ± 0.01	0.00 ± 0.00	0.02 ± 0.00	0.11 ± 0.01	0.12 ± 0.01	0.03 ± 0.01
DistilGPT-2	GA†	0.86 ± 0.00	0.58 ± 0.10	-0.02 ± 0.11	0.05 ± 0.02	0.09 ± 0.10	0.13 ± 0.08	0.08 ± 0.07
DistilGPT-2	GD	0.86 ± 0.00	0.60 ± 0.02	-0.01 ± 0.02	0.03 ± 0.02	0.10 ± 0.02	0.10 ± 0.01	0.06 ± 0.02
DistilGPT-2	NPO	0.86 ± 0.00	0.55 ± 0.04	-0.06 ± 0.04	0.01 ± 0.01	0.05 ± 0.04	0.03 ± 0.02	0.04 ± 0.02
GPT-2	Retrain	0.91 ± 0.00	0.61 ± 0.00	0.00 ± 0.00	0.03 ± 0.00	0.11 ± 0.00	0.11 ± 0.01	0.04 ± 0.01
GPT-2	GA†	0.91 ± 0.00	0.59 ± 0.10	-0.02 ± 0.10	0.03 ± 0.05	0.10 ± 0.09	0.11 ± 0.10	0.11 ± 0.06
GPT-2	GD	0.91 ± 0.00	0.59 ± 0.08	-0.02 ± 0.08	0.03 ± 0.01	0.09 ± 0.06	0.09 ± 0.06	0.09 ± 0.04
GPT-2	NPO	0.91 ± 0.00	0.50 ± 0.14	-0.11 ± 0.14	0.02 ± 0.01	0.08 ± 0.09	0.06 ± 0.10	0.08 ± 0.11
GPT-2-medium	Retrain	0.95 ± 0.00	0.60 ± 0.01	0.00 ± 0.00	0.03 ± 0.01	0.10 ± 0.01	0.10 ± 0.01	0.01 ± 0.01
GPT-2-medium	GA†	0.95 ± 0.00	0.58 ± 0.11	-0.02 ± 0.12	0.03 ± 0.02	0.10 ± 0.07	0.13 ± 0.09	0.08 ± 0.08
GPT-2-medium	GD	0.95 ± 0.00	0.05 ± 0.05	-0.54 ± 0.05	0.00 ± 0.00	0.45 ± 0.05	0.44 ± 0.06	0.46 ± 0.04
GPT-2-medium	NPO	0.95 ± 0.00	0.14 ± 0.14	-0.46 ± 0.13	0.00 ± 0.00	0.36 ± 0.14	0.36 ± 0.13	0.37 ± 0.14

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Table S6 (continued)

Model	Method	MIA AUC		Gap	TPR@1%	Post-unlearning advantage		
		Pre	Post			All	Typical	Outlier
Pythia-1B (LoRA)	Retrain	0.87 ± 0.01	0.62 ± 0.01	0.00 ± 0.00	0.05 ± 0.02	0.12 ± 0.01	0.12 ± 0.01	0.06 ± 0.01
Pythia-1B (LoRA)	GA†	0.87 ± 0.01	0.54 ± 0.27	-0.08 ± 0.26	0.04 ± 0.04	0.17 ± 0.14	0.18 ± 0.13	0.15 ± 0.14
Pythia-1B (LoRA)	GD†	0.87 ± 0.01	0.09 ± 0.06	-0.53 ± 0.07	0.00 ± 0.01	0.41 ± 0.06	0.39 ± 0.07	0.43 ± 0.05
Pythia-1B (LoRA)	NPO	0.87 ± 0.01	0.07 ± 0.06	-0.55 ± 0.07	0.00 ± 0.00	0.43 ± 0.06	0.43 ± 0.04	0.43 ± 0.10

Mean ± half-width of the 95% t-interval over seeds (up to 5); a single value means one seed. †Model collapsed: retain loss rose more than threefold in 5/5 seeds. GA: all four central models; GD: Pythia-1B (LoRA).

Table S7: Membership inference before and after federated unlearning (Loss thr., primary non-members). Gap is the post-unlearning AUC minus the retrained model's (same seed); advantage is $|\text{AUC} - 0.5|$; outliers by base-model difficulty, stratum-matched.

Model	Setting	Stage	MIA AUC		Gap	TPR@1%	Post-unlearning advantage			Update AUC	
			Pre	Post			All	Typical	Outlier	Score chg.	Grad. align.
GPT-2	by_author, $K = 5$	Retrain	0.49 ± 0.03	0.44 ± 0.03	0.00 ± 0.00	0.01 ± 0.01	0.06 ± 0.03	0.03 ± 0.07	0.09 ± 0.08	0.97 ± 0.01	0.97 ± 0.01
GPT-2	by_author, $K = 5$	Client GA, upload†	0.49 ± 0.03	0.42 ± 0.07	-0.02 ± 0.10	0.01 ± 0.01	0.08 ± 0.07	0.08 ± 0.08	0.08 ± 0.18	0.57 ± 0.07	0.64 ± 0.05
GPT-2	by_author, $K = 5$	Client GA, global	0.49 ± 0.03	0.44 ± 0.01	0.00 ± 0.03	0.01 ± 0.00	0.06 ± 0.01	0.03 ± 0.07	0.07 ± 0.06	0.80 ± 0.20	0.75 ± 0.08
GPT-2	iid, $K = 3$	Retrain	0.48 ± 0.03	0.44 ± 0.02	0.00 ± 0.00	0.02 ± 0.00	0.06 ± 0.02	0.04 ± 0.06	0.09 ± 0.06	0.92 ± 0.02	0.92 ± 0.01
GPT-2	iid, $K = 3$	Client GA, upload†	0.48 ± 0.03	0.43 ± 0.01	-0.01 ± 0.03	0.01 ± 0.01	0.07 ± 0.01	0.07 ± 0.05	0.08 ± 0.03	0.57 ± 0.01	0.63 ± 0.07
GPT-2	iid, $K = 3$	Client GA, global	0.48 ± 0.03	0.42 ± 0.03	-0.02 ± 0.02	0.01 ± 0.01	0.08 ± 0.03	0.06 ± 0.08	0.11 ± 0.05	0.87 ± 0.15	0.79 ± 0.12
GPT-2	iid, $K = 10$	Retrain	0.49 ± 0.03	0.45 ± 0.03	0.00 ± 0.00	0.01 ± 0.01	0.05 ± 0.03	0.02 ± 0.05	0.09 ± 0.07	0.99 ± 0.01	0.99 ± 0.01
GPT-2	iid, $K = 10$	Client GA, upload†	0.49 ± 0.03	0.39 ± 0.12	-0.06 ± 0.14	0.01 ± 0.02	0.11 ± 0.12	0.11 ± 0.15	0.11 ± 0.13	0.61 ± 0.10	0.56 ± 0.05
GPT-2	iid, $K = 10$	Client GA, global	0.49 ± 0.03	0.49 ± 0.04	0.04 ± 0.01	0.01 ± 0.01	0.01 ± 0.03	0.03 ± 0.07	0.02 ± 0.04	0.66 ± 0.06	0.65 ± 0.06
DistilGPT-2	iid, $K = 5$	Retrain	0.49 ± 0.01	0.45 ± 0.02	0.00 ± 0.00	0.01 ± 0.01	0.05 ± 0.02	0.02 ± 0.02	0.06 ± 0.05	0.96 ± 0.01	0.96 ± 0.01
DistilGPT-2	iid, $K = 5$	Client GA, upload†	0.49 ± 0.01	0.43 ± 0.05	-0.02 ± 0.06	0.01 ± 0.01	0.07 ± 0.05	0.07 ± 0.05	0.08 ± 0.08	0.56 ± 0.05	0.62 ± 0.02
DistilGPT-2	iid, $K = 5$	Client GA, global	0.49 ± 0.01	0.43 ± 0.02	-0.02 ± 0.01	0.01 ± 0.01	0.07 ± 0.02	0.05 ± 0.02	0.07 ± 0.06	0.74 ± 0.08	0.67 ± 0.04
GPT-2	iid, $K = 5$	Retrain	0.49 ± 0.02	0.45 ± 0.02	0.00 ± 0.00	0.01 ± 0.01	0.05 ± 0.02	0.03 ± 0.03	0.08 ± 0.04	0.97 ± 0.01	0.97 ± 0.01
GPT-2	iid, $K = 5$	Client GA, upload†	0.49 ± 0.02	0.41 ± 0.04	-0.03 ± 0.05	0.01 ± 0.01	0.09 ± 0.04	0.09 ± 0.02	0.09 ± 0.08	0.58 ± 0.04	0.64 ± 0.02
GPT-2	iid, $K = 5$	Client GA, global	0.49 ± 0.02	0.45 ± 0.02	0.01 ± 0.01	0.01 ± 0.01	0.05 ± 0.02	0.02 ± 0.03	0.05 ± 0.03	0.77 ± 0.11	0.74 ± 0.07
GPT-2-medium	iid, $K = 5$	Retrain	0.52 ± 0.02	0.43 ± 0.02	0.00 ± 0.00	0.01 ± 0.00	0.07 ± 0.02	0.04 ± 0.02	0.05 ± 0.03	0.98 ± 0.00	0.98 ± 0.00
GPT-2-medium	iid, $K = 5$	Client GA, upload†	0.52 ± 0.02	0.46 ± 0.04	0.03 ± 0.05	0.00 ± 0.00	0.04 ± 0.04	0.05 ± 0.04	0.05 ± 0.05	0.54 ± 0.03	0.64 ± 0.02
GPT-2-medium	iid, $K = 5$	Client GA, global	0.52 ± 0.02	0.48 ± 0.03	0.04 ± 0.02	0.01 ± 0.01	0.03 ± 0.02	0.02 ± 0.02	0.04 ± 0.05	0.84 ± 0.06	0.80 ± 0.05

Mean ± half-width of the 95% t-interval over seeds (up to 5); a single value means one seed. †Model collapsed: retain loss rose more than threefold in every seed. client-GA upload: all six federated runs (intermediate model before recovery; the released global model is not collapsed).

Table S8: Membership inference before and after federated unlearning (Loss thr., holdout10 non-members). Gap is the post-unlearning AUC minus the retrained model's (same seed); advantage is |AUC − 0.5|; outliers by base-model difficulty, stratum-matched.

			MIA AUC				Post-unlearning advantage			Update AUC	
Model	**Setting**	**Stage**	**Pre**	**Post**	**Gap**	**TPR@1%**	**All**	**Typical**	**Outlier**	**Score chg.**	**Grad. align.**
GPT-2	by_author, $K = 5$	Retrain	0.47 ± 0.00	0.43 ± 0.00	0.00 ± 0.00	0.02 ± 0.00	0.07 ± 0.00	0.07 ± 0.00	0.08 ± 0.01	0.97 ± 0.02	0.96 ± 0.02
GPT-2	by_author, $K = 5$	Client GA, upload†	0.47 ± 0.00	0.49 ± 0.27	0.07 ± 0.28	0.02 ± 0.04	0.09 ± 0.08	0.13 ± 0.16	0.07 ± 0.06	0.49 ± 0.28	0.55 ± 0.14
GPT-2	by_author, $K = 5$	Client GA, global	0.47 ± 0.00	0.43 ± 0.03	0.01 ± 0.02	0.02 ± 0.01	0.07 ± 0.03	0.08 ± 0.03	0.06 ± 0.05	0.78 ± 0.21	0.68 ± 0.05
GPT-2	iid, $K = 3$	Retrain	0.47 ± 0.01	0.43 ± 0.01	0.00 ± 0.00	0.02 ± 0.00	0.07 ± 0.01	0.07 ± 0.02	0.08 ± 0.01	0.93 ± 0.01	0.92 ± 0.02
GPT-2	iid, $K = 3$	Client GA, upload†	0.47 ± 0.01	0.57 ± 0.15	0.14 ± 0.14	0.02 ± 0.02	0.07 ± 0.15	0.09 ± 0.21	0.10 ± 0.19	0.43 ± 0.16	0.54 ± 0.14
GPT-2	iid, $K = 3$	Client GA, global	0.47 ± 0.01	0.41 ± 0.02	-0.01 ± 0.02	0.02 ± 0.00	0.09 ± 0.02	0.05 ± 0.04	0.09 ± 0.03	0.85 ± 0.16	0.72 ± 0.11
GPT-2	iid, $K = 10$	Retrain	0.47 ± 0.00	0.42 ± 0.00	0.00 ± 0.00	0.02 ± 0.00	0.08 ± 0.00	0.07 ± 0.00	0.08 ± 0.00	0.99 ± 0.00	0.99 ± 0.00
GPT-2	iid, $K = 10$	Client GA, upload†	0.47 ± 0.00	0.46 ± 0.22	0.04 ± 0.22	0.01 ± 0.01	0.08 ± 0.05	0.11 ± 0.08	0.07 ± 0.10	0.53 ± 0.22	0.50 ± 0.11
GPT-2	iid, $K = 10$	Client GA, global	0.47 ± 0.00	0.46 ± 0.00	0.04 ± 0.00	0.02 ± 0.01	0.04 ± 0.00	0.12 ± 0.00	0.01 ± 0.00	0.60 ± 0.02	0.60 ± 0.02
DistilGPT-2	iid, $K = 5$	Retrain	0.47 ± 0.00	0.43 ± 0.00	0.00 ± 0.00	0.02 ± 0.00	0.07 ± 0.00	0.07 ± 0.00	0.08 ± 0.00	0.96 ± 0.01	0.95 ± 0.01
DistilGPT-2	iid, $K = 5$	Client GA, upload†	0.47 ± 0.00	0.47 ± 0.08	0.04 ± 0.08	0.01 ± 0.01	0.06 ± 0.03	0.07 ± 0.04	0.08 ± 0.05	0.52 ± 0.07	0.60 ± 0.04
DistilGPT-2	iid, $K = 5$	Client GA, global	0.47 ± 0.00	0.42 ± 0.01	-0.01 ± 0.01	0.01 ± 0.00	0.08 ± 0.01	0.05 ± 0.02	0.08 ± 0.03	0.72 ± 0.06	0.65 ± 0.03
GPT-2	iid, $K = 5$	Retrain	0.47 ± 0.00	0.42 ± 0.00	0.00 ± 0.00	0.02 ± 0.00	0.08 ± 0.00	0.07 ± 0.00	0.08 ± 0.01	0.97 ± 0.01	0.96 ± 0.01
GPT-2	iid, $K = 5$	Client GA, upload†	0.47 ± 0.00	0.47 ± 0.08	0.04 ± 0.08	0.01 ± 0.01	0.06 ± 0.04	0.08 ± 0.06	0.07 ± 0.04	0.52 ± 0.07	0.58 ± 0.03
GPT-2	iid, $K = 5$	Client GA, global	0.47 ± 0.00	0.43 ± 0.01	0.01 ± 0.01	0.02 ± 0.00	0.07 ± 0.01	0.08 ± 0.01	0.07 ± 0.02	0.76 ± 0.09	0.70 ± 0.04
GPT-2-medium	iid, $K = 5$	Retrain	0.50 ± 0.00	0.41 ± 0.00	0.00 ± 0.00	0.02 ± 0.00	0.09 ± 0.00	0.05 ± 0.00	0.11 ± 0.01	0.98 ± 0.00	0.98 ± 0.00
GPT-2-medium	iid, $K = 5$	Client GA, upload†	0.50 ± 0.00	0.50 ± 0.08	0.09 ± 0.08	0.01 ± 0.01	0.06 ± 0.03	0.09 ± 0.06	0.07 ± 0.04	0.50 ± 0.08	0.59 ± 0.05
GPT-2-medium	iid, $K = 5$	Client GA, global	0.50 ± 0.00	0.46 ± 0.02	0.05 ± 0.02	0.02 ± 0.00	0.04 ± 0.02	0.10 ± 0.02	0.02 ± 0.02	0.84 ± 0.06	0.76 ± 0.04

Mean ± half-width of the 95% t-interval over seeds (up to 5); a single value means one seed. †Model collapsed: retain loss rose more than threefold in every seed. client-GA upload: all six federated runs (intermediate model before recovery; the released global model is not collapsed).

Table S9: Membership inference before and after federated unlearning (Min-K%, primary non-members). Gap is the post-unlearning AUC minus the retrained model's (same seed); advantage is |AUC − 0.5|; outliers by base-model difficulty, stratum-matched.

			MIA AUC				Post-unlearning advantage			Update AUC	
Model	**Setting**	**Stage**	**Pre**	**Post**	**Gap**	**TPR@1%**	**All**	**Typical**	**Outlier**	**Score chg.**	**Grad. align.**
GPT-2	by_author, $K = 5$	Retrain	0.49 ± 0.01	0.44 ± 0.01	0.00 ± 0.00	0.01 ± 0.01	0.06 ± 0.01	0.06 ± 0.05	0.02 ± 0.05	0.94 ± 0.03	0.97 ± 0.01
GPT-2	by_author, $K = 5$	Client GA, upload†	0.49 ± 0.01	0.39 ± 0.12	-0.05 ± 0.13	0.00 ± 0.00	0.11 ± 0.12	0.12 ± 0.12	0.11 ± 0.21	0.60 ± 0.11	0.64 ± 0.05
GPT-2	by_author, $K = 5$	Client GA, global	0.49 ± 0.01	0.44 ± 0.02	0.00 ± 0.03	0.01 ± 0.01	0.06 ± 0.02	0.05 ± 0.07	0.02 ± 0.05	0.75 ± 0.22	0.75 ± 0.08
GPT-2	iid, $K = 3$	Retrain	0.48 ± 0.01	0.43 ± 0.01	0.00 ± 0.00	0.01 ± 0.02	0.07 ± 0.01	0.06 ± 0.05	0.02 ± 0.05	0.87 ± 0.02	0.92 ± 0.01
GPT-2	iid, $K = 3$	Client GA, upload†	0.48 ± 0.01	0.40 ± 0.11	-0.03 ± 0.11	0.00 ± 0.00	0.10 ± 0.11	0.09 ± 0.15	0.13 ± 0.08	0.58 ± 0.08	0.63 ± 0.07
GPT-2	iid, $K = 3$	Client GA, global	0.48 ± 0.01	0.42 ± 0.02	-0.01 ± 0.02	0.01 ± 0.01	0.08 ± 0.02	0.07 ± 0.07	0.04 ± 0.08	0.80 ± 0.19	0.79 ± 0.12
GPT-2	iid, $K = 10$	Retrain	0.51 ± 0.01	0.45 ± 0.01	0.00 ± 0.00	0.01 ± 0.00	0.05 ± 0.01	0.04 ± 0.04	0.03 ± 0.07	0.98 ± 0.01	0.99 ± 0.01
GPT-2	iid, $K = 10$	Client GA, upload†	0.51 ± 0.01	0.38 ± 0.08	-0.07 ± 0.08	0.00 ± 0.01	0.12 ± 0.08	0.13 ± 0.13	0.10 ± 0.13	0.61 ± 0.08	0.56 ± 0.05
GPT-2	iid, $K = 10$	Client GA, global	0.51 ± 0.01	0.50 ± 0.01	0.05 ± 0.01	0.01 ± 0.00	0.00 ± 0.01	0.02 ± 0.03	0.06 ± 0.09	0.62 ± 0.08	0.65 ± 0.06
DistilGPT-2	iid, $K = 5$	Retrain	0.49 ± 0.00	0.45 ± 0.00	0.00 ± 0.00	0.01 ± 0.00	0.05 ± 0.00	0.04 ± 0.01	0.03 ± 0.04	0.92 ± 0.01	0.96 ± 0.01

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Table S9 (continued)

			MIA AUC				Post-unlearning advantage			Update AUC	
Model	**Setting**	**Stage**	**Pre**	**Post**	**Gap**	**TPR@1%**	**All**	**Typical**	**Outlier**	**Score chg.**	**Grad. align.**
DistilGPT-2	iid, $K = 5$	Client GA, upload†	0.49 ± 0.00	0.41 ± 0.06	-0.04 ± 0.06	0.01 ± 0.00	0.09 ± 0.06	0.09 ± 0.07	0.10 ± 0.06	0.57 ± 0.05	0.62 ± 0.02
DistilGPT-2	iid, $K = 5$	Client GA, global	0.49 ± 0.00	0.44 ± 0.01	-0.01 ± 0.01	0.01 ± 0.00	0.06 ± 0.01	0.05 ± 0.01	0.03 ± 0.05	0.69 ± 0.10	0.67 ± 0.04
GPT-2	iid, $K = 5$	Retrain	0.49 ± 0.01	0.44 ± 0.01	0.00 ± 0.00	0.01 ± 0.01	0.06 ± 0.01	0.05 ± 0.02	0.02 ± 0.02	0.94 ± 0.00	0.97 ± 0.01
GPT-2	iid, $K = 5$	Client GA, upload†	0.49 ± 0.01	0.39 ± 0.03	-0.04 ± 0.03	0.00 ± 0.00	0.11 ± 0.03	0.11 ± 0.02	0.10 ± 0.07	0.59 ± 0.03	0.64 ± 0.02
GPT-2	iid, $K = 5$	Client GA, global	0.49 ± 0.01	0.45 ± 0.02	0.01 ± 0.01	0.01 ± 0.00	0.05 ± 0.02	0.05 ± 0.03	0.02 ± 0.02	0.72 ± 0.09	0.74 ± 0.07
GPT-2-medium	iid, $K = 5$	Retrain	0.50 ± 0.01	0.41 ± 0.01	0.00 ± 0.00	0.01 ± 0.00	0.09 ± 0.01	0.07 ± 0.02	0.06 ± 0.02	0.97 ± 0.00	0.98 ± 0.00
GPT-2-medium	iid, $K = 5$	Client GA, upload†	0.50 ± 0.01	0.44 ± 0.07	0.02 ± 0.07	0.00 ± 0.00	0.06 ± 0.07	0.07 ± 0.08	0.06 ± 0.04	0.56 ± 0.06	0.64 ± 0.02
GPT-2-medium	iid, $K = 5$	Client GA, global	0.50 ± 0.01	0.46 ± 0.02	0.05 ± 0.02	0.01 ± 0.01	0.04 ± 0.02	0.02 ± 0.02	0.02 ± 0.02	0.79 ± 0.05	0.80 ± 0.05

Mean ± half-width of the 95% t-interval over seeds (up to 5); a single value means one seed. †Model collapsed: retain loss rose more than threefold in every seed. client-GA upload: all six federated runs (intermediate model before recovery; the released global model is not collapsed).

Table S10: Membership inference before and after federated unlearning (Min-K%, holdout10 non-members). Gap is the post-unlearning AUC minus the retrained model's (same seed); advantage is $|\text{AUC} - 0.5|$; outliers by base-model difficulty, stratum-matched.

			MIA AUC				Post-unlearning advantage			Update AUC	
Model	**Setting**	**Stage**	**Pre**	**Post**	**Gap**	**TPR@1%**	**All**	**Typical**	**Outlier**	**Score chg.**	**Grad. align.**
GPT-2	by_author, $K = 5$	Retrain	0.46 ± 0.00	0.41 ± 0.00	0.00 ± 0.00	0.01 ± 0.00	0.09 ± 0.00	0.02 ± 0.00	0.06 ± 0.00	0.94 ± 0.04	0.96 ± 0.02
GPT-2	by_author, $K = 5$	Client GA, upload†	0.46 ± 0.00	0.47 ± 0.19	0.06 ± 0.20	0.01 ± 0.01	0.07 ± 0.05	0.11 ± 0.11	0.04 ± 0.10	0.50 ± 0.16	0.55 ± 0.14
GPT-2	by_author, $K = 5$	Client GA, global	0.46 ± 0.00	0.42 ± 0.03	0.01 ± 0.03	0.01 ± 0.00	0.08 ± 0.03	0.03 ± 0.05	0.05 ± 0.05	0.72 ± 0.20	0.68 ± 0.05
GPT-2	iid, $K = 3$	Retrain	0.46 ± 0.02	0.41 ± 0.01	0.00 ± 0.00	0.02 ± 0.00	0.09 ± 0.01	0.01 ± 0.02	0.08 ± 0.01	0.88 ± 0.02	0.92 ± 0.02
GPT-2	iid, $K = 3$	Client GA, upload†	0.46 ± 0.02	0.54 ± 0.16	0.13 ± 0.16	0.02 ± 0.05	0.06 ± 0.05	0.10 ± 0.11	0.06 ± 0.07	0.44 ± 0.14	0.54 ± 0.14
GPT-2	iid, $K = 3$	Client GA, global	0.46 ± 0.02	0.40 ± 0.02	-0.01 ± 0.01	0.02 ± 0.00	0.10 ± 0.02	0.01 ± 0.01	0.09 ± 0.03	0.79 ± 0.16	0.72 ± 0.11
GPT-2	iid, $K = 10$	Retrain	0.46 ± 0.00	0.41 ± 0.00	0.00 ± 0.00	0.02 ± 0.00	0.09 ± 0.00	0.01 ± 0.00	0.05 ± 0.01	0.98 ± 0.00	0.99 ± 0.00
GPT-2	iid, $K = 10$	Client GA, upload†	0.46 ± 0.00	0.46 ± 0.13	0.05 ± 0.13	0.00 ± 0.01	0.05 ± 0.08	0.09 ± 0.03	0.05 ± 0.10	0.52 ± 0.12	0.50 ± 0.11
GPT-2	iid, $K = 10$	Client GA, global	0.46 ± 0.00	0.46 ± 0.00	0.05 ± 0.00	0.02 ± 0.00	0.04 ± 0.00	0.07 ± 0.00	0.02 ± 0.01	0.59 ± 0.03	0.60 ± 0.02
DistilGPT-2	iid, $K = 5$	Retrain	0.47 ± 0.00	0.42 ± 0.00	0.00 ± 0.00	0.01 ± 0.00	0.08 ± 0.00	0.03 ± 0.00	0.07 ± 0.01	0.91 ± 0.01	0.95 ± 0.01
DistilGPT-2	iid, $K = 5$	Client GA, upload†	0.47 ± 0.00	0.47 ± 0.06	0.05 ± 0.06	0.02 ± 0.01	0.05 ± 0.01	0.06 ± 0.02	0.09 ± 0.02	0.50 ± 0.05	0.60 ± 0.04
DistilGPT-2	iid, $K = 5$	Client GA, global	0.47 ± 0.00	0.43 ± 0.01	0.00 ± 0.01	0.01 ± 0.00	0.07 ± 0.01	0.02 ± 0.01	0.06 ± 0.03	0.65 ± 0.08	0.65 ± 0.03
GPT-2	iid, $K = 5$	Retrain	0.46 ± 0.00	0.41 ± 0.00	0.00 ± 0.00	0.01 ± 0.00	0.09 ± 0.00	0.02 ± 0.00	0.07 ± 0.01	0.93 ± 0.01	0.96 ± 0.01
GPT-2	iid, $K = 5$	Client GA, upload†	0.46 ± 0.00	0.46 ± 0.09	0.05 ± 0.09	0.02 ± 0.01	0.07 ± 0.03	0.10 ± 0.06	0.08 ± 0.05	0.51 ± 0.08	0.58 ± 0.03
GPT-2	iid, $K = 5$	Client GA, global	0.46 ± 0.00	0.42 ± 0.01	0.01 ± 0.01	0.01 ± 0.00	0.08 ± 0.01	0.03 ± 0.02	0.06 ± 0.02	0.70 ± 0.08	0.70 ± 0.04
GPT-2-medium	iid, $K = 5$	Retrain	0.48 ± 0.00	0.38 ± 0.00	0.00 ± 0.00	0.02 ± 0.00	0.12 ± 0.00	0.01 ± 0.00	0.12 ± 0.01	0.97 ± 0.00	0.98 ± 0.00
GPT-2-medium	iid, $K = 5$	Client GA, upload†	0.48 ± 0.00	0.51 ± 0.08	0.13 ± 0.08	0.01 ± 0.01	0.05 ± 0.03	0.09 ± 0.05	0.05 ± 0.05	0.48 ± 0.07	0.59 ± 0.05
GPT-2-medium	iid, $K = 5$	Client GA, global	0.48 ± 0.00	0.43 ± 0.02	0.05 ± 0.02	0.02 ± 0.00	0.07 ± 0.02	0.05 ± 0.02	0.04 ± 0.02	0.78 ± 0.06	0.76 ± 0.04

Mean ± half-width of the 95% t-interval over seeds (up to 5); a single value means one seed. †Model collapsed: retain loss rose more than threefold in every seed. client-GA upload: all six federated runs (intermediate model before recovery; the released global model is not collapsed).

Table S11: Membership inference before and after federated unlearning (Ref.-calib., primary non-members). Gap is the post-unlearning AUC minus the retrained model's (same seed); advantage is $|\text{AUC} - 0.5|$; outliers by base-model difficulty, stratum-matched.

			MIA AUC				Post-unlearning advantage			Update AUC	
Model	**Setting**	**Stage**	**Pre**	**Post**	**Gap**	**TPR@1%**	**All**	**Typical**	**Outlier**	**Score chg.**	**Grad. align.**
GPT-2	by_author, $K = 5$	Retrain	0.55 ± 0.02	0.45 ± 0.02	0.00 ± 0.00	0.01 ± 0.00	0.05 ± 0.02	0.06 ± 0.01	0.07 ± 0.07	–	0.97 ± 0.01
GPT-2	by_author, $K = 5$	Client GA, upload†	0.55 ± 0.02	0.44 ± 0.07	-0.01 ± 0.09	0.01 ± 0.01	0.06 ± 0.07	0.07 ± 0.08	0.08 ± 0.17	–	0.64 ± 0.05
GPT-2	by_author, $K = 5$	Client GA, global	0.55 ± 0.02	0.45 ± 0.04	0.00 ± 0.06	0.00 ± 0.00	0.05 ± 0.04	0.05 ± 0.05	0.05 ± 0.02	–	0.75 ± 0.08
GPT-2	iid, $K = 3$	Retrain	0.53 ± 0.00	0.44 ± 0.01	0.00 ± 0.00	0.01 ± 0.01	0.06 ± 0.01	0.06 ± 0.02	0.07 ± 0.05	–	0.92 ± 0.01
GPT-2	iid, $K = 3$	Client GA, upload†	0.53 ± 0.00	0.43 ± 0.01	-0.01 ± 0.01	0.01 ± 0.02	0.07 ± 0.01	0.07 ± 0.05	0.08 ± 0.03	–	0.63 ± 0.07
GPT-2	iid, $K = 3$	Client GA, global	0.53 ± 0.00	0.41 ± 0.02	-0.04 ± 0.03	0.01 ± 0.00	0.09 ± 0.02	0.10 ± 0.04	0.09 ± 0.03	–	0.79 ± 0.12
GPT-2	iid, $K = 10$	Retrain	0.58 ± 0.04	0.45 ± 0.03	0.00 ± 0.00	0.01 ± 0.00	0.05 ± 0.03	0.05 ± 0.02	0.07 ± 0.05	–	0.99 ± 0.01
GPT-2	iid, $K = 10$	Client GA, upload†	0.58 ± 0.04	0.40 ± 0.11	-0.05 ± 0.11	0.00 ± 0.01	0.10 ± 0.11	0.11 ± 0.16	0.11 ± 0.11	–	0.56 ± 0.05
GPT-2	iid, $K = 10$	Client GA, global	0.58 ± 0.04	0.56 ± 0.03	0.11 ± 0.01	0.01 ± 0.00	0.06 ± 0.03	0.05 ± 0.02	0.06 ± 0.09	–	0.65 ± 0.06
DistilGPT-2	iid, $K = 5$	Retrain	0.55 ± 0.03	0.46 ± 0.03	0.00 ± 0.00	0.01 ± 0.00	0.04 ± 0.03	0.04 ± 0.02	0.05 ± 0.04	–	0.96 ± 0.01
DistilGPT-2	iid, $K = 5$	Client GA, upload†	0.55 ± 0.03	0.45 ± 0.05	-0.02 ± 0.04	0.01 ± 0.01	0.05 ± 0.05	0.06 ± 0.05	0.07 ± 0.09	–	0.62 ± 0.02
DistilGPT-2	iid, $K = 5$	Client GA, global	0.55 ± 0.03	0.43 ± 0.02	-0.03 ± 0.01	0.01 ± 0.00	0.07 ± 0.02	0.07 ± 0.02	0.06 ± 0.05	–	0.67 ± 0.04
GPT-2	iid, $K = 5$	Retrain	0.55 ± 0.01	0.45 ± 0.02	0.00 ± 0.00	0.01 ± 0.01	0.05 ± 0.02	0.05 ± 0.02	0.07 ± 0.02	–	0.97 ± 0.01
GPT-2	iid, $K = 5$	Client GA, upload†	0.55 ± 0.01	0.43 ± 0.04	-0.02 ± 0.05	0.00 ± 0.01	0.07 ± 0.04	0.09 ± 0.02	0.09 ± 0.08	–	0.64 ± 0.02
GPT-2	iid, $K = 5$	Client GA, global	0.55 ± 0.01	0.46 ± 0.02	0.01 ± 0.02	0.01 ± 0.00	0.04 ± 0.02	0.04 ± 0.03	0.05 ± 0.02	–	0.74 ± 0.07
GPT-2-medium	iid, $K = 5$	Retrain	0.62 ± 0.02	0.45 ± 0.01	0.00 ± 0.00	0.01 ± 0.00	0.05 ± 0.01	0.06 ± 0.02	0.04 ± 0.04	–	0.98 ± 0.00
GPT-2-medium	iid, $K = 5$	Client GA, upload†	0.62 ± 0.02	0.47 ± 0.03	0.02 ± 0.04	0.01 ± 0.01	0.03 ± 0.03	0.04 ± 0.04	0.05 ± 0.05	–	0.64 ± 0.02
GPT-2-medium	iid, $K = 5$	Client GA, global	0.62 ± 0.02	0.55 ± 0.04	0.09 ± 0.03	0.01 ± 0.01	0.05 ± 0.04	0.03 ± 0.04	0.06 ± 0.07	–	0.80 ± 0.05

Mean ± half-width of the 95% t-interval over seeds (up to 5); a single value means one seed. †Model collapsed: retain loss rose more than threefold in every seed. client-GA upload: all six federated runs (intermediate model before recovery; the released global model is not collapsed).

Table S12: Membership inference before and after federated unlearning (Ref.-calib., holdout10 non-members). Gap is the post-unlearning AUC minus the retrained model's (same seed); advantage is $|\text{AUC} - 0.5|$; outliers by base-model difficulty, stratum-matched.

			MIA AUC				Post-unlearning advantage			Update AUC	
Model	**Setting**	**Stage**	**Pre**	**Post**	**Gap**	**TPR@1%**	**All**	**Typical**	**Outlier**	**Score chg.**	**Grad. align.**
GPT-2	by_author, $K = 5$	Retrain	0.72 ± 0.01	0.62 ± 0.01	0.00 ± 0.00	0.02 ± 0.00	0.12 ± 0.01	0.12 ± 0.01	0.05 ± 0.01	–	0.96 ± 0.02
GPT-2	by_author, $K = 5$	Client GA, upload†	0.72 ± 0.01	0.54 ± 0.28	-0.08 ± 0.28	0.04 ± 0.07	0.07 ± 0.20	0.13 ± 0.17	0.06 ± 0.12	–	0.55 ± 0.14
GPT-2	by_author, $K = 5$	Client GA, global	0.72 ± 0.01	0.62 ± 0.09	-0.00 ± 0.08	0.02 ± 0.01	0.12 ± 0.09	0.12 ± 0.06	0.05 ± 0.10	–	0.68 ± 0.05
GPT-2	iid, $K = 3$	Retrain	0.69 ± 0.02	0.60 ± 0.02	0.00 ± 0.00	0.02 ± 0.00	0.10 ± 0.02	0.10 ± 0.02	0.03 ± 0.02	–	0.92 ± 0.02
GPT-2	iid, $K = 3$	Client GA, upload†	0.69 ± 0.02	0.60 ± 0.17	0.00 ± 0.15	0.03 ± 0.04	0.10 ± 0.17	0.09 ± 0.22	0.11 ± 0.24	–	0.54 ± 0.14
GPT-2	iid, $K = 3$	Client GA, global	0.69 ± 0.02	0.56 ± 0.04	-0.04 ± 0.03	0.02 ± 0.00	0.06 ± 0.04	0.06 ± 0.06	0.01 ± 0.03	–	0.72 ± 0.11
GPT-2	iid, $K = 10$	Retrain	0.75 ± 0.01	0.63 ± 0.00	0.00 ± 0.00	0.02 ± 0.01	0.13 ± 0.00	0.13 ± 0.01	0.06 ± 0.00	–	0.99 ± 0.00
GPT-2	iid, $K = 10$	Client GA, upload†	0.75 ± 0.01	0.50 ± 0.22	-0.13 ± 0.22	0.01 ± 0.01	0.07 ± 0.07	0.11 ± 0.08	0.07 ± 0.07	–	0.50 ± 0.11
GPT-2	iid, $K = 10$	Client GA, global	0.75 ± 0.01	0.73 ± 0.00	0.10 ± 0.00	0.05 ± 0.01	0.23 ± 0.00	0.22 ± 0.00	0.18 ± 0.01	–	0.60 ± 0.02
DistilGPT-2	iid, $K = 5$	Retrain	0.70 ± 0.00	0.62 ± 0.00	0.00 ± 0.00	0.03 ± 0.01	0.12 ± 0.00	0.13 ± 0.01	0.03 ± 0.01	–	0.95 ± 0.01

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Table S12 (continued)

			MIA AUC				Post-unlearning advantage			Update AUC	
Model	**Setting**	**Stage**	**Pre**	**Post**	**Gap**	**TPR@1%**	**All**	**Typical**	**Outlier**	**Score chg.**	**Grad. align.**
DistilGPT-2	iid, $K = 5$	Client GA, upload†	0.70 ± 0.00	0.51 ± 0.07	-0.11 ± 0.08	0.02 ± 0.03	0.04 ± 0.05	0.07 ± 0.04	0.07 ± 0.05	–	0.60 ± 0.04
DistilGPT-2	iid, $K = 5$	Client GA, global	0.70 ± 0.00	0.57 ± 0.04	-0.05 ± 0.04	0.02 ± 0.01	0.07 ± 0.04	0.08 ± 0.03	0.02 ± 0.02	–	0.65 ± 0.03
GPT-2	iid, $K = 5$	Retrain	0.72 ± 0.00	0.62 ± 0.00	0.00 ± 0.00	0.02 ± 0.01	0.12 ± 0.00	0.12 ± 0.00	0.04 ± 0.01	–	0.96 ± 0.01
GPT-2	iid, $K = 5$	Client GA, upload†	0.72 ± 0.00	0.50 ± 0.07	-0.11 ± 0.08	0.02 ± 0.02	0.05 ± 0.03	0.07 ± 0.06	0.06 ± 0.06	–	0.58 ± 0.03
GPT-2	iid, $K = 5$	Client GA, global	0.72 ± 0.00	0.62 ± 0.03	0.00 ± 0.03	0.03 ± 0.01	0.12 ± 0.03	0.12 ± 0.02	0.05 ± 0.03	–	0.70 ± 0.04
GPT-2-medium	iid, $K = 5$	Retrain	0.77 ± 0.00	0.61 ± 0.00	0.00 ± 0.00	0.03 ± 0.00	0.11 ± 0.00	0.11 ± 0.00	0.03 ± 0.01	–	0.98 ± 0.00
GPT-2-medium	iid, $K = 5$	Client GA, upload†	0.77 ± 0.00	0.53 ± 0.07	-0.08 ± 0.07	0.02 ± 0.02	0.05 ± 0.04	0.08 ± 0.05	0.06 ± 0.05	–	0.59 ± 0.05
GPT-2-medium	iid, $K = 5$	Client GA, global	0.77 ± 0.00	0.70 ± 0.03	0.09 ± 0.03	0.05 ± 0.01	0.20 ± 0.03	0.19 ± 0.04	0.13 ± 0.03	–	0.76 ± 0.04

Mean ± half-width of the 95% t-interval over seeds (up to 5); a single value means one seed. †Model collapsed: retain loss rose more than threefold in every seed. client-GA upload: all six federated runs (intermediate model before recovery; the released global model is not collapsed).

2 Hypotheses

Table S13: H1: post-unlearning AUC minus the retrained model's AUC, paired by seed.

Model	**Run**	**Method**	**Attack**	**Non-members**	**Gap**	**Verdict**
DistilGPT-2	central	GA†	Loss thr.	primary	0.027 ± 0.033	inconclusive
DistilGPT-2	central	GA†	Loss thr.	holdout10	0.115 ± 0.109	above retrain
DistilGPT-2	central	GA†	Min-K%	primary	0.012 ± 0.035	inconclusive
DistilGPT-2	central	GA†	Min-K%	holdout10	0.095 ± 0.063	above retrain
DistilGPT-2	central	GA†	Ref.-calib.	primary	0.029 ± 0.049	inconclusive
DistilGPT-2	central	GA†	Ref.-calib.	holdout10	-0.025 ± 0.106	inconclusive
DistilGPT-2	central	GD	Loss thr.	primary	-0.045 ± 0.020	below retrain
DistilGPT-2	central	GD	Loss thr.	holdout10	0.056 ± 0.015	above retrain
DistilGPT-2	central	GD	Min-K%	primary	-0.040 ± 0.017	below retrain
DistilGPT-2	central	GD	Min-K%	holdout10	0.088 ± 0.017	above retrain
DistilGPT-2	central	GD	Ref.-calib.	primary	-0.053 ± 0.036	below retrain
DistilGPT-2	central	GD	Ref.-calib.	holdout10	-0.011 ± 0.020	inconclusive
DistilGPT-2	central	NPO	Loss thr.	primary	-0.056 ± 0.012	below retrain
DistilGPT-2	central	NPO	Loss thr.	holdout10	0.001 ± 0.039	inconclusive
DistilGPT-2	central	NPO	Min-K%	primary	-0.030 ± 0.026	below retrain
DistilGPT-2	central	NPO	Min-K%	holdout10	0.057 ± 0.034	above retrain
DistilGPT-2	central	NPO	Ref.-calib.	primary	-0.052 ± 0.015	below retrain
DistilGPT-2	central	NPO	Ref.-calib.	holdout10	-0.056 ± 0.042	below retrain
GPT-2	central	GA†	Loss thr.	primary	0.023 ± 0.046	inconclusive
GPT-2	central	GA†	Loss thr.	holdout10	0.123 ± 0.099	above retrain
GPT-2	central	GA†	Min-K%	primary	0.022 ± 0.041	inconclusive

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Table S13 (continued)

Model	Run	Method	Attack	Non-members	Gap	Verdict
GPT-2	central	GA†	Min-K%	holdout10	0.137 ± 0.031	above retrain
GPT-2	central	GA†	Ref.-calib.	primary	0.028 ± 0.047	inconclusive
GPT-2	central	GA†	Ref.-calib.	holdout10	-0.019 ± 0.101	inconclusive
GPT-2	central	GD	Loss thr.	primary	-0.050 ± 0.038	below retrain
GPT-2	central	GD	Loss thr.	holdout10	0.048 ± 0.056	inconclusive
GPT-2	central	GD	Min-K%	primary	-0.053 ± 0.047	below retrain
GPT-2	central	GD	Min-K%	holdout10	0.070 ± 0.059	above retrain
GPT-2	central	GD	Ref.-calib.	primary	-0.060 ± 0.053	below retrain
GPT-2	central	GD	Ref.-calib.	holdout10	-0.023 ± 0.082	inconclusive
GPT-2	central	NPO	Loss thr.	primary	-0.086 ± 0.085	below retrain
GPT-2	central	NPO	Loss thr.	holdout10	-0.030 ± 0.097	inconclusive
GPT-2	central	NPO	Min-K%	primary	-0.083 ± 0.106	inconclusive
GPT-2	central	NPO	Min-K%	holdout10	-0.007 ± 0.116	inconclusive
GPT-2	central	NPO	Ref.-calib.	primary	-0.096 ± 0.107	inconclusive
GPT-2	central	NPO	Ref.-calib.	holdout10	-0.111 ± 0.144	inconclusive
GPT-2-medium	central	GA†	Loss thr.	primary	0.069 ± 0.040	above retrain
GPT-2-medium	central	GA†	Loss thr.	holdout10	0.140 ± 0.116	above retrain
GPT-2-medium	central	GA†	Min-K%	primary	0.059 ± 0.027	above retrain
GPT-2-medium	central	GA†	Min-K%	holdout10	0.135 ± 0.121	above retrain
GPT-2-medium	central	GA†	Ref.-calib.	primary	0.056 ± 0.043	above retrain
GPT-2-medium	central	GA†	Ref.-calib.	holdout10	-0.016 ± 0.116	inconclusive
GPT-2-medium	central	GD	Loss thr.	primary	-0.399 ± 0.051	below retrain
GPT-2-medium	central	GD	Loss thr.	holdout10	-0.360 ± 0.050	below retrain
GPT-2-medium	central	GD	Min-K%	primary	-0.379 ± 0.041	below retrain
GPT-2-medium	central	GD	Min-K%	holdout10	-0.323 ± 0.057	below retrain
GPT-2-medium	central	GD	Ref.-calib.	primary	-0.419 ± 0.047	below retrain
GPT-2-medium	central	GD	Ref.-calib.	holdout10	-0.544 ± 0.048	below retrain
GPT-2-medium	central	NPO	Loss thr.	primary	-0.338 ± 0.117	below retrain
GPT-2-medium	central	NPO	Loss thr.	holdout10	-0.283 ± 0.120	below retrain
GPT-2-medium	central	NPO	Min-K%	primary	-0.321 ± 0.110	below retrain
GPT-2-medium	central	NPO	Min-K%	holdout10	-0.251 ± 0.120	below retrain
GPT-2-medium	central	NPO	Ref.-calib.	primary	-0.358 ± 0.117	below retrain
GPT-2-medium	central	NPO	Ref.-calib.	holdout10	-0.456 ± 0.130	below retrain
GPT-2	by_author_K5	Client GA, upload†	Loss thr.	primary	-0.016 ± 0.101	inconclusive
GPT-2	by_author_K5	Client GA, upload†	Loss thr.	holdout10	0.067 ± 0.276	inconclusive
GPT-2	by_author_K5	Client GA, upload†	Min-K%	primary	-0.052 ± 0.127	inconclusive
GPT-2	by_author_K5	Client GA, upload†	Min-K%	holdout10	0.061 ± 0.197	inconclusive
GPT-2	by_author_K5	Client GA, upload†	Ref.-calib.	primary	-0.007 ± 0.086	inconclusive
GPT-2	by_author_K5	Client GA, upload†	Ref.-calib.	holdout10	-0.084 ± 0.280	inconclusive
GPT-2	by_author_K5	Client GA, global	Loss thr.	primary	0.000 ± 0.032	inconclusive
GPT-2	by_author_K5	Client GA, global	Loss thr.	holdout10	0.006 ± 0.023	inconclusive
GPT-2	by_author_K5	Client GA, global	Min-K%	primary	0.004 ± 0.030	inconclusive

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Table S13 (continued)

Model	Run	Method	Attack	Non-members	Gap	Verdict
GPT-2	by_author_K5	Client GA, global	Min-K%	holdout10	0.014 ± 0.026	inconclusive
GPT-2	by_author_K5	Client GA, global	Ref.-calib.	primary	0.005 ± 0.061	inconclusive
GPT-2	by_author_K5	Client GA, global	Ref.-calib.	holdout10	-0.005 ± 0.082	inconclusive
GPT-2	iid_K3	Client GA, upload†	Loss thr.	primary	-0.008 ± 0.029	inconclusive
GPT-2	iid_K3	Client GA, upload†	Loss thr.	holdout10	0.139 ± 0.136	above retrain
GPT-2	iid_K3	Client GA, upload†	Min-K%	primary	-0.032 ± 0.110	inconclusive
GPT-2	iid_K3	Client GA, upload†	Min-K%	holdout10	0.131 ± 0.157	inconclusive
GPT-2	iid_K3	Client GA, upload†	Ref.-calib.	primary	-0.009 ± 0.013	inconclusive
GPT-2	iid_K3	Client GA, upload†	Ref.-calib.	holdout10	0.001 ± 0.155	inconclusive
GPT-2	iid_K3	Client GA, global	Loss thr.	primary	-0.019 ± 0.015	below retrain
GPT-2	iid_K3	Client GA, global	Loss thr.	holdout10	-0.013 ± 0.023	inconclusive
GPT-2	iid_K3	Client GA, global	Min-K%	primary	-0.010 ± 0.016	inconclusive
GPT-2	iid_K3	Client GA, global	Min-K%	holdout10	-0.006 ± 0.009	inconclusive
GPT-2	iid_K3	Client GA, global	Ref.-calib.	primary	-0.036 ± 0.028	below retrain
GPT-2	iid_K3	Client GA, global	Ref.-calib.	holdout10	-0.036 ± 0.030	below retrain
GPT-2	iid_K10	Client GA, upload†	Loss thr.	primary	-0.057 ± 0.136	inconclusive
GPT-2	iid_K10	Client GA, upload†	Loss thr.	holdout10	0.044 ± 0.224	inconclusive
GPT-2	iid_K10	Client GA, upload†	Min-K%	primary	-0.068 ± 0.078	inconclusive
GPT-2	iid_K10	Client GA, upload†	Min-K%	holdout10	0.052 ± 0.129	inconclusive
GPT-2	iid_K10	Client GA, upload†	Ref.-calib.	primary	-0.051 ± 0.108	inconclusive
GPT-2	iid_K10	Client GA, upload†	Ref.-calib.	holdout10	-0.132 ± 0.220	inconclusive
GPT-2	iid_K10	Client GA, global	Loss thr.	primary	0.042 ± 0.008	above retrain
GPT-2	iid_K10	Client GA, global	Loss thr.	holdout10	0.045 ± 0.001	above retrain
GPT-2	iid_K10	Client GA, global	Min-K%	primary	0.048 ± 0.010	above retrain
GPT-2	iid_K10	Client GA, global	Min-K%	holdout10	0.050 ± 0.002	above retrain
GPT-2	iid_K10	Client GA, global	Ref.-calib.	primary	0.110 ± 0.011	above retrain
GPT-2	iid_K10	Client GA, global	Ref.-calib.	holdout10	0.104 ± 0.001	above retrain
DistilGPT-2	iid_K5	Client GA, upload†	Loss thr.	primary	-0.020 ± 0.056	inconclusive
DistilGPT-2	iid_K5	Client GA, upload†	Loss thr.	holdout10	0.038 ± 0.077	inconclusive
DistilGPT-2	iid_K5	Client GA, upload†	Min-K%	primary	-0.039 ± 0.059	inconclusive
DistilGPT-2	iid_K5	Client GA, upload†	Min-K%	holdout10	0.050 ± 0.058	inconclusive
DistilGPT-2	iid_K5	Client GA, upload†	Ref.-calib.	primary	-0.016 ± 0.043	inconclusive
DistilGPT-2	iid_K5	Client GA, upload†	Ref.-calib.	holdout10	-0.107 ± 0.076	below retrain
DistilGPT-2	iid_K5	Client GA, global	Loss thr.	primary	-0.016 ± 0.005	below retrain
DistilGPT-2	iid_K5	Client GA, global	Loss thr.	holdout10	-0.013 ± 0.012	below retrain
DistilGPT-2	iid_K5	Client GA, global	Min-K%	primary	-0.009 ± 0.008	below retrain
DistilGPT-2	iid_K5	Client GA, global	Min-K%	holdout10	0.001 ± 0.006	inconclusive
DistilGPT-2	iid_K5	Client GA, global	Ref.-calib.	primary	-0.030 ± 0.013	below retrain
DistilGPT-2	iid_K5	Client GA, global	Ref.-calib.	holdout10	-0.047 ± 0.040	below retrain
GPT-2	iid_K5	Client GA, upload†	Loss thr.	primary	-0.031 ± 0.051	inconclusive
GPT-2	iid_K5	Client GA, upload†	Loss thr.	holdout10	0.041 ± 0.083	inconclusive
GPT-2	iid_K5	Client GA, upload†	Min-K%	primary	-0.044 ± 0.032	below retrain

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Table S13 (continued)

Model	Run	Method	Attack	Non-members	Gap	Verdict
GPT-2	iid_K5	Client GA, upload†	Min-K%	holdout10	0.054 ± 0.087	inconclusive
GPT-2	iid_K5	Client GA, upload†	Ref.-calib.	primary	-0.023 ± 0.050	inconclusive
GPT-2	iid_K5	Client GA, upload†	Ref.-calib.	holdout10	-0.112 ± 0.075	below retrain
GPT-2	iid_K5	Client GA, global	Loss thr.	primary	0.005 ± 0.009	inconclusive
GPT-2	iid_K5	Client GA, global	Loss thr.	holdout10	0.009 ± 0.010	inconclusive
GPT-2	iid_K5	Client GA, global	Min-K%	primary	0.010 ± 0.013	inconclusive
GPT-2	iid_K5	Client GA, global	Min-K%	holdout10	0.015 ± 0.011	above retrain
GPT-2	iid_K5	Client GA, global	Ref.-calib.	primary	0.010 ± 0.018	inconclusive
GPT-2	iid_K5	Client GA, global	Ref.-calib.	holdout10	0.000 ± 0.028	inconclusive
GPT-2-medium	iid_K5	Client GA, upload†	Loss thr.	primary	0.026 ± 0.051	inconclusive
GPT-2-medium	iid_K5	Client GA, upload†	Loss thr.	holdout10	0.086 ± 0.084	above retrain
GPT-2-medium	iid_K5	Client GA, upload†	Min-K%	primary	0.021 ± 0.067	inconclusive
GPT-2-medium	iid_K5	Client GA, upload†	Min-K%	holdout10	0.130 ± 0.076	above retrain
GPT-2-medium	iid_K5	Client GA, upload†	Ref.-calib.	primary	0.019 ± 0.038	inconclusive
GPT-2-medium	iid_K5	Client GA, upload†	Ref.-calib.	holdout10	-0.080 ± 0.072	below retrain
GPT-2-medium	iid_K5	Client GA, global	Loss thr.	primary	0.044 ± 0.015	above retrain
GPT-2-medium	iid_K5	Client GA, global	Loss thr.	holdout10	0.047 ± 0.017	above retrain
GPT-2-medium	iid_K5	Client GA, global	Min-K%	primary	0.049 ± 0.015	above retrain
GPT-2-medium	iid_K5	Client GA, global	Min-K%	holdout10	0.052 ± 0.017	above retrain
GPT-2-medium	iid_K5	Client GA, global	Ref.-calib.	primary	0.092 ± 0.034	above retrain
GPT-2-medium	iid_K5	Client GA, global	Ref.-calib.	holdout10	0.091 ± 0.032	above retrain
Pythia-1B (LoRA)	central	GA†	Loss thr.	primary	-0.010 ± 0.169	inconclusive
Pythia-1B (LoRA)	central	GA†	Loss thr.	holdout10	0.080 ± 0.240	inconclusive
Pythia-1B (LoRA)	central	GA†	Min-K%	primary	0.024 ± 0.174	inconclusive
Pythia-1B (LoRA)	central	GA†	Min-K%	holdout10	0.038 ± 0.199	inconclusive
Pythia-1B (LoRA)	central	GA†	Ref.-calib.	primary	-0.018 ± 0.186	inconclusive
Pythia-1B (LoRA)	central	GA†	Ref.-calib.	holdout10	-0.082 ± 0.263	inconclusive
Pythia-1B (LoRA)	central	GD†	Loss thr.	primary	-0.383 ± 0.051	below retrain
Pythia-1B (LoRA)	central	GD†	Loss thr.	holdout10	-0.321 ± 0.061	below retrain
Pythia-1B (LoRA)	central	GD†	Min-K%	primary	-0.366 ± 0.047	below retrain
Pythia-1B (LoRA)	central	GD†	Min-K%	holdout10	-0.294 ± 0.063	below retrain
Pythia-1B (LoRA)	central	GD†	Ref.-calib.	primary	-0.404 ± 0.053	below retrain
Pythia-1B (LoRA)	central	GD†	Ref.-calib.	holdout10	-0.525 ± 0.065	below retrain
Pythia-1B (LoRA)	central	NPO	Loss thr.	primary	-0.401 ± 0.040	below retrain
Pythia-1B (LoRA)	central	NPO	Loss thr.	holdout10	-0.346 ± 0.049	below retrain
Pythia-1B (LoRA)	central	NPO	Min-K%	primary	-0.385 ± 0.032	below retrain
Pythia-1B (LoRA)	central	NPO	Min-K%	holdout10	-0.320 ± 0.046	below retrain
Pythia-1B (LoRA)	central	NPO	Ref.-calib.	primary	-0.421 ± 0.056	below retrain
Pythia-1B (LoRA)	central	NPO	Ref.-calib.	holdout10	-0.547 ± 0.072	below retrain

Mean ± half-width of the 95% t-interval over seeds (up to 5); a single value means one seed. Verdict: above retrain (CI lower bound > 0, supports H1), below retrain (CI upper bound < 0, contradicts H1) or inconclusive. †Model collapsed: retain loss rose more than threefold in every seed. GA: all four central models; GD: Pythia-1B (LoRA); client-GA upload: all six federated runs (intermediate model before recovery; the released global model is not collapsed).

Table S14: Exploratory (post hoc): two-sided distance $|\mathrm{AUC} - \mathrm{AUC}_{\mathrm{retrain}}|$ to the retrained model of the same seed (federated: the federated retrain), overall and per stratum, added after seeing the GPT-2 results; next to MUSE's PrivLeak from Shi et al. (ICLR 2025), $(\mathrm{AUC} - \mathrm{AUC}_{\mathrm{retrain}})/\mathrm{AUC}_{\mathrm{retrain}}$, and TOFU forget quality. H1 (Table S13) is the pre-registered one-sided test.

Model	Run	Method	Attack	Non-members	All	Typical	Outlier	Below	PrivLeak (%)	Forget quality
DistilGPT-2	central	GA†	Loss thr.	primary	0.029 ± 0.030	0.023 ± 0.030	0.064 ± 0.060	1/5	6.2 ± 7.4	0.000 ± 0.000
DistilGPT-2	central	GA†	Loss thr.	holdout10	0.115 ± 0.109	0.117 ± 0.050	0.130 ± 0.101	0/5	26.6 ± 25.3	0.000 ± 0.000
DistilGPT-2	central	GA†	Min-K%	primary	0.025 ± 0.015	0.017 ± 0.020	0.032 ± 0.018	2/5	2.6 ± 7.9	0.000 ± 0.000
DistilGPT-2	central	GA†	Min-K%	holdout10	0.095 ± 0.063	0.073 ± 0.049	0.094 ± 0.069	0/5	22.7 ± 15.0	0.000 ± 0.000
DistilGPT-2	central	GA†	Ref.-calib.	primary	0.034 ± 0.042	0.030 ± 0.047	0.047 ± 0.059	1/5	6.6 ± 11.0	0.000 ± 0.000
DistilGPT-2	central	GA†	Ref.-calib.	holdout10	0.070 ± 0.056	0.112 ± 0.096	0.077 ± 0.038	2/5	-4.0 ± 17.3	0.000 ± 0.000
DistilGPT-2	central	GD	Loss thr.	primary	0.045 ± 0.020	0.056 ± 0.027	0.029 ± 0.030	5/5	-10.0 ± 4.3	0.085 ± 0.064
DistilGPT-2	central	GD	Loss thr.	holdout10	0.056 ± 0.015	0.017 ± 0.014	0.072 ± 0.022	0/5	13.0 ± 3.5	0.085 ± 0.064
DistilGPT-2	central	GD	Min-K%	primary	0.040 ± 0.017	0.062 ± 0.018	0.033 ± 0.021	5/5	-9.1 ± 3.7	0.085 ± 0.064
DistilGPT-2	central	GD	Min-K%	holdout10	0.088 ± 0.017	0.037 ± 0.013	0.117 ± 0.027	0/5	20.9 ± 4.0	0.085 ± 0.064
DistilGPT-2	central	GD	Ref.-calib.	primary	0.053 ± 0.036	0.059 ± 0.043	0.031 ± 0.027	5/5	-11.4 ± 7.3	0.085 ± 0.064
DistilGPT-2	central	GD	Ref.-calib.	holdout10	0.016 ± 0.012	0.016 ± 0.012	0.030 ± 0.027	4/5	-1.8 ± 3.3	0.085 ± 0.064
DistilGPT-2	central	NPO	Loss thr.	primary	0.056 ± 0.012	0.083 ± 0.011	0.029 ± 0.020	5/5	-12.5 ± 2.9	0.051 ± 0.048
DistilGPT-2	central	NPO	Loss thr.	holdout10	0.023 ± 0.021	0.052 ± 0.042	0.044 ± 0.028	2/5	0.2 ± 8.9	0.051 ± 0.048
DistilGPT-2	central	NPO	Min-K%	primary	0.030 ± 0.026	0.061 ± 0.027	0.052 ± 0.052	5/5	-6.7 ± 5.8	0.051 ± 0.048
DistilGPT-2	central	NPO	Min-K%	holdout10	0.057 ± 0.034	0.022 ± 0.025	0.119 ± 0.036	0/5	13.4 ± 8.2	0.051 ± 0.048
DistilGPT-2	central	NPO	Ref.-calib.	primary	0.052 ± 0.015	0.088 ± 0.024	0.033 ± 0.018	5/5	-11.5 ± 3.2	0.051 ± 0.048
DistilGPT-2	central	NPO	Ref.-calib.	holdout10	0.056 ± 0.042	0.101 ± 0.043	0.025 ± 0.021	5/5	-9.2 ± 6.9	0.051 ± 0.048
GPT-2	central	GA†	Loss thr.	primary	0.032 ± 0.035	0.022 ± 0.025	0.068 ± 0.085	2/5	5.4 ± 10.7	0.000 ± 0.000
GPT-2	central	GA†	Loss thr.	holdout10	0.123 ± 0.099	0.092 ± 0.072	0.146 ± 0.064	0/5	29.0 ± 23.3	0.000 ± 0.000
GPT-2	central	GA†	Min-K%	primary	0.031 ± 0.028	0.031 ± 0.046	0.043 ± 0.037	2/5	5.0 ± 9.3	0.000 ± 0.000
GPT-2	central	GA†	Min-K%	holdout10	0.137 ± 0.031	0.072 ± 0.047	0.083 ± 0.062	0/5	33.5 ± 7.9	0.000 ± 0.000
GPT-2	central	GA†	Ref.-calib.	primary	0.031 ± 0.044	0.031 ± 0.045	0.054 ± 0.076	2/5	6.4 ± 10.9	0.000 ± 0.000
GPT-2	central	GA†	Ref.-calib.	holdout10	0.065 ± 0.053	0.101 ± 0.072	0.090 ± 0.078	3/5	-3.1 ± 16.5	0.000 ± 0.000
GPT-2	central	GD	Loss thr.	primary	0.050 ± 0.038	0.060 ± 0.041	0.054 ± 0.055	5/5	-11.2 ± 8.9	0.001 ± 0.001
GPT-2	central	GD	Loss thr.	holdout10	0.058 ± 0.034	0.033 ± 0.030	0.089 ± 0.047	1/5	11.3 ± 13.2	0.001 ± 0.001
GPT-2	central	GD	Min-K%	primary	0.054 ± 0.045	0.062 ± 0.047	0.060 ± 0.059	4/5	-12.2 ± 11.0	0.001 ± 0.001
GPT-2	central	GD	Min-K%	holdout10	0.072 ± 0.054	0.045 ± 0.028	0.090 ± 0.047	1/5	17.3 ± 14.5	0.001 ± 0.001
GPT-2	central	GD	Ref.-calib.	primary	0.060 ± 0.053	0.057 ± 0.043	0.062 ± 0.069	5/5	-13.4 ± 12.2	0.001 ± 0.001
GPT-2	central	GD	Ref.-calib.	holdout10	0.045 ± 0.062	0.035 ± 0.049	0.067 ± 0.055	3/5	-3.8 ± 13.5	0.001 ± 0.001
GPT-2	central	NPO	Loss thr.	primary	0.086 ± 0.085	0.111 ± 0.097	0.066 ± 0.086	5/5	-19.6 ± 20.4	0.004 ± 0.007
GPT-2	central	NPO	Loss thr.	holdout10	0.047 ± 0.082	0.089 ± 0.114	0.067 ± 0.086	2/5	-7.0 ± 22.7	0.004 ± 0.007
GPT-2	central	NPO	Min-K%	primary	0.085 ± 0.104	0.103 ± 0.111	0.074 ± 0.115	4/5	-19.2 ± 24.8	0.004 ± 0.007
GPT-2	central	NPO	Min-K%	holdout10	0.064 ± 0.074	0.074 ± 0.125	0.077 ± 0.076	2/5	-1.8 ± 28.1	0.004 ± 0.007
GPT-2	central	NPO	Ref.-calib.	primary	0.096 ± 0.107	0.116 ± 0.101	0.076 ± 0.102	5/5	-21.8 ± 25.0	0.004 ± 0.007
GPT-2	central	NPO	Ref.-calib.	holdout10	0.111 ± 0.144	0.138 ± 0.129	0.075 ± 0.126	5/5	-18.2 ± 23.6	0.004 ± 0.007
GPT-2-medium	central	GA†	Loss thr.	primary	0.069 ± 0.040	0.051 ± 0.045	0.079 ± 0.065	0/5	16.1 ± 9.5	0.000 ± 0.000
GPT-2-medium	central	GA†	Loss thr.	holdout10	0.140 ± 0.115	0.117 ± 0.058	0.143 ± 0.125	1/5	34.1 ± 28.3	0.000 ± 0.000
GPT-2-medium	central	GA†	Min-K%	primary	0.059 ± 0.027	0.049 ± 0.056	0.058 ± 0.039	0/5	14.3 ± 6.7	0.000 ± 0.000

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Table S14 (continued)

Model	Run	Method	Attack	Non-members	All	Typical	Outlier	Below	PrivLeak (%)	Forget quality
GPT-2-medium	central	GA†	Min-K%	holdout10	0.136 ± 0.118	0.141 ± 0.046	0.160 ± 0.073	1/5	35.3 ± 31.7	0.000 ± 0.000
GPT-2-medium	central	GA†	Ref.-calib.	primary	0.056 ± 0.043	0.073 ± 0.041	0.069 ± 0.069	0/5	12.5 ± 9.7	0.000 ± 0.000
GPT-2-medium	central	GA†	Ref.-calib.	holdout10	0.070 ± 0.067	0.111 ± 0.075	0.084 ± 0.067	2/5	-2.7 ± 19.4	0.000 ± 0.000
GPT-2-medium	central	GD	Loss thr.	primary	0.399 ± 0.051	0.422 ± 0.047	0.418 ± 0.062	5/5	-92.5 ± 8.8	0.000 ± 0.000
GPT-2-medium	central	GD	Loss thr.	holdout10	0.360 ± 0.050	0.494 ± 0.056	0.348 ± 0.036	5/5	-87.8 ± 12.4	0.000 ± 0.000
GPT-2-medium	central	GD	Min-K%	primary	0.379 ± 0.041	0.391 ± 0.042	0.405 ± 0.038	5/5	-91.8 ± 9.1	0.000 ± 0.000
GPT-2-medium	central	GD	Min-K%	holdout10	0.323 ± 0.057	0.425 ± 0.062	0.330 ± 0.041	5/5	-84.4 ± 15.6	0.000 ± 0.000
GPT-2-medium	central	GD	Ref.-calib.	primary	0.419 ± 0.047	0.404 ± 0.043	0.429 ± 0.067	5/5	-92.9 ± 8.6	0.000 ± 0.000
GPT-2-medium	central	GD	Ref.-calib.	holdout10	0.544 ± 0.048	0.542 ± 0.053	0.471 ± 0.036	5/5	-91.4 ± 8.8	0.000 ± 0.000
GPT-2-medium	central	NPO	Loss thr.	primary	0.338 ± 0.117	0.362 ± 0.102	0.352 ± 0.142	5/5	-78.0 ± 24.6	0.000 ± 0.001
GPT-2-medium	central	NPO	Loss thr.	holdout10	0.283 ± 0.120	0.410 ± 0.126	0.266 ± 0.126	5/5	-69.2 ± 29.4	0.000 ± 0.001
GPT-2-medium	central	NPO	Min-K%	primary	0.321 ± 0.110	0.338 ± 0.091	0.334 ± 0.135	5/5	-77.6 ± 25.3	0.000 ± 0.001
GPT-2-medium	central	NPO	Min-K%	holdout10	0.251 ± 0.120	0.356 ± 0.115	0.243 ± 0.140	5/5	-65.5 ± 31.7	0.000 ± 0.001
GPT-2-medium	central	NPO	Ref.-calib.	primary	0.358 ± 0.117	0.345 ± 0.118	0.366 ± 0.124	5/5	-79.3 ± 24.0	0.000 ± 0.001
GPT-2-medium	central	NPO	Ref.-calib.	holdout10	0.456 ± 0.130	0.459 ± 0.120	0.381 ± 0.137	5/5	-76.8 ± 22.6	0.000 ± 0.001
GPT-2	by_author_K5	Client GA, upload†	Loss thr.	primary	0.028 ± 0.073	0.059 ± 0.091	0.071 ± 0.089	2/3	-3.4 ± 22.5	–
GPT-2	by_author_K5	Client GA, upload†	Loss thr.	holdout10	0.075 ± 0.256	0.153 ± 0.141	0.071 ± 0.224	1/3	15.7 ± 64.7	–
GPT-2	by_author_K5	Client GA, upload†	Min-K%	primary	0.052 ± 0.127	0.068 ± 0.151	0.091 ± 0.179	3/3	-11.7 ± 28.6	–
GPT-2	by_author_K5	Client GA, upload†	Min-K%	holdout10	0.063 ± 0.190	0.116 ± 0.118	0.054 ± 0.080	1/3	15.0 ± 48.2	–
GPT-2	by_author_K5	Client GA, upload†	Ref.-calib.	primary	0.026 ± 0.041	0.030 ± 0.025	0.082 ± 0.057	2/3	-1.5 ± 19.5	–
GPT-2	by_author_K5	Client GA, upload†	Ref.-calib.	holdout10	0.114 ± 0.151	0.166 ± 0.220	0.080 ± 0.019	2/3	-13.6 ± 45.0	–
GPT-2	by_author_K5	Client GA, global	Loss thr.	primary	0.009 ± 0.013	0.013 ± 0.013	0.016 ± 0.040	2/3	0.1 ± 7.2	0.725 ± 1.170
GPT-2	by_author_K5	Client GA, global	Loss thr.	holdout10	0.007 ± 0.021	0.008 ± 0.023	0.016 ± 0.025	1/3	1.4 ± 5.5	0.725 ± 1.170
GPT-2	by_author_K5	Client GA, global	Min-K%	primary	0.008 ± 0.021	0.012 ± 0.020	0.011 ± 0.025	2/3	0.9 ± 6.9	0.725 ± 1.170
GPT-2	by_author_K5	Client GA, global	Min-K%	holdout10	0.014 ± 0.026	0.014 ± 0.036	0.017 ± 0.012	0/3	3.5 ± 6.3	0.725 ± 1.170
GPT-2	by_author_K5	Client GA, global	Ref.-calib.	primary	0.017 ± 0.035	0.016 ± 0.027	0.017 ± 0.046	2/3	1.2 ± 13.8	0.725 ± 1.170
GPT-2	by_author_K5	Client GA, global	Ref.-calib.	holdout10	0.024 ± 0.041	0.017 ± 0.022	0.028 ± 0.030	2/3	-0.8 ± 13.2	0.725 ± 1.170
GPT-2	iid_K3	Client GA, upload†	Loss thr.	primary	0.011 ± 0.021	0.030 ± 0.074	0.016 ± 0.022	2/3	-1.9 ± 6.6	–
GPT-2	iid_K3	Client GA, upload†	Loss thr.	holdout10	0.139 ± 0.136	0.064 ± 0.112	0.162 ± 0.240	0/3	32.4 ± 31.1	–
GPT-2	iid_K3	Client GA, upload†	Min-K%	primary	0.039 ± 0.087	0.041 ± 0.116	0.101 ± 0.033	2/3	-7.4 ± 25.5	–
GPT-2	iid_K3	Client GA, upload†	Min-K%	holdout10	0.131 ± 0.157	0.098 ± 0.091	0.087 ± 0.179	0/3	32.1 ± 38.7	–
GPT-2	iid_K3	Client GA, upload†	Ref.-calib.	primary	0.009 ± 0.013	0.012 ± 0.022	0.009 ± 0.018	3/3	-2.1 ± 3.0	–
GPT-2	iid_K3	Client GA, upload†	Ref.-calib.	holdout10	0.043 ± 0.083	0.070 ± 0.110	0.093 ± 0.149	2/3	0.1 ± 25.7	–
GPT-2	iid_K3	Client GA, global	Loss thr.	primary	0.019 ± 0.015	0.027 ± 0.024	0.017 ± 0.013	3/3	-4.4 ± 3.6	0.954 ± 0.196
GPT-2	iid_K3	Client GA, global	Loss thr.	holdout10	0.013 ± 0.023	0.022 ± 0.034	0.012 ± 0.021	3/3	-3.0 ± 5.3	0.954 ± 0.196
GPT-2	iid_K3	Client GA, global	Min-K%	primary	0.010 ± 0.016	0.011 ± 0.023	0.020 ± 0.034	3/3	-2.4 ± 3.9	0.954 ± 0.196
GPT-2	iid_K3	Client GA, global	Min-K%	holdout10	0.006 ± 0.009	0.009 ± 0.020	0.008 ± 0.018	3/3	-1.5 ± 2.3	0.954 ± 0.196
GPT-2	iid_K3	Client GA, global	Ref.-calib.	primary	0.036 ± 0.028	0.042 ± 0.043	0.022 ± 0.017	3/3	-8.0 ± 6.3	0.954 ± 0.196
GPT-2	iid_K3	Client GA, global	Ref.-calib.	holdout10	0.036 ± 0.030	0.040 ± 0.054	0.028 ± 0.027	3/3	-6.0 ± 5.0	0.954 ± 0.196
GPT-2	iid_K10	Client GA, upload†	Loss thr.	primary	0.057 ± 0.136	0.088 ± 0.206	0.054 ± 0.016	3/3	-12.8 ± 29.5	–
GPT-2	iid_K10	Client GA, upload†	Loss thr.	holdout10	0.057 ± 0.193	0.132 ± 0.153	0.056 ± 0.132	1/3	10.4 ± 53.5	–
GPT-2	iid_K10	Client GA, upload†	Min-K%	primary	0.068 ± 0.078	0.085 ± 0.173	0.085 ± 0.127	3/3	-15.0 ± 17.5	–

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Table S14 (continued)

Model	Run	Method	Attack	Non-members	All	Typical	Outlier	Below	PrivLeak (%)	Forget quality
GPT-2	iid_K10	Client GA, upload†	Min-K%	holdout10	0.052 ± 0.129	0.091 ± 0.055	0.028 ± 0.040	0/3	12.7 ± 31.7	–
GPT-2	iid_K10	Client GA, upload†	Ref.-calib.	primary	0.051 ± 0.108	0.060 ± 0.147	0.062 ± 0.084	3/3	-11.4 ± 23.8	–
GPT-2	iid_K10	Client GA, upload†	Ref.-calib.	holdout10	0.132 ± 0.220	0.153 ± 0.299	0.090 ± 0.195	3/3	-20.9 ± 35.0	–
GPT-2	iid_K10	Client GA, global	Loss thr.	primary	0.042 ± 0.008	0.052 ± 0.008	0.093 ± 0.027	0/3	9.4 ± 1.4	0.527 ± 0.279
GPT-2	iid_K10	Client GA, global	Loss thr.	holdout10	0.045 ± 0.001	0.056 ± 0.001	0.092 ± 0.002	0/3	10.7 ± 0.2	0.527 ± 0.279
GPT-2	iid_K10	Client GA, global	Min-K%	primary	0.048 ± 0.010	0.053 ± 0.012	0.076 ± 0.020	0/3	10.7 ± 2.3	0.527 ± 0.279
GPT-2	iid_K10	Client GA, global	Min-K%	holdout10	0.050 ± 0.002	0.058 ± 0.001	0.068 ± 0.010	0/3	12.2 ± 0.5	0.527 ± 0.279
GPT-2	iid_K10	Client GA, global	Ref.-calib.	primary	0.110 ± 0.011	0.103 ± 0.003	0.124 ± 0.043	0/3	24.6 ± 2.6	0.527 ± 0.279
GPT-2	iid_K10	Client GA, global	Ref.-calib.	holdout10	0.104 ± 0.001	0.098 ± 0.004	0.118 ± 0.006	0/3	16.6 ± 0.2	0.527 ± 0.279
DistilGPT-2	iid_K5	Client GA, upload†	Loss thr.	primary	0.034 ± 0.041	0.051 ± 0.052	0.055 ± 0.081	2/5	-4.3 ± 12.2	–
DistilGPT-2	iid_K5	Client GA, upload†	Loss thr.	holdout10	0.043 ± 0.071	0.117 ± 0.070	0.062 ± 0.079	1/5	8.7 ± 18.0	–
DistilGPT-2	iid_K5	Client GA, upload†	Min-K%	primary	0.050 ± 0.037	0.061 ± 0.063	0.070 ± 0.040	4/5	-8.7 ± 13.2	–
DistilGPT-2	iid_K5	Client GA, upload†	Min-K%	holdout10	0.050 ± 0.058	0.079 ± 0.016	0.053 ± 0.061	0/5	11.8 ± 13.7	–
DistilGPT-2	iid_K5	Client GA, upload†	Ref.-calib.	primary	0.027 ± 0.031	0.028 ± 0.033	0.057 ± 0.060	3/5	-3.4 ± 9.5	–
DistilGPT-2	iid_K5	Client GA, upload†	Ref.-calib.	holdout10	0.107 ± 0.076	0.154 ± 0.096	0.085 ± 0.045	5/5	-17.3 ± 12.3	–
DistilGPT-2	iid_K5	Client GA, global	Loss thr.	primary	0.016 ± 0.005	0.026 ± 0.011	0.008 ± 0.009	5/5	-3.6 ± 1.2	0.780 ± 0.381
DistilGPT-2	iid_K5	Client GA, global	Loss thr.	holdout10	0.013 ± 0.012	0.025 ± 0.022	0.016 ± 0.007	5/5	-3.1 ± 2.8	0.780 ± 0.381
DistilGPT-2	iid_K5	Client GA, global	Min-K%	primary	0.010 ± 0.006	0.014 ± 0.008	0.009 ± 0.005	4/5	-2.0 ± 1.8	0.780 ± 0.381
DistilGPT-2	iid_K5	Client GA, global	Min-K%	holdout10	0.005 ± 0.001	0.009 ± 0.011	0.015 ± 0.018	2/5	0.2 ± 1.5	0.780 ± 0.381
DistilGPT-2	iid_K5	Client GA, global	Ref.-calib.	primary	0.030 ± 0.013	0.035 ± 0.014	0.011 ± 0.007	5/5	-6.5 ± 2.5	0.780 ± 0.381
DistilGPT-2	iid_K5	Client GA, global	Ref.-calib.	holdout10	0.047 ± 0.040	0.044 ± 0.029	0.028 ± 0.027	5/5	-7.6 ± 6.6	0.780 ± 0.381
GPT-2	iid_K5	Client GA, upload†	Loss thr.	primary	0.040 ± 0.038	0.066 ± 0.040	0.068 ± 0.077	4/5	-6.9 ± 11.2	–
GPT-2	iid_K5	Client GA, upload†	Loss thr.	holdout10	0.050 ± 0.074	0.123 ± 0.105	0.060 ± 0.097	2/5	9.8 ± 19.7	–
GPT-2	iid_K5	Client GA, upload†	Min-K%	primary	0.044 ± 0.032	0.058 ± 0.040	0.077 ± 0.057	5/5	-10.1 ± 7.3	–
GPT-2	iid_K5	Client GA, upload†	Min-K%	holdout10	0.058 ± 0.082	0.108 ± 0.053	0.057 ± 0.054	1/5	13.3 ± 21.5	–
GPT-2	iid_K5	Client GA, upload†	Ref.-calib.	primary	0.036 ± 0.030	0.039 ± 0.027	0.059 ± 0.074	4/5	-4.9 ± 10.9	–
GPT-2	iid_K5	Client GA, upload†	Ref.-calib.	holdout10	0.112 ± 0.075	0.162 ± 0.100	0.081 ± 0.034	5/5	-18.1 ± 12.2	–
GPT-2	iid_K5	Client GA, global	Loss thr.	primary	0.007 ± 0.006	0.008 ± 0.006	0.021 ± 0.016	1/5	1.1 ± 2.0	0.706 ± 0.509
GPT-2	iid_K5	Client GA, global	Loss thr.	holdout10	0.011 ± 0.006	0.012 ± 0.007	0.015 ± 0.014	1/5	2.2 ± 2.3	0.706 ± 0.509
GPT-2	iid_K5	Client GA, global	Min-K%	primary	0.011 ± 0.012	0.012 ± 0.011	0.019 ± 0.020	1/5	2.3 ± 2.9	0.706 ± 0.509
GPT-2	iid_K5	Client GA, global	Min-K%	holdout10	0.015 ± 0.011	0.016 ± 0.014	0.011 ± 0.009	0/5	3.6 ± 2.6	0.706 ± 0.509
GPT-2	iid_K5	Client GA, global	Ref.-calib.	primary	0.014 ± 0.012	0.014 ± 0.011	0.022 ± 0.019	1/5	2.1 ± 4.1	0.706 ± 0.509
GPT-2	iid_K5	Client GA, global	Ref.-calib.	holdout10	0.018 ± 0.013	0.015 ± 0.013	0.017 ± 0.013	2/5	0.1 ± 4.6	0.706 ± 0.509
GPT-2-medium	iid_K5	Client GA, upload†	Loss thr.	primary	0.041 ± 0.028	0.044 ± 0.025	0.044 ± 0.044	2/5	6.2 ± 11.9	–
GPT-2-medium	iid_K5	Client GA, upload†	Loss thr.	holdout10	0.086 ± 0.084	0.092 ± 0.093	0.103 ± 0.103	0/5	20.9 ± 20.6	–
GPT-2-medium	iid_K5	Client GA, upload†	Min-K%	primary	0.048 ± 0.029	0.054 ± 0.053	0.016 ± 0.017	1/5	5.1 ± 16.2	–
GPT-2-medium	iid_K5	Client GA, upload†	Min-K%	holdout10	0.130 ± 0.076	0.085 ± 0.059	0.145 ± 0.086	0/5	33.9 ± 19.8	–
GPT-2-medium	iid_K5	Client GA, upload†	Ref.-calib.	primary	0.031 ± 0.018	0.034 ± 0.043	0.054 ± 0.037	2/5	4.2 ± 8.5	–
GPT-2-medium	iid_K5	Client GA, upload†	Ref.-calib.	holdout10	0.080 ± 0.072	0.106 ± 0.128	0.068 ± 0.034	5/5	-13.2 ± 11.8	–
GPT-2-medium	iid_K5	Client GA, global	Loss thr.	primary	0.044 ± 0.015	0.050 ± 0.020	0.087 ± 0.030	0/5	10.0 ± 3.3	0.745 ± 0.236
GPT-2-medium	iid_K5	Client GA, global	Loss thr.	holdout10	0.047 ± 0.017	0.054 ± 0.022	0.093 ± 0.030	0/5	11.5 ± 4.2	0.745 ± 0.236
GPT-2-medium	iid_K5	Client GA, global	Min-K%	primary	0.049 ± 0.015	0.053 ± 0.020	0.074 ± 0.018	0/5	11.7 ± 3.6	0.745 ± 0.236

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Table S14 (continued)

Model	Run	Method	Attack	Non-members	All	Typical	Outlier	Below	PrivLeak (%)	Forget quality
GPT-2-medium	iid_K5	Client GA, global	Min-K%	holdout10	0.052 ± 0.017	0.056 ± 0.021	0.077 ± 0.022	0/5	13.6 ± 4.5	0.745 ± 0.236
GPT-2-medium	iid_K5	Client GA, global	Ref.-calib.	primary	0.092 ± 0.034	0.088 ± 0.036	0.103 ± 0.031	0/5	20.3 ± 7.0	0.745 ± 0.236
GPT-2-medium	iid_K5	Client GA, global	Ref.-calib.	holdout10	0.091 ± 0.032	0.088 ± 0.035	0.095 ± 0.032	0/5	14.9 ± 5.2	0.745 ± 0.236
Pythia-1B (LoRA)	central	GA†	Loss thr.	primary	0.090 ± 0.115	0.077 ± 0.139	0.113 ± 0.112	1/5	-2.4 ± 39.2	0.000 ± 0.000
Pythia-1B (LoRA)	central	GA†	Loss thr.	holdout10	0.166 ± 0.129	0.186 ± 0.114	0.176 ± 0.086	1/5	19.3 ± 57.8	0.000 ± 0.000
Pythia-1B (LoRA)	central	GA†	Min-K%	primary	0.112 ± 0.084	0.100 ± 0.097	0.113 ± 0.116	1/5	5.5 ± 41.9	0.000 ± 0.000
Pythia-1B (LoRA)	central	GA†	Min-K%	holdout10	0.133 ± 0.090	0.107 ± 0.159	0.116 ± 0.096	1/5	9.9 ± 51.7	0.000 ± 0.000
Pythia-1B (LoRA)	central	GA†	Ref.-calib.	primary	0.096 ± 0.132	0.093 ± 0.128	0.102 ± 0.149	1/5	-3.6 ± 39.9	0.000 ± 0.000
Pythia-1B (LoRA)	central	GA†	Ref.-calib.	holdout10	0.163 ± 0.176	0.199 ± 0.171	0.132 ± 0.162	3/5	-13.4 ± 42.4	0.000 ± 0.000
Pythia-1B (LoRA)	central	GD†	Loss thr.	primary	0.383 ± 0.051	0.402 ± 0.073	0.378 ± 0.080	5/5	-87.3 ± 9.3	0.001 ± 0.002
Pythia-1B (LoRA)	central	GD†	Loss thr.	holdout10	0.321 ± 0.061	0.432 ± 0.069	0.341 ± 0.058	5/5	-77.7 ± 14.3	0.001 ± 0.002
Pythia-1B (LoRA)	central	GD†	Min-K%	primary	0.366 ± 0.047	0.369 ± 0.070	0.406 ± 0.052	5/5	-87.4 ± 10.3	0.001 ± 0.002
Pythia-1B (LoRA)	central	GD†	Min-K%	holdout10	0.294 ± 0.063	0.368 ± 0.074	0.332 ± 0.054	5/5	-76.7 ± 15.5	0.001 ± 0.002
Pythia-1B (LoRA)	central	GD†	Ref.-calib.	primary	0.404 ± 0.053	0.381 ± 0.064	0.432 ± 0.047	5/5	-88.2 ± 9.1	0.001 ± 0.002
Pythia-1B (LoRA)	central	GD†	Ref.-calib.	holdout10	0.525 ± 0.065	0.514 ± 0.073	0.492 ± 0.057	5/5	-84.7 ± 9.7	0.001 ± 0.002
Pythia-1B (LoRA)	central	NPO	Loss thr.	primary	0.401 ± 0.040	0.428 ± 0.043	0.384 ± 0.100	5/5	-91.4 ± 7.5	0.000 ± 0.001
Pythia-1B (LoRA)	central	NPO	Loss thr.	holdout10	0.346 ± 0.049	0.466 ± 0.042	0.342 ± 0.097	5/5	-83.7 ± 11.0	0.000 ± 0.001
Pythia-1B (LoRA)	central	NPO	Min-K%	primary	0.385 ± 0.032	0.396 ± 0.036	0.412 ± 0.074	5/5	-92.0 ± 7.1	0.000 ± 0.001
Pythia-1B (LoRA)	central	NPO	Min-K%	holdout10	0.320 ± 0.046	0.404 ± 0.042	0.333 ± 0.089	5/5	-83.6 ± 10.6	0.000 ± 0.001
Pythia-1B (LoRA)	central	NPO	Ref.-calib.	primary	0.421 ± 0.056	0.407 ± 0.041	0.438 ± 0.082	5/5	-91.9 ± 9.5	0.000 ± 0.001
Pythia-1B (LoRA)	central	NPO	Ref.-calib.	holdout10	0.547 ± 0.072	0.549 ± 0.055	0.488 ± 0.111	5/5	-88.2 ± 10.2	0.000 ± 0.001

Mean ± half-width of the 95% t-interval over seeds (up to 5); a single value means one seed. Strata by base-model difficulty, stratum-matched. Below: seeds whose AUC is under the retrained model's. PrivLeak: per seed, over all records, in percent (MUSE's convention; near 0 for good unlearning). †Model collapsed: retain loss rose more than threefold in every seed. GA: all four central models; GD: Pythia-1B (LoRA); client-GA upload: all six federated runs (intermediate model before recovery; the released global model is not collapsed).

Table S15: H2: retrain-relative gap and relative advantage drop $(\mathrm{adv}_{pre} - \mathrm{adv}_{post})/\mathrm{adv}_{pre}$ across the GPT-2 series; slope of the gap per decade of parameters, fitted within each seed (mean and t-interval over seeds).

Method	Attack	Non-members	Gap DistilGPT-2	Gap GPT-2	Gap GPT-2-medium	Drop DistilGPT-2	Drop GPT-2	Drop GPT-2-medium	Slope
GA†	Loss thr.	primary	0.03 ± 0.03	0.02 ± 0.05	0.07 ± 0.04	0.66 ± 0.17	0.72 ± 0.30	0.91 ± 0.11	0.072 ± 0.065
GA†	Loss thr.	holdout10	0.11 ± 0.11	0.12 ± 0.10	0.14 ± 0.12	0.17 ± 0.43	0.44 ± 0.42	0.66 ± 0.24	0.038 ± 0.291
GA†	Min-K%	primary	0.01 ± 0.03	0.02 ± 0.04	0.06 ± 0.03	0.65 ± 0.27	0.73 ± 0.28	0.88 ± 0.08	0.075 ± 0.064
GA†	Min-K%	holdout10	0.10 ± 0.06	0.14 ± 0.03	0.13 ± 0.12	0.66 ± 0.18	0.70 ± 0.18	0.68 ± 0.12	0.049 ± 0.227
GA†	Ref.-calib.	primary	0.03 ± 0.05	0.03 ± 0.05	0.06 ± 0.04	0.91 ± 0.05	0.90 ± 0.11	0.94 ± 0.06	0.047 ± 0.094
GA†	Ref.-calib.	holdout10	-0.02 ± 0.11	-0.02 ± 0.10	-0.02 ± 0.12	0.76 ± 0.26	0.76 ± 0.21	0.77 ± 0.15	0.012 ± 0.288
GD	Loss thr.	primary	-0.04 ± 0.02	-0.05 ± 0.04	-0.40 ± 0.05	-0.04 ± 0.31	0.17 ± 0.38	-1.02 ± 0.12	-0.598 ± 0.083
GD	Loss thr.	holdout10	0.06 ± 0.01	0.05 ± 0.06	-0.36 ± 0.05	0.89 ± 0.14	0.74 ± 0.34	-0.80 ± 0.19	-0.701 ± 0.105
GD	Min-K%	primary	-0.04 ± 0.02	-0.05 ± 0.05	-0.38 ± 0.04	0.26 ± 0.16	0.26 ± 0.34	-0.88 ± 0.06	-0.568 ± 0.076
GD	Min-K%	holdout10	0.09 ± 0.02	0.07 ± 0.06	-0.32 ± 0.06	0.90 ± 0.10	0.78 ± 0.29	-0.70 ± 0.22	-0.689 ± 0.119
GD	Ref.-calib.	primary	-0.05 ± 0.04	-0.06 ± 0.05	-0.42 ± 0.05	0.62 ± 0.09	0.62 ± 0.20	-0.20 ± 0.05	-0.616 ± 0.094

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Table S15 (continued)

Method	Attack	Non-members	Gap DistilGPT-2	Gap GPT-2	Gap GPT-2-medium	Drop DistilGPT-2	Drop GPT-2	Drop GPT-2-medium	Slope
GD	Ref.-calib.	holdout10	-0.01 ± 0.02	-0.02 ± 0.08	-0.54 ± 0.05	0.73 ± 0.06	0.77 ± 0.15	0.01 ± 0.11	-0.897 ± 0.117
NPO	Loss thr.	primary	-0.06 ± 0.01	-0.09 ± 0.09	-0.34 ± 0.12	-0.16 ± 0.46	-0.13 ± 0.85	-0.76 ± 0.46	-0.465 ± 0.234
NPO	Loss thr.	holdout10	0.00 ± 0.04	-0.03 ± 0.10	-0.28 ± 0.12	0.34 ± 0.40	0.23 ± 0.71	-0.50 ± 0.49	-0.468 ± 0.269
NPO	Min-K%	primary	-0.03 ± 0.03	-0.08 ± 0.11	-0.32 ± 0.11	0.34 ± 0.26	0.06 ± 0.78	-0.64 ± 0.41	-0.470 ± 0.237
NPO	Min-K%	holdout10	0.06 ± 0.03	-0.01 ± 0.12	-0.25 ± 0.12	0.82 ± 0.25	0.34 ± 0.75	-0.42 ± 0.47	-0.493 ± 0.268
NPO	Ref.-calib.	primary	-0.05 ± 0.01	-0.10 ± 0.11	-0.36 ± 0.12	0.61 ± 0.14	0.50 ± 0.41	-0.04 ± 0.26	-0.499 ± 0.234
NPO	Ref.-calib.	holdout10	-0.06 ± 0.04	-0.11 ± 0.14	-0.46 ± 0.13	0.85 ± 0.12	0.80 ± 0.23	0.20 ± 0.30	-0.654 ± 0.306

Mean ± half-width of the 95% t-interval over seeds (up to 5); a single value means one seed. †Model collapsed: retain loss rose more than threefold in 5/5 seeds. GA: all three central models.

Table S16: H3: retrain-relative gap of outliers minus typical records, paired by seed.

Model	Method	Attack	Non-members	Definition	Gap out.	Gap typ.	Difference	Verdict	Adv. out.	Adv. typ.
DistilGPT-2	GA†	Loss thr.	primary	basediff	0.04	0.01	0.035 ± 0.076	inconclusive	0.03	0.04
DistilGPT-2	GA†	Loss thr.	primary	rarity	0.04	0.02	0.019 ± 0.055	inconclusive	0.04	0.04
DistilGPT-2	GA†	Loss thr.	holdout10	basediff	0.13	−0.00	0.134 ± 0.077	outliers higher	0.08	0.13
DistilGPT-2	GA†	Loss thr.	holdout10	rarity	0.20	0.11	0.091 ± 0.087	outliers higher	0.14	0.08
DistilGPT-2	GA†	Min-K%	primary	basediff	−0.01	0.00	-0.014 ± 0.025	inconclusive	0.05	0.05
DistilGPT-2	GA†	Min-K%	primary	rarity	0.03	−0.00	0.035 ± 0.042	inconclusive	0.03	0.06
DistilGPT-2	GA†	Min-K%	holdout10	basediff	0.09	0.01	0.084 ± 0.088	inconclusive	0.05	0.08
DistilGPT-2	GA†	Min-K%	holdout10	rarity	0.20	0.09	0.110 ± 0.116	inconclusive	0.12	0.02
DistilGPT-2	GA†	Ref.-calib.	primary	basediff	0.03	0.02	0.009 ± 0.079	inconclusive	0.03	0.04
DistilGPT-2	GA†	Ref.-calib.	primary	rarity	0.02	0.02	0.007 ± 0.096	inconclusive	0.03	0.03
DistilGPT-2	GA†	Ref.-calib.	holdout10	basediff	0.04	−0.04	0.084 ± 0.080	outliers higher	0.08	0.13
DistilGPT-2	GA†	Ref.-calib.	holdout10	rarity	−0.04	−0.00	-0.030 ± 0.092	inconclusive	0.15	0.08
DistilGPT-2	GD	Loss thr.	primary	basediff	−0.03	−0.06	0.030 ± 0.033	inconclusive	0.10	0.09
DistilGPT-2	GD	Loss thr.	primary	rarity	−0.03	−0.05	0.025 ± 0.020	outliers higher	0.08	0.11
DistilGPT-2	GD	Loss thr.	holdout10	basediff	0.07	0.02	0.056 ± 0.017	outliers higher	0.02	0.09
DistilGPT-2	GD	Loss thr.	holdout10	rarity	0.15	0.03	0.113 ± 0.040	outliers higher	0.03	0.04
DistilGPT-2	GD	Min-K%	primary	basediff	0.01	−0.06	0.074 ± 0.047	outliers higher	0.03	0.11
DistilGPT-2	GD	Min-K%	primary	rarity	0.01	−0.06	0.068 ± 0.026	outliers higher	0.04	0.12
DistilGPT-2	GD	Min-K%	holdout10	basediff	0.12	0.04	0.080 ± 0.018	outliers higher	0.04	0.06
DistilGPT-2	GD	Min-K%	holdout10	rarity	0.21	0.05	0.160 ± 0.027	outliers higher	0.08	0.03
DistilGPT-2	GD	Ref.-calib.	primary	basediff	−0.03	−0.06	0.029 ± 0.050	inconclusive	0.09	0.10
DistilGPT-2	GD	Ref.-calib.	primary	rarity	−0.03	−0.07	0.037 ± 0.053	inconclusive	0.06	0.11
DistilGPT-2	GD	Ref.-calib.	holdout10	basediff	0.03	−0.02	0.046 ± 0.021	outliers higher	0.06	0.10
DistilGPT-2	GD	Ref.-calib.	holdout10	rarity	0.04	−0.03	0.067 ± 0.042	outliers higher	0.19	0.05
DistilGPT-2	NPO	Loss thr.	primary	basediff	−0.00	−0.08	0.083 ± 0.051	outliers higher	0.07	0.11
DistilGPT-2	NPO	Loss thr.	primary	rarity	−0.02	−0.07	0.048 ± 0.021	outliers higher	0.07	0.12
DistilGPT-2	NPO	Loss thr.	holdout10	basediff	0.04	−0.05	0.093 ± 0.027	outliers higher	0.04	0.03

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Table S16 (continued)

Model	Method	Attack	Non-members	Definition	Gap out.	Gap typ.	Difference	Verdict	Adv. out.	Adv. typ.
DistilGPT-2	NPO	Loss thr.	holdout10	rarity	0.09	−0.03	0.115 ± 0.031	outliers higher	0.04	0.10
DistilGPT-2	NPO	Min-K%	primary	basediff	0.05	−0.06	0.111 ± 0.059	outliers higher	0.04	0.11
DistilGPT-2	NPO	Min-K%	primary	rarity	0.02	−0.05	0.067 ± 0.033	outliers higher	0.04	0.10
DistilGPT-2	NPO	Min-K%	holdout10	basediff	0.12	−0.01	0.126 ± 0.016	outliers higher	0.04	0.03
DistilGPT-2	NPO	Min-K%	holdout10	rarity	0.17	0.02	0.145 ± 0.036	outliers higher	0.05	0.06
DistilGPT-2	NPO	Ref.-calib.	primary	basediff	0.01	−0.09	0.098 ± 0.065	outliers higher	0.06	0.13
DistilGPT-2	NPO	Ref.-calib.	primary	rarity	−0.01	−0.08	0.064 ± 0.026	outliers higher	0.04	0.12
DistilGPT-2	NPO	Ref.-calib.	holdout10	basediff	0.01	−0.10	0.108 ± 0.024	outliers higher	0.04	0.03
DistilGPT-2	NPO	Ref.-calib.	holdout10	rarity	0.00	−0.08	0.084 ± 0.036	outliers higher	0.16	0.02
GPT-2	GA†	Loss thr.	primary	basediff	0.04	0.00	0.033 ± 0.102	inconclusive	0.07	0.04
GPT-2	GA†	Loss thr.	primary	rarity	0.02	0.03	-0.007 ± 0.099	inconclusive	0.06	0.04
GPT-2	GA†	Loss thr.	holdout10	basediff	0.11	0.00	0.107 ± 0.079	outliers higher	0.10	0.11
GPT-2	GA†	Loss thr.	holdout10	rarity	0.20	0.11	0.084 ± 0.096	inconclusive	0.13	0.06
GPT-2	GA†	Min-K%	primary	basediff	−0.04	0.03	-0.073 ± 0.061	typical higher	0.08	0.05
GPT-2	GA†	Min-K%	primary	rarity	−0.01	0.02	-0.029 ± 0.069	inconclusive	0.05	0.05
GPT-2	GA†	Min-K%	holdout10	basediff	0.06	0.07	-0.013 ± 0.072	inconclusive	0.07	0.09
GPT-2	GA†	Min-K%	holdout10	rarity	0.22	0.13	0.089 ± 0.116	inconclusive	0.10	0.03
GPT-2	GA†	Ref.-calib.	primary	basediff	0.03	0.03	0.001 ± 0.094	inconclusive	0.07	0.04
GPT-2	GA†	Ref.-calib.	primary	rarity	0.02	0.02	-0.001 ± 0.110	inconclusive	0.05	0.03
GPT-2	GA†	Ref.-calib.	holdout10	basediff	0.02	−0.03	0.047 ± 0.083	inconclusive	0.11	0.11
GPT-2	GA†	Ref.-calib.	holdout10	rarity	−0.04	−0.00	-0.036 ± 0.090	inconclusive	0.15	0.08
GPT-2	GD	Loss thr.	primary	basediff	−0.04	−0.06	0.016 ± 0.038	inconclusive	0.13	0.09
GPT-2	GD	Loss thr.	primary	rarity	−0.03	−0.06	0.030 ± 0.033	inconclusive	0.07	0.12
GPT-2	GD	Loss thr.	holdout10	basediff	0.07	0.01	0.061 ± 0.039	outliers higher	0.05	0.08
GPT-2	GD	Loss thr.	holdout10	rarity	0.14	0.02	0.113 ± 0.012	outliers higher	0.04	0.05
GPT-2	GD	Min-K%	primary	basediff	−0.04	−0.06	0.024 ± 0.062	inconclusive	0.07	0.12
GPT-2	GD	Min-K%	primary	rarity	−0.02	−0.07	0.055 ± 0.036	outliers higher	0.06	0.14
GPT-2	GD	Min-K%	holdout10	basediff	0.07	0.03	0.041 ± 0.046	inconclusive	0.05	0.05
GPT-2	GD	Min-K%	holdout10	rarity	0.19	0.04	0.146 ± 0.027	outliers higher	0.07	0.06
GPT-2	GD	Ref.-calib.	primary	basediff	−0.05	−0.06	0.007 ± 0.065	inconclusive	0.12	0.11
GPT-2	GD	Ref.-calib.	primary	rarity	−0.03	−0.08	0.047 ± 0.050	inconclusive	0.07	0.13
GPT-2	GD	Ref.-calib.	holdout10	basediff	0.02	−0.02	0.035 ± 0.045	inconclusive	0.09	0.09
GPT-2	GD	Ref.-calib.	holdout10	rarity	0.03	−0.04	0.065 ± 0.022	outliers higher	0.19	0.06
GPT-2	NPO	Loss thr.	primary	basediff	−0.05	−0.11	0.066 ± 0.058	outliers higher	0.13	0.14
GPT-2	NPO	Loss thr.	primary	rarity	−0.05	−0.10	0.055 ± 0.035	outliers higher	0.09	0.16
GPT-2	NPO	Loss thr.	holdout10	basediff	−0.00	−0.09	0.087 ± 0.059	outliers higher	0.09	0.05
GPT-2	NPO	Loss thr.	holdout10	rarity	0.05	−0.05	0.106 ± 0.034	outliers higher	0.07	0.13
GPT-2	NPO	Min-K%	primary	basediff	−0.06	−0.10	0.045 ± 0.066	inconclusive	0.09	0.16
GPT-2	NPO	Min-K%	primary	rarity	−0.04	−0.10	0.062 ± 0.057	outliers higher	0.08	0.17
GPT-2	NPO	Min-K%	holdout10	basediff	0.01	−0.07	0.086 ± 0.067	outliers higher	0.09	0.07
GPT-2	NPO	Min-K%	holdout10	rarity	0.10	−0.03	0.131 ± 0.055	outliers higher	0.07	0.13
GPT-2	NPO	Ref.-calib.	primary	basediff	−0.05	−0.12	0.063 ± 0.072	inconclusive	0.13	0.17

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Table S16 (continued)

Model	Method	Attack	Non-members	Definition	Gap out.	Gap typ.	Difference	Verdict	Adv. out.	Adv. typ.
GPT-2	NPO	Ref.-calib.	primary	rarity	−0.05	−0.12	0.074 ± 0.055	outliers higher	0.09	0.17
GPT-2	NPO	Ref.-calib.	holdout10	basediff	−0.05	−0.14	0.085 ± 0.061	outliers higher	0.08	0.06
GPT-2	NPO	Ref.-calib.	holdout10	rarity	−0.06	−0.13	0.072 ± 0.065	outliers higher	0.16	0.07
GPT-2-medium	GA†	Loss thr.	primary	basediff	0.05	0.05	-0.002 ± 0.134	inconclusive	0.05	0.02
GPT-2-medium	GA†	Loss thr.	primary	rarity	0.08	0.07	0.011 ± 0.027	inconclusive	0.03	0.03
GPT-2-medium	GA†	Loss thr.	holdout10	basediff	0.14	0.03	0.118 ± 0.067	outliers higher	0.08	0.13
GPT-2-medium	GA†	Loss thr.	holdout10	rarity	0.22	0.13	0.092 ± 0.138	inconclusive	0.16	0.06
GPT-2-medium	GA†	Min-K%	primary	basediff	0.03	0.05	-0.014 ± 0.119	inconclusive	0.05	0.04
GPT-2-medium	GA†	Min-K%	primary	rarity	0.07	0.05	0.016 ± 0.038	inconclusive	0.02	0.05
GPT-2-medium	GA†	Min-K%	holdout10	basediff	0.16	0.04	0.120 ± 0.127	inconclusive	0.06	0.14
GPT-2-medium	GA†	Min-K%	holdout10	rarity	0.23	0.12	0.113 ± 0.093	outliers higher	0.15	0.07
GPT-2-medium	GA†	Ref.-calib.	primary	basediff	0.04	0.07	-0.035 ± 0.120	inconclusive	0.05	0.02
GPT-2-medium	GA†	Ref.-calib.	primary	rarity	0.08	0.04	0.035 ± 0.028	outliers higher	0.04	0.03
GPT-2-medium	GA†	Ref.-calib.	holdout10	basediff	0.04	−0.01	0.050 ± 0.069	inconclusive	0.08	0.13
GPT-2-medium	GA†	Ref.-calib.	holdout10	rarity	−0.00	−0.01	0.004 ± 0.133	inconclusive	0.17	0.07
GPT-2-medium	GD	Loss thr.	primary	basediff	−0.42	−0.42	0.004 ± 0.033	inconclusive	0.47	0.47
GPT-2-medium	GD	Loss thr.	primary	rarity	−0.44	−0.39	-0.052 ± 0.022	typical higher	0.48	0.46
GPT-2-medium	GD	Loss thr.	holdout10	basediff	−0.35	−0.49	0.145 ± 0.029	outliers higher	0.46	0.44
GPT-2-medium	GD	Loss thr.	holdout10	rarity	−0.31	−0.36	0.048 ± 0.027	outliers higher	0.45	0.45
GPT-2-medium	GD	Min-K%	primary	basediff	−0.41	−0.39	-0.015 ± 0.027	inconclusive	0.47	0.46
GPT-2-medium	GD	Min-K%	primary	rarity	−0.43	−0.36	-0.070 ± 0.011	typical higher	0.48	0.46
GPT-2-medium	GD	Min-K%	holdout10	basediff	−0.33	−0.43	0.095 ± 0.023	outliers higher	0.46	0.43
GPT-2-medium	GD	Min-K%	holdout10	rarity	−0.27	−0.32	0.051 ± 0.023	outliers higher	0.43	0.45
GPT-2-medium	GD	Ref.-calib.	primary	basediff	−0.43	−0.40	-0.026 ± 0.038	inconclusive	0.47	0.47
GPT-2-medium	GD	Ref.-calib.	primary	rarity	−0.44	−0.42	-0.024 ± 0.042	inconclusive	0.48	0.46
GPT-2-medium	GD	Ref.-calib.	holdout10	basediff	−0.47	−0.54	0.071 ± 0.027	outliers higher	0.46	0.44
GPT-2-medium	GD	Ref.-calib.	holdout10	rarity	−0.56	−0.52	-0.044 ± 0.018	typical higher	0.45	0.45
GPT-2-medium	NPO	Loss thr.	primary	basediff	−0.35	−0.36	0.010 ± 0.047	inconclusive	0.41	0.41
GPT-2-medium	NPO	Loss thr.	primary	rarity	−0.37	−0.33	-0.044 ± 0.046	inconclusive	0.41	0.41
GPT-2-medium	NPO	Loss thr.	holdout10	basediff	−0.27	−0.41	0.144 ± 0.033	outliers higher	0.38	0.36
GPT-2-medium	NPO	Loss thr.	holdout10	rarity	−0.23	−0.28	0.047 ± 0.043	outliers higher	0.37	0.38
GPT-2-medium	NPO	Min-K%	primary	basediff	−0.33	−0.34	0.003 ± 0.047	inconclusive	0.40	0.41
GPT-2-medium	NPO	Min-K%	primary	rarity	−0.37	−0.31	-0.062 ± 0.037	typical higher	0.41	0.41
GPT-2-medium	NPO	Min-K%	holdout10	basediff	−0.24	−0.36	0.114 ± 0.041	outliers higher	0.37	0.36
GPT-2-medium	NPO	Min-K%	holdout10	rarity	−0.21	−0.25	0.038 ± 0.041	inconclusive	0.36	0.37
GPT-2-medium	NPO	Ref.-calib.	primary	basediff	−0.37	−0.35	-0.021 ± 0.055	inconclusive	0.41	0.41
GPT-2-medium	NPO	Ref.-calib.	primary	rarity	−0.38	−0.36	-0.020 ± 0.046	inconclusive	0.42	0.40
GPT-2-medium	NPO	Ref.-calib.	holdout10	basediff	−0.38	−0.46	0.078 ± 0.033	outliers higher	0.37	0.36
GPT-2-medium	NPO	Ref.-calib.	holdout10	rarity	−0.47	−0.43	-0.036 ± 0.037	inconclusive	0.35	0.37
GPT-2	Client GA, upload†	Loss thr.	primary	basediff	0.00	−0.05	0.049 ± 0.316	inconclusive	0.08	0.08
GPT-2	Client GA, upload†	Loss thr.	primary	rarity	−0.02	−0.02	-0.002 ± 0.024	inconclusive	0.06	0.08
GPT-2	Client GA, upload†	Loss thr.	holdout10	basediff	0.07	−0.08	0.151 ± 0.225	inconclusive	0.07	0.13

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Table S16 (continued)

Model	Method	Attack	Non-members	Definition	Gap out.	Gap typ.	Difference	Verdict	Adv. out.	Adv. typ.
GPT-2	Client GA, upload†	Loss thr.	holdout10	rarity	0.12	0.05	0.071 ± 0.210	inconclusive	0.13	0.09
GPT-2	Client GA, upload†	Min-K%	primary	basediff	−0.09	−0.06	-0.028 ± 0.257	inconclusive	0.11	0.12
GPT-2	Client GA, upload†	Min-K%	primary	rarity	−0.07	−0.05	-0.019 ± 0.054	inconclusive	0.09	0.12
GPT-2	Client GA, upload†	Min-K%	holdout10	basediff	0.04	−0.05	0.086 ± 0.290	inconclusive	0.04	0.11
GPT-2	Client GA, upload†	Min-K%	holdout10	rarity	0.13	0.05	0.076 ± 0.181	inconclusive	0.11	0.07
GPT-2	Client GA, upload†	Ref.-calib.	primary	basediff	−0.01	−0.02	0.005 ± 0.282	inconclusive	0.08	0.07
GPT-2	Client GA, upload†	Ref.-calib.	primary	rarity	−0.00	−0.02	0.020 ± 0.051	inconclusive	0.05	0.07
GPT-2	Client GA, upload†	Ref.-calib.	holdout10	basediff	−0.03	−0.12	0.088 ± 0.217	inconclusive	0.06	0.13
GPT-2	Client GA, upload†	Ref.-calib.	holdout10	rarity	−0.12	−0.07	-0.046 ± 0.206	inconclusive	0.11	0.08
GPT-2	Client GA, global	Loss thr.	primary	basediff	0.02	−0.00	0.018 ± 0.009	outliers higher	0.07	0.03
GPT-2	Client GA, global	Loss thr.	primary	rarity	0.00	−0.00	0.006 ± 0.018	inconclusive	0.03	0.07
GPT-2	Client GA, global	Loss thr.	holdout10	basediff	0.01	0.01	0.006 ± 0.035	inconclusive	0.06	0.08
GPT-2	Client GA, global	Loss thr.	holdout10	rarity	0.01	0.00	0.012 ± 0.020	inconclusive	0.11	0.07
GPT-2	Client GA, global	Min-K%	primary	basediff	0.01	0.00	0.009 ± 0.021	inconclusive	0.02	0.05
GPT-2	Client GA, global	Min-K%	primary	rarity	0.00	0.00	-0.001 ± 0.020	inconclusive	0.02	0.07
GPT-2	Client GA, global	Min-K%	holdout10	basediff	0.01	0.01	-0.004 ± 0.052	inconclusive	0.05	0.03
GPT-2	Client GA, global	Min-K%	holdout10	rarity	0.02	0.01	0.010 ± 0.042	inconclusive	0.11	0.09
GPT-2	Client GA, global	Ref.-calib.	primary	basediff	0.01	0.00	0.010 ± 0.021	inconclusive	0.05	0.05
GPT-2	Client GA, global	Ref.-calib.	primary	rarity	0.02	−0.00	0.021 ± 0.042	inconclusive	0.03	0.05
GPT-2	Client GA, global	Ref.-calib.	holdout10	basediff	0.00	0.00	0.004 ± 0.069	inconclusive	0.05	0.12
GPT-2	Client GA, global	Ref.-calib.	holdout10	rarity	0.01	−0.01	0.024 ± 0.016	outliers higher	0.18	0.08
GPT-2	Client GA, upload†	Loss thr.	primary	basediff	0.01	−0.03	0.041 ± 0.110	inconclusive	0.08	0.07
GPT-2	Client GA, upload†	Loss thr.	primary	rarity	−0.01	−0.01	-0.001 ± 0.089	inconclusive	0.05	0.08
GPT-2	Client GA, upload†	Loss thr.	holdout10	basediff	0.16	0.01	0.150 ± 0.036	outliers higher	0.10	0.09
GPT-2	Client GA, upload†	Loss thr.	holdout10	rarity	0.19	0.12	0.070 ± 0.096	inconclusive	0.08	0.05
GPT-2	Client GA, upload†	Min-K%	primary	basediff	−0.10	−0.02	-0.080 ± 0.128	inconclusive	0.13	0.09
GPT-2	Client GA, upload†	Min-K%	primary	rarity	−0.05	−0.04	-0.006 ± 0.162	inconclusive	0.07	0.12
GPT-2	Client GA, upload†	Min-K%	holdout10	basediff	0.08	0.06	0.025 ± 0.100	inconclusive	0.06	0.10
GPT-2	Client GA, upload†	Min-K%	holdout10	rarity	0.23	0.12	0.107 ± 0.096	outliers higher	0.09	0.06
GPT-2	Client GA, upload†	Ref.-calib.	primary	basediff	−0.01	−0.01	0.003 ± 0.020	inconclusive	0.08	0.07
GPT-2	Client GA, upload†	Ref.-calib.	primary	rarity	0.00	−0.02	0.029 ± 0.105	inconclusive	0.05	0.07
GPT-2	Client GA, upload†	Ref.-calib.	holdout10	basediff	0.07	−0.02	0.092 ± 0.056	outliers higher	0.11	0.09
GPT-2	Client GA, upload†	Ref.-calib.	holdout10	rarity	−0.02	0.01	-0.032 ± 0.080	inconclusive	0.12	0.08
GPT-2	Client GA, global	Loss thr.	primary	basediff	−0.02	−0.03	0.009 ± 0.023	inconclusive	0.11	0.06
GPT-2	Client GA, global	Loss thr.	primary	rarity	−0.02	−0.02	0.002 ± 0.025	inconclusive	0.06	0.09
GPT-2	Client GA, global	Loss thr.	holdout10	basediff	−0.01	−0.02	0.010 ± 0.034	inconclusive	0.09	0.05
GPT-2	Client GA, global	Loss thr.	holdout10	rarity	−0.01	−0.02	0.014 ± 0.005	outliers higher	0.12	0.10
GPT-2	Client GA, global	Min-K%	primary	basediff	−0.02	−0.01	-0.009 ± 0.047	inconclusive	0.04	0.07
GPT-2	Client GA, global	Min-K%	primary	rarity	−0.02	−0.01	-0.006 ± 0.022	inconclusive	0.04	0.09
GPT-2	Client GA, global	Min-K%	holdout10	basediff	−0.01	−0.01	0.003 ± 0.035	inconclusive	0.09	0.01
GPT-2	Client GA, global	Min-K%	holdout10	rarity	−0.00	−0.01	0.007 ± 0.018	inconclusive	0.14	0.11
GPT-2	Client GA, global	Ref.-calib.	primary	basediff	−0.02	−0.04	0.021 ± 0.046	inconclusive	0.09	0.10

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Table S16 (continued)

Model	Method	Attack	Non-members	Definition	Gap out.	Gap typ.	Difference	Verdict	Adv. out.	Adv. typ.
GPT-2	Client GA, global	Ref.-calib.	primary	rarity	−0.04	−0.04	0.004 ± 0.047	inconclusive	0.09	0.09
GPT-2	Client GA, global	Ref.-calib.	holdout10	basediff	−0.03	−0.04	0.012 ± 0.059	inconclusive	0.01	0.06
GPT-2	Client GA, global	Ref.-calib.	holdout10	rarity	−0.03	−0.04	0.015 ± 0.012	outliers higher	0.12	0.03
GPT-2	Client GA, upload†	Loss thr.	primary	basediff	−0.02	−0.09	0.064 ± 0.267	inconclusive	0.11	0.11
GPT-2	Client GA, upload†	Loss thr.	primary	rarity	−0.06	−0.06	0.003 ± 0.090	inconclusive	0.09	0.13
GPT-2	Client GA, upload†	Loss thr.	holdout10	basediff	0.03	−0.09	0.118 ± 0.188	inconclusive	0.07	0.11
GPT-2	Client GA, upload†	Loss thr.	holdout10	rarity	0.13	0.03	0.101 ± 0.116	inconclusive	0.10	0.08
GPT-2	Client GA, upload†	Min-K%	primary	basediff	−0.08	−0.09	0.000 ± 0.293	inconclusive	0.10	0.13
GPT-2	Client GA, upload†	Min-K%	primary	rarity	−0.09	−0.07	-0.020 ± 0.100	inconclusive	0.09	0.13
GPT-2	Client GA, upload†	Min-K%	holdout10	basediff	0.00	−0.05	0.050 ± 0.149	inconclusive	0.05	0.09
GPT-2	Client GA, upload†	Min-K%	holdout10	rarity	0.12	0.04	0.077 ± 0.211	inconclusive	0.09	0.05
GPT-2	Client GA, upload†	Ref.-calib.	primary	basediff	−0.04	−0.06	0.015 ± 0.187	inconclusive	0.11	0.11
GPT-2	Client GA, upload†	Ref.-calib.	primary	rarity	−0.05	−0.07	0.019 ± 0.124	inconclusive	0.09	0.11
GPT-2	Client GA, upload†	Ref.-calib.	holdout10	basediff	−0.09	−0.14	0.053 ± 0.188	inconclusive	0.07	0.11
GPT-2	Client GA, upload†	Ref.-calib.	holdout10	rarity	−0.15	−0.10	-0.043 ± 0.136	inconclusive	0.09	0.07
GPT-2	Client GA, global	Loss thr.	primary	basediff	0.09	0.05	0.041 ± 0.019	outliers higher	0.02	0.03
GPT-2	Client GA, global	Loss thr.	primary	rarity	0.05	0.04	0.011 ± 0.009	outliers higher	0.02	0.02
GPT-2	Client GA, global	Loss thr.	holdout10	basediff	0.09	0.06	0.036 ± 0.002	outliers higher	0.01	0.12
GPT-2	Client GA, global	Loss thr.	holdout10	rarity	0.05	0.04	0.011 ± 0.007	outliers higher	0.08	0.04
GPT-2	Client GA, global	Min-K%	primary	basediff	0.08	0.05	0.023 ± 0.008	outliers higher	0.06	0.02
GPT-2	Client GA, global	Min-K%	primary	rarity	0.06	0.05	0.007 ± 0.011	inconclusive	0.06	0.02
GPT-2	Client GA, global	Min-K%	holdout10	basediff	0.07	0.06	0.010 ± 0.009	outliers higher	0.02	0.07
GPT-2	Client GA, global	Min-K%	holdout10	rarity	0.06	0.05	0.016 ± 0.007	outliers higher	0.07	0.05
GPT-2	Client GA, global	Ref.-calib.	primary	basediff	0.12	0.10	0.020 ± 0.042	inconclusive	0.06	0.05
GPT-2	Client GA, global	Ref.-calib.	primary	rarity	0.13	0.10	0.025 ± 0.014	outliers higher	0.09	0.06
GPT-2	Client GA, global	Ref.-calib.	holdout10	basediff	0.12	0.10	0.020 ± 0.009	outliers higher	0.18	0.22
GPT-2	Client GA, global	Ref.-calib.	holdout10	rarity	0.12	0.10	0.022 ± 0.019	outliers higher	0.31	0.19
DistilGPT-2	Client GA, upload†	Loss thr.	primary	basediff	−0.01	−0.05	0.031 ± 0.113	inconclusive	0.08	0.07
DistilGPT-2	Client GA, upload†	Loss thr.	primary	rarity	−0.00	−0.03	0.025 ± 0.069	inconclusive	0.06	0.08
DistilGPT-2	Client GA, upload†	Loss thr.	holdout10	basediff	0.03	−0.11	0.138 ± 0.029	outliers higher	0.08	0.07
DistilGPT-2	Client GA, upload†	Loss thr.	holdout10	rarity	0.08	0.03	0.052 ± 0.081	inconclusive	0.10	0.06
DistilGPT-2	Client GA, upload†	Min-K%	primary	basediff	−0.07	−0.05	-0.017 ± 0.072	inconclusive	0.10	0.09
DistilGPT-2	Client GA, upload†	Min-K%	primary	rarity	−0.02	−0.05	0.026 ± 0.045	inconclusive	0.08	0.10
DistilGPT-2	Client GA, upload†	Min-K%	holdout10	basediff	0.00	−0.05	0.057 ± 0.024	outliers higher	0.09	0.06
DistilGPT-2	Client GA, upload†	Min-K%	holdout10	rarity	0.09	0.05	0.034 ± 0.058	inconclusive	0.09	0.05
DistilGPT-2	Client GA, upload†	Ref.-calib.	primary	basediff	−0.02	−0.03	0.007 ± 0.087	inconclusive	0.07	0.06
DistilGPT-2	Client GA, upload†	Ref.-calib.	primary	rarity	−0.02	−0.02	0.002 ± 0.072	inconclusive	0.05	0.06
DistilGPT-2	Client GA, upload†	Ref.-calib.	holdout10	basediff	−0.05	−0.15	0.102 ± 0.029	outliers higher	0.07	0.07
DistilGPT-2	Client GA, upload†	Ref.-calib.	holdout10	rarity	−0.15	−0.09	-0.063 ± 0.071	inconclusive	0.06	0.04
DistilGPT-2	Client GA, global	Loss thr.	primary	basediff	−0.00	−0.03	0.022 ± 0.022	inconclusive	0.07	0.05
DistilGPT-2	Client GA, global	Loss thr.	primary	rarity	−0.02	−0.02	-0.001 ± 0.013	inconclusive	0.06	0.07
DistilGPT-2	Client GA, global	Loss thr.	holdout10	basediff	−0.00	−0.03	0.021 ± 0.030	inconclusive	0.08	0.05

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Table S16 (continued)

Model	Method	Attack	Non-members	Definition	Gap out.	Gap typ.	Difference	Verdict	Adv. out.	Adv. typ.
DistilGPT-2	Client GA, global	Loss thr.	holdout10	rarity	−0.00	−0.02	0.017 ± 0.008	outliers higher	0.12	0.09
DistilGPT-2	Client GA, global	Min-K%	primary	basediff	−0.01	−0.01	0.009 ± 0.007	outliers higher	0.03	0.05
DistilGPT-2	Client GA, global	Min-K%	primary	rarity	−0.01	−0.01	-0.002 ± 0.029	inconclusive	0.05	0.06
DistilGPT-2	Client GA, global	Min-K%	holdout10	basediff	0.00	−0.01	0.013 ± 0.036	inconclusive	0.06	0.02
DistilGPT-2	Client GA, global	Min-K%	holdout10	rarity	0.02	−0.01	0.025 ± 0.020	outliers higher	0.11	0.09
DistilGPT-2	Client GA, global	Ref.-calib.	primary	basediff	−0.01	−0.03	0.024 ± 0.017	outliers higher	0.06	0.07
DistilGPT-2	Client GA, global	Ref.-calib.	primary	rarity	−0.03	−0.03	-0.002 ± 0.029	inconclusive	0.04	0.07
DistilGPT-2	Client GA, global	Ref.-calib.	holdout10	basediff	−0.02	−0.04	0.022 ± 0.035	inconclusive	0.02	0.08
DistilGPT-2	Client GA, global	Ref.-calib.	holdout10	rarity	−0.04	−0.05	0.017 ± 0.017	outliers higher	0.13	0.04
GPT-2	Client GA, upload†	Loss thr.	primary	basediff	−0.01	−0.07	0.057 ± 0.108	inconclusive	0.09	0.09
GPT-2	Client GA, upload†	Loss thr.	primary	rarity	−0.03	−0.03	0.002 ± 0.072	inconclusive	0.07	0.09
GPT-2	Client GA, upload†	Loss thr.	holdout10	basediff	0.06	−0.12	0.178 ± 0.057	outliers higher	0.07	0.08
GPT-2	Client GA, upload†	Loss thr.	holdout10	rarity	0.08	0.02	0.055 ± 0.048	outliers higher	0.08	0.07
GPT-2	Client GA, upload†	Min-K%	primary	basediff	−0.08	−0.06	-0.020 ± 0.065	inconclusive	0.10	0.11
GPT-2	Client GA, upload†	Min-K%	primary	rarity	−0.06	−0.04	-0.019 ± 0.065	inconclusive	0.09	0.11
GPT-2	Client GA, upload†	Min-K%	holdout10	basediff	0.02	−0.06	0.077 ± 0.106	inconclusive	0.08	0.10
GPT-2	Client GA, upload†	Min-K%	holdout10	rarity	0.11	0.04	0.064 ± 0.047	outliers higher	0.08	0.08
GPT-2	Client GA, upload†	Ref.-calib.	primary	basediff	−0.01	−0.04	0.027 ± 0.087	inconclusive	0.09	0.09
GPT-2	Client GA, upload†	Ref.-calib.	primary	rarity	−0.02	−0.03	0.005 ± 0.056	inconclusive	0.06	0.08
GPT-2	Client GA, upload†	Ref.-calib.	holdout10	basediff	−0.04	−0.16	0.120 ± 0.051	outliers higher	0.06	0.07
GPT-2	Client GA, upload†	Ref.-calib.	holdout10	rarity	−0.16	−0.10	-0.062 ± 0.060	typical higher	0.06	0.05
GPT-2	Client GA, global	Loss thr.	primary	basediff	0.02	0.00	0.018 ± 0.014	outliers higher	0.05	0.02
GPT-2	Client GA, global	Loss thr.	primary	rarity	0.01	0.00	0.009 ± 0.011	inconclusive	0.02	0.06
GPT-2	Client GA, global	Loss thr.	holdout10	basediff	0.01	0.01	0.005 ± 0.011	inconclusive	0.07	0.08
GPT-2	Client GA, global	Loss thr.	holdout10	rarity	0.02	0.01	0.012 ± 0.014	inconclusive	0.10	0.07
GPT-2	Client GA, global	Min-K%	primary	basediff	0.02	0.01	0.011 ± 0.013	inconclusive	0.02	0.05
GPT-2	Client GA, global	Min-K%	primary	rarity	0.01	0.01	0.002 ± 0.012	inconclusive	0.01	0.06
GPT-2	Client GA, global	Min-K%	holdout10	basediff	0.01	0.02	-0.008 ± 0.004	typical higher	0.06	0.03
GPT-2	Client GA, global	Min-K%	holdout10	rarity	0.02	0.01	0.011 ± 0.015	inconclusive	0.11	0.09
GPT-2	Client GA, global	Ref.-calib.	primary	basediff	0.02	0.01	0.015 ± 0.017	inconclusive	0.05	0.04
GPT-2	Client GA, global	Ref.-calib.	primary	rarity	0.02	0.00	0.020 ± 0.022	inconclusive	0.03	0.04
GPT-2	Client GA, global	Ref.-calib.	holdout10	basediff	0.01	0.01	-0.001 ± 0.018	inconclusive	0.05	0.12
GPT-2	Client GA, global	Ref.-calib.	holdout10	rarity	0.02	−0.00	0.023 ± 0.024	inconclusive	0.19	0.08
GPT-2-medium	Client GA, upload†	Loss thr.	primary	basediff	0.01	−0.00	0.010 ± 0.056	inconclusive	0.05	0.05
GPT-2-medium	Client GA, upload†	Loss thr.	primary	rarity	−0.01	0.04	-0.047 ± 0.059	inconclusive	0.05	0.04
GPT-2-medium	Client GA, upload†	Loss thr.	holdout10	basediff	0.10	−0.06	0.162 ± 0.066	outliers higher	0.07	0.09
GPT-2-medium	Client GA, upload†	Loss thr.	holdout10	rarity	0.13	0.08	0.046 ± 0.094	inconclusive	0.10	0.04
GPT-2-medium	Client GA, upload†	Min-K%	primary	basediff	−0.00	0.01	-0.009 ± 0.068	inconclusive	0.06	0.07
GPT-2-medium	Client GA, upload†	Min-K%	primary	rarity	−0.02	0.03	-0.045 ± 0.061	inconclusive	0.06	0.07
GPT-2-medium	Client GA, upload†	Min-K%	holdout10	basediff	0.14	0.02	0.125 ± 0.055	outliers higher	0.05	0.09
GPT-2-medium	Client GA, upload†	Min-K%	holdout10	rarity	0.20	0.11	0.086 ± 0.078	outliers higher	0.08	0.05
GPT-2-medium	Client GA, upload†	Ref.-calib.	primary	basediff	−0.00	0.02	-0.025 ± 0.071	inconclusive	0.05	0.04

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Table S16 (continued)

Model	Method	Attack	Non-members	Definition	Gap out.	Gap typ.	Difference	Verdict	Adv. out.	Adv. typ.
GPT-2-medium	Client GA, upload†	Ref.-calib.	primary	rarity	−0.01	0.02	-0.032 ± 0.067	inconclusive	0.04	0.02
GPT-2-medium	Client GA, upload†	Ref.-calib.	holdout10	basediff	−0.02	−0.11	0.090 ± 0.067	outliers higher	0.06	0.08
GPT-2-medium	Client GA, upload†	Ref.-calib.	holdout10	rarity	−0.10	−0.06	-0.045 ± 0.093	inconclusive	0.10	0.04
GPT-2-medium	Client GA, global	Loss thr.	primary	basediff	0.09	0.05	0.038 ± 0.011	outliers higher	0.04	0.02
GPT-2-medium	Client GA, global	Loss thr.	primary	rarity	0.06	0.04	0.017 ± 0.007	outliers higher	0.03	0.03
GPT-2-medium	Client GA, global	Loss thr.	holdout10	basediff	0.09	0.05	0.039 ± 0.013	outliers higher	0.02	0.10
GPT-2-medium	Client GA, global	Loss thr.	holdout10	rarity	0.06	0.04	0.015 ± 0.006	outliers higher	0.08	0.05
GPT-2-medium	Client GA, global	Min-K%	primary	basediff	0.07	0.05	0.021 ± 0.008	outliers higher	0.02	0.02
GPT-2-medium	Client GA, global	Min-K%	primary	rarity	0.06	0.05	0.010 ± 0.005	outliers higher	0.02	0.05
GPT-2-medium	Client GA, global	Min-K%	holdout10	basediff	0.08	0.06	0.021 ± 0.010	outliers higher	0.04	0.05
GPT-2-medium	Client GA, global	Min-K%	holdout10	rarity	0.06	0.05	0.010 ± 0.011	inconclusive	0.09	0.07
GPT-2-medium	Client GA, global	Ref.-calib.	primary	basediff	0.10	0.09	0.015 ± 0.006	outliers higher	0.06	0.03
GPT-2-medium	Client GA, global	Ref.-calib.	primary	rarity	0.12	0.09	0.030 ± 0.010	outliers higher	0.09	0.04
GPT-2-medium	Client GA, global	Ref.-calib.	holdout10	basediff	0.10	0.09	0.007 ± 0.019	inconclusive	0.13	0.19
GPT-2-medium	Client GA, global	Ref.-calib.	holdout10	rarity	0.12	0.09	0.039 ± 0.015	outliers higher	0.26	0.17
Pythia-1B (LoRA)	GA†	Loss thr.	primary	basediff	0.02	−0.04	0.063 ± 0.080	inconclusive	0.11	0.09
Pythia-1B (LoRA)	GA†	Loss thr.	primary	rarity	−0.04	−0.01	-0.032 ± 0.023	typical higher	0.11	0.07
Pythia-1B (LoRA)	GA†	Loss thr.	holdout10	basediff	0.10	−0.06	0.151 ± 0.099	outliers higher	0.13	0.18
Pythia-1B (LoRA)	GA†	Loss thr.	holdout10	rarity	0.17	0.06	0.110 ± 0.047	outliers higher	0.13	0.15
Pythia-1B (LoRA)	GA†	Min-K%	primary	basediff	0.01	0.01	-0.000 ± 0.063	inconclusive	0.09	0.09
Pythia-1B (LoRA)	GA†	Min-K%	primary	rarity	−0.03	0.04	-0.065 ± 0.042	typical higher	0.13	0.08
Pythia-1B (LoRA)	GA†	Min-K%	holdout10	basediff	0.02	−0.05	0.070 ± 0.033	outliers higher	0.10	0.12
Pythia-1B (LoRA)	GA†	Min-K%	holdout10	rarity	0.09	0.03	0.066 ± 0.127	inconclusive	0.10	0.12
Pythia-1B (LoRA)	GA†	Ref.-calib.	primary	basediff	−0.02	−0.02	-0.006 ± 0.090	inconclusive	0.11	0.09
Pythia-1B (LoRA)	GA†	Ref.-calib.	primary	rarity	−0.06	−0.01	-0.052 ± 0.041	typical higher	0.09	0.08
Pythia-1B (LoRA)	GA†	Ref.-calib.	holdout10	basediff	−0.03	−0.12	0.097 ± 0.091	outliers higher	0.15	0.18
Pythia-1B (LoRA)	GA†	Ref.-calib.	holdout10	rarity	−0.06	−0.09	0.028 ± 0.049	inconclusive	0.19	0.17
Pythia-1B (LoRA)	GD†	Loss thr.	primary	basediff	−0.38	−0.40	0.024 ± 0.077	inconclusive	0.47	0.43
Pythia-1B (LoRA)	GD†	Loss thr.	primary	rarity	−0.38	−0.38	0.001 ± 0.015	inconclusive	0.46	0.44
Pythia-1B (LoRA)	GD†	Loss thr.	holdout10	basediff	−0.34	−0.43	0.091 ± 0.018	outliers higher	0.43	0.39
Pythia-1B (LoRA)	GD†	Loss thr.	holdout10	rarity	−0.27	−0.34	0.072 ± 0.017	outliers higher	0.42	0.41
Pythia-1B (LoRA)	GD†	Min-K%	primary	basediff	−0.41	−0.37	-0.038 ± 0.061	inconclusive	0.47	0.44
Pythia-1B (LoRA)	GD†	Min-K%	primary	rarity	−0.38	−0.37	-0.009 ± 0.019	inconclusive	0.46	0.44
Pythia-1B (LoRA)	GD†	Min-K%	holdout10	basediff	−0.33	−0.37	0.036 ± 0.026	outliers higher	0.43	0.40
Pythia-1B (LoRA)	GD†	Min-K%	holdout10	rarity	−0.23	−0.31	0.082 ± 0.024	outliers higher	0.42	0.41
Pythia-1B (LoRA)	GD†	Ref.-calib.	primary	basediff	−0.43	−0.38	-0.050 ± 0.042	typical higher	0.47	0.44
Pythia-1B (LoRA)	GD†	Ref.-calib.	primary	rarity	−0.43	−0.40	-0.032 ± 0.035	inconclusive	0.46	0.44
Pythia-1B (LoRA)	GD†	Ref.-calib.	holdout10	basediff	−0.49	−0.51	0.022 ± 0.023	inconclusive	0.43	0.39
Pythia-1B (LoRA)	GD†	Ref.-calib.	holdout10	rarity	−0.55	−0.51	-0.038 ± 0.027	typical higher	0.41	0.40
Pythia-1B (LoRA)	NPO	Loss thr.	primary	basediff	−0.38	−0.43	0.044 ± 0.080	inconclusive	0.47	0.46
Pythia-1B (LoRA)	NPO	Loss thr.	primary	rarity	−0.39	−0.41	0.017 ± 0.027	inconclusive	0.47	0.46
Pythia-1B (LoRA)	NPO	Loss thr.	holdout10	basediff	−0.34	−0.47	0.125 ± 0.066	outliers higher	0.43	0.43

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Table S16 (continued)

Model	Method	Attack	Non-members	Definition	Gap out.	Gap typ.	Difference	Verdict	Adv. out.	Adv. typ.
Pythia-1B (LoRA)	NPO	Loss thr.	holdout10	rarity	−0.29	−0.36	0.078 ± 0.011	outliers higher	0.44	0.43
Pythia-1B (LoRA)	NPO	Min-K%	primary	basediff	−0.41	−0.40	-0.016 ± 0.067	inconclusive	0.48	0.46
Pythia-1B (LoRA)	NPO	Min-K%	primary	rarity	−0.39	−0.39	0.003 ± 0.019	inconclusive	0.47	0.47
Pythia-1B (LoRA)	NPO	Min-K%	holdout10	basediff	−0.33	−0.40	0.071 ± 0.063	outliers higher	0.43	0.43
Pythia-1B (LoRA)	NPO	Min-K%	holdout10	rarity	−0.26	−0.34	0.084 ± 0.015	outliers higher	0.44	0.44
Pythia-1B (LoRA)	NPO	Ref.-calib.	primary	basediff	−0.44	−0.41	-0.031 ± 0.051	inconclusive	0.47	0.46
Pythia-1B (LoRA)	NPO	Ref.-calib.	primary	rarity	−0.44	−0.42	-0.016 ± 0.047	inconclusive	0.47	0.46
Pythia-1B (LoRA)	NPO	Ref.-calib.	holdout10	basediff	−0.49	−0.55	0.061 ± 0.073	inconclusive	0.43	0.43
Pythia-1B (LoRA)	NPO	Ref.-calib.	holdout10	rarity	−0.57	−0.54	-0.030 ± 0.011	typical higher	0.43	0.43

Mean ± half-width of the 95% t-interval over seeds (up to 5); a single value means one seed. Verdict: outliers higher (CI lower bound > 0), typical higher (CI upper bound < 0) or inconclusive. †Model collapsed: retain loss rose more than threefold in every seed. GA: all four central models; GD: Pythia-1B (LoRA); client-GA upload: all six federated runs (intermediate model before recovery; the released global model is not collapsed).

Table S17: H3 ablation (appendix): retrain-relative gap of outliers minus typical records, paired by seed.

Model	Method	Attack	Non-members	Definition	Gap out.	Gap typ.	Difference	Verdict	Adv. out.	Adv. typ.
DistilGPT-2	GA†	Loss thr.	primary	legacy	−0.24	0.14	-0.388 ± 0.363	typical higher	0.19	0.08
DistilGPT-2	GA†	Loss thr.	primary	legacy_pooled	0.16	0.11	0.049 ± 0.110	inconclusive	0.07	0.05
DistilGPT-2	GA†	Loss thr.	holdout10	legacy	−0.22	0.26	-0.480 ± 0.378	typical higher	0.28	0.01
DistilGPT-2	GA†	Loss thr.	holdout10	legacy_pooled	0.21	0.26	-0.058 ± 0.046	typical higher	0.05	0.04
DistilGPT-2	GA†	Min-K%	primary	legacy	−0.17	0.09	-0.261 ± 0.296	inconclusive	0.14	0.09
DistilGPT-2	GA†	Min-K%	primary	legacy_pooled	0.10	0.05	0.052 ± 0.071	inconclusive	0.08	0.05
DistilGPT-2	GA†	Min-K%	holdout10	legacy	−0.12	0.19	-0.302 ± 0.314	inconclusive	0.21	0.03
DistilGPT-2	GA†	Min-K%	holdout10	legacy_pooled	0.17	0.17	0.000 ± 0.079	inconclusive	0.05	0.02
DistilGPT-2	GA†	Ref.-calib.	primary	legacy	−0.03	0.05	-0.083 ± 0.398	inconclusive	0.20	0.09
DistilGPT-2	GA†	Ref.-calib.	primary	legacy_pooled	0.06	0.03	0.030 ± 0.112	inconclusive	0.05	0.04
DistilGPT-2	GA†	Ref.-calib.	holdout10	legacy	−0.09	0.01	-0.100 ± 0.415	inconclusive	0.29	0.08
DistilGPT-2	GA†	Ref.-calib.	holdout10	legacy_pooled	0.02	−0.01	0.031 ± 0.046	inconclusive	0.06	0.10
DistilGPT-2	GD	Loss thr.	primary	legacy	−0.09	−0.03	-0.065 ± 0.031	typical higher	0.21	0.23
DistilGPT-2	GD	Loss thr.	primary	legacy_pooled	0.02	−0.03	0.049 ± 0.021	outliers higher	0.21	0.17
DistilGPT-2	GD	Loss thr.	holdout10	legacy	−0.05	0.10	-0.156 ± 0.027	typical higher	0.29	0.14
DistilGPT-2	GD	Loss thr.	holdout10	legacy_pooled	0.06	0.14	-0.079 ± 0.029	typical higher	0.12	0.09
DistilGPT-2	GD	Min-K%	primary	legacy	−0.10	−0.01	-0.087 ± 0.013	typical higher	0.10	0.18
DistilGPT-2	GD	Min-K%	primary	legacy_pooled	−0.02	−0.02	-0.003 ± 0.020	inconclusive	0.20	0.12
DistilGPT-2	GD	Min-K%	holdout10	legacy	−0.02	0.13	-0.152 ± 0.017	typical higher	0.19	0.07
DistilGPT-2	GD	Min-K%	holdout10	legacy_pooled	0.07	0.16	-0.087 ± 0.024	typical higher	0.12	0.01
DistilGPT-2	GD	Ref.-calib.	primary	legacy	−0.03	−0.06	0.037 ± 0.035	outliers higher	0.02	0.15
DistilGPT-2	GD	Ref.-calib.	primary	legacy_pooled	−0.05	−0.06	0.010 ± 0.016	inconclusive	0.16	0.10
DistilGPT-2	GD	Ref.-calib.	holdout10	legacy	−0.00	−0.01	0.013 ± 0.034	inconclusive	0.19	0.06
DistilGPT-2	GD	Ref.-calib.	holdout10	legacy_pooled	−0.07	0.01	-0.073 ± 0.040	typical higher	0.03	0.12

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Table S17 (continued)

Model	Method	Attack	Non-members	Definition	Gap out.	Gap typ.	Difference	Verdict	Adv. out.	Adv. typ.
DistilGPT-2	NPO	Loss thr.	primary	legacy	−0.16	−0.01	-0.147 ± 0.044	typical higher	0.14	0.22
DistilGPT-2	NPO	Loss thr.	primary	legacy_pooled	−0.02	−0.02	-0.005 ± 0.033	inconclusive	0.25	0.16
DistilGPT-2	NPO	Loss thr.	holdout10	legacy	−0.15	0.07	-0.217 ± 0.033	typical higher	0.19	0.18
DistilGPT-2	NPO	Loss thr.	holdout10	legacy_pooled	−0.03	0.10	-0.137 ± 0.025	typical higher	0.22	0.13
DistilGPT-2	NPO	Min-K%	primary	legacy	−0.15	0.02	-0.175 ± 0.030	typical higher	0.05	0.15
DistilGPT-2	NPO	Min-K%	primary	legacy_pooled	−0.03	0.02	-0.050 ± 0.015	typical higher	0.21	0.09
DistilGPT-2	NPO	Min-K%	holdout10	legacy	−0.10	0.12	-0.221 ± 0.035	typical higher	0.11	0.08
DistilGPT-2	NPO	Min-K%	holdout10	legacy_pooled	0.01	0.15	-0.137 ± 0.020	typical higher	0.19	0.02
DistilGPT-2	NPO	Ref.-calib.	primary	legacy	−0.18	0.00	-0.179 ± 0.032	typical higher	0.13	0.08
DistilGPT-2	NPO	Ref.-calib.	primary	legacy_pooled	−0.07	−0.02	-0.050 ± 0.038	typical higher	0.18	0.06
DistilGPT-2	NPO	Ref.-calib.	holdout10	legacy	−0.18	−0.00	-0.174 ± 0.028	typical higher	0.04	0.07
DistilGPT-2	NPO	Ref.-calib.	holdout10	legacy_pooled	−0.14	0.00	-0.146 ± 0.021	typical higher	0.10	0.11
GPT-2	GA†	Loss thr.	primary	legacy	−0.26	0.14	-0.402 ± 0.269	typical higher	0.13	0.07
GPT-2	GA†	Loss thr.	primary	legacy_pooled	0.22	0.12	0.102 ± 0.115	inconclusive	0.06	0.06
GPT-2	GA†	Loss thr.	holdout10	legacy	−0.22	0.27	-0.489 ± 0.278	typical higher	0.21	0.02
GPT-2	GA†	Loss thr.	holdout10	legacy_pooled	0.30	0.29	0.010 ± 0.065	inconclusive	0.05	0.05
GPT-2	GA†	Min-K%	primary	legacy	−0.16	0.10	-0.255 ± 0.227	typical higher	0.11	0.08
GPT-2	GA†	Min-K%	primary	legacy_pooled	0.17	0.06	0.108 ± 0.091	outliers higher	0.09	0.06
GPT-2	GA†	Min-K%	holdout10	legacy	−0.07	0.23	-0.297 ± 0.240	typical higher	0.16	0.04
GPT-2	GA†	Min-K%	holdout10	legacy_pooled	0.26	0.22	0.046 ± 0.125	inconclusive	0.07	0.02
GPT-2	GA†	Ref.-calib.	primary	legacy	−0.04	0.06	-0.090 ± 0.285	inconclusive	0.15	0.06
GPT-2	GA†	Ref.-calib.	primary	legacy_pooled	0.10	0.03	0.069 ± 0.091	inconclusive	0.06	0.05
GPT-2	GA†	Ref.-calib.	holdout10	legacy	−0.10	0.02	-0.114 ± 0.285	inconclusive	0.21	0.09
GPT-2	GA†	Ref.-calib.	holdout10	legacy_pooled	0.06	−0.00	0.056 ± 0.079	inconclusive	0.09	0.10
GPT-2	GD	Loss thr.	primary	legacy	−0.09	−0.03	-0.058 ± 0.023	typical higher	0.21	0.24
GPT-2	GD	Loss thr.	primary	legacy_pooled	0.03	−0.03	0.062 ± 0.027	outliers higher	0.24	0.20
GPT-2	GD	Loss thr.	holdout10	legacy	−0.05	0.09	-0.136 ± 0.017	typical higher	0.29	0.16
GPT-2	GD	Loss thr.	holdout10	legacy_pooled	0.10	0.13	-0.029 ± 0.042	inconclusive	0.14	0.13
GPT-2	GD	Min-K%	primary	legacy	−0.10	−0.03	-0.062 ± 0.018	typical higher	0.10	0.21
GPT-2	GD	Min-K%	primary	legacy_pooled	−0.02	−0.04	0.022 ± 0.040	inconclusive	0.23	0.16
GPT-2	GD	Min-K%	holdout10	legacy	−0.01	0.10	-0.113 ± 0.013	typical higher	0.19	0.11
GPT-2	GD	Min-K%	holdout10	legacy_pooled	0.09	0.13	-0.034 ± 0.051	inconclusive	0.13	0.07
GPT-2	GD	Ref.-calib.	primary	legacy	0.00	−0.09	0.086 ± 0.037	outliers higher	0.03	0.17
GPT-2	GD	Ref.-calib.	primary	legacy_pooled	−0.06	−0.07	0.013 ± 0.046	inconclusive	0.19	0.13
GPT-2	GD	Ref.-calib.	holdout10	legacy	0.02	−0.04	0.058 ± 0.050	outliers higher	0.20	0.07
GPT-2	GD	Ref.-calib.	holdout10	legacy_pooled	−0.06	−0.02	-0.047 ± 0.066	inconclusive	0.03	0.10
GPT-2	NPO	Loss thr.	primary	legacy	−0.20	−0.04	-0.166 ± 0.061	typical higher	0.12	0.25
GPT-2	NPO	Loss thr.	primary	legacy_pooled	−0.00	−0.03	0.028 ± 0.044	inconclusive	0.27	0.20
GPT-2	NPO	Loss thr.	holdout10	legacy	−0.19	0.04	-0.229 ± 0.063	typical higher	0.15	0.21
GPT-2	NPO	Loss thr.	holdout10	legacy_pooled	0.02	0.08	-0.060 ± 0.065	inconclusive	0.23	0.18
GPT-2	NPO	Min-K%	primary	legacy	−0.21	−0.03	-0.182 ± 0.073	typical higher	0.08	0.20
GPT-2	NPO	Min-K%	primary	legacy_pooled	−0.03	−0.04	0.011 ± 0.047	inconclusive	0.25	0.16

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Table S17 (continued)

Model	Method	Attack	Non-members	Definition	Gap out.	Gap typ.	Difference	Verdict	Adv. out.	Adv. typ.
GPT-2	NPO	Min-K%	holdout10	legacy	−0.17	0.06	-0.227 ± 0.080	typical higher	0.09	0.15
GPT-2	NPO	Min-K%	holdout10	legacy_pooled	0.02	0.08	-0.058 ± 0.069	inconclusive	0.21	0.12
GPT-2	NPO	Ref.-calib.	primary	legacy	−0.17	−0.07	-0.103 ± 0.076	typical higher	0.15	0.15
GPT-2	NPO	Ref.-calib.	primary	legacy_pooled	−0.10	−0.07	-0.034 ± 0.075	inconclusive	0.23	0.13
GPT-2	NPO	Ref.-calib.	holdout10	legacy	−0.19	−0.08	-0.108 ± 0.086	typical higher	0.07	0.09
GPT-2	NPO	Ref.-calib.	holdout10	legacy_pooled	−0.16	−0.07	-0.087 ± 0.094	inconclusive	0.12	0.11
GPT-2-medium	GA†	Loss thr.	primary	legacy	−0.19	0.18	-0.365 ± 0.349	typical higher	0.17	0.10
GPT-2-medium	GA†	Loss thr.	primary	legacy_pooled	0.30	0.18	0.122 ± 0.077	outliers higher	0.09	0.07
GPT-2-medium	GA†	Loss thr.	holdout10	legacy	−0.17	0.27	-0.444 ± 0.347	typical higher	0.27	0.03
GPT-2-medium	GA†	Loss thr.	holdout10	legacy_pooled	0.34	0.32	0.024 ± 0.104	inconclusive	0.05	0.05
GPT-2-medium	GA†	Min-K%	primary	legacy	−0.12	0.14	-0.261 ± 0.318	inconclusive	0.18	0.08
GPT-2-medium	GA†	Min-K%	primary	legacy_pooled	0.20	0.13	0.068 ± 0.057	outliers higher	0.11	0.06
GPT-2-medium	GA†	Min-K%	holdout10	legacy	−0.09	0.23	-0.319 ± 0.336	inconclusive	0.26	0.01
GPT-2-medium	GA†	Min-K%	holdout10	legacy_pooled	0.24	0.25	-0.016 ± 0.046	inconclusive	0.06	0.02
GPT-2-medium	GA†	Ref.-calib.	primary	legacy	0.04	0.06	-0.030 ± 0.358	inconclusive	0.17	0.10
GPT-2-medium	GA†	Ref.-calib.	primary	legacy_pooled	0.11	0.05	0.063 ± 0.071	inconclusive	0.08	0.07
GPT-2-medium	GA†	Ref.-calib.	holdout10	legacy	−0.05	−0.00	-0.044 ± 0.350	inconclusive	0.27	0.07
GPT-2-medium	GA†	Ref.-calib.	holdout10	legacy_pooled	0.03	−0.00	0.031 ± 0.104	inconclusive	0.05	0.07
GPT-2-medium	GD	Loss thr.	primary	legacy	−0.70	−0.27	-0.431 ± 0.066	typical higher	0.44	0.48
GPT-2-medium	GD	Loss thr.	primary	legacy_pooled	−0.11	−0.27	0.158 ± 0.053	outliers higher	0.46	0.48
GPT-2-medium	GD	Loss thr.	holdout10	legacy	−0.71	−0.21	-0.494 ± 0.058	typical higher	0.41	0.46
GPT-2-medium	GD	Loss thr.	holdout10	legacy_pooled	−0.11	−0.16	0.052 ± 0.021	outliers higher	0.45	0.46
GPT-2-medium	GD	Min-K%	primary	legacy	−0.61	−0.28	-0.325 ± 0.066	typical higher	0.44	0.48
GPT-2-medium	GD	Min-K%	primary	legacy_pooled	−0.15	−0.31	0.157 ± 0.063	outliers higher	0.46	0.48
GPT-2-medium	GD	Min-K%	holdout10	legacy	−0.58	−0.21	-0.363 ± 0.058	typical higher	0.41	0.45
GPT-2-medium	GD	Min-K%	holdout10	legacy_pooled	−0.13	−0.20	0.070 ± 0.028	outliers higher	0.43	0.45
GPT-2-medium	GD	Ref.-calib.	primary	legacy	−0.45	−0.40	-0.047 ± 0.059	inconclusive	0.45	0.48
GPT-2-medium	GD	Ref.-calib.	primary	legacy_pooled	−0.33	−0.42	0.094 ± 0.063	outliers higher	0.46	0.48
GPT-2-medium	GD	Ref.-calib.	holdout10	legacy	−0.57	−0.53	-0.038 ± 0.062	inconclusive	0.42	0.46
GPT-2-medium	GD	Ref.-calib.	holdout10	legacy_pooled	−0.47	−0.54	0.069 ± 0.026	outliers higher	0.44	0.46
GPT-2-medium	NPO	Loss thr.	primary	legacy	−0.65	−0.21	-0.442 ± 0.042	typical higher	0.39	0.42
GPT-2-medium	NPO	Loss thr.	primary	legacy_pooled	−0.06	−0.20	0.143 ± 0.032	outliers higher	0.41	0.42
GPT-2-medium	NPO	Loss thr.	holdout10	legacy	−0.64	−0.13	-0.510 ± 0.043	typical higher	0.35	0.38
GPT-2-medium	NPO	Loss thr.	holdout10	legacy_pooled	−0.07	−0.08	0.012 ± 0.020	inconclusive	0.40	0.38
GPT-2-medium	NPO	Min-K%	primary	legacy	−0.58	−0.21	-0.366 ± 0.046	typical higher	0.41	0.41
GPT-2-medium	NPO	Min-K%	primary	legacy_pooled	−0.10	−0.24	0.139 ± 0.039	outliers higher	0.41	0.41
GPT-2-medium	NPO	Min-K%	holdout10	legacy	−0.55	−0.12	-0.421 ± 0.041	typical higher	0.37	0.37
GPT-2-medium	NPO	Min-K%	holdout10	legacy_pooled	−0.10	−0.11	0.013 ± 0.021	inconclusive	0.39	0.36
GPT-2-medium	NPO	Ref.-calib.	primary	legacy	−0.42	−0.33	-0.093 ± 0.039	typical higher	0.42	0.40
GPT-2-medium	NPO	Ref.-calib.	primary	legacy_pooled	−0.27	−0.35	0.080 ± 0.047	outliers higher	0.40	0.41
GPT-2-medium	NPO	Ref.-calib.	holdout10	legacy	−0.52	−0.43	-0.097 ± 0.048	typical higher	0.37	0.36
GPT-2-medium	NPO	Ref.-calib.	holdout10	legacy_pooled	−0.41	−0.43	0.016 ± 0.024	inconclusive	0.39	0.35

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Table S17 (continued)

Model	Method	Attack	Non-members	Definition	Gap out.	Gap typ.	Difference	Verdict	Adv. out.	Adv. typ.
GPT-2	Client GA, upload†	Loss thr.	primary	legacy	−0.38	0.14	-0.523 ± 0.688	inconclusive	0.17	0.08
GPT-2	Client GA, upload†	Loss thr.	primary	legacy_pooled	0.02	0.03	-0.007 ± 0.153	inconclusive	0.09	0.08
GPT-2	Client GA, upload†	Loss thr.	holdout10	legacy	−0.36	0.25	-0.609 ± 0.672	inconclusive	0.23	0.03
GPT-2	Client GA, upload†	Loss thr.	holdout10	legacy_pooled	0.07	0.18	-0.102 ± 0.154	inconclusive	0.09	0.07
GPT-2	Client GA, upload†	Min-K%	primary	legacy	−0.35	0.08	-0.423 ± 0.525	inconclusive	0.16	0.11
GPT-2	Client GA, upload†	Min-K%	primary	legacy_pooled	−0.03	−0.03	-0.001 ± 0.034	inconclusive	0.13	0.11
GPT-2	Client GA, upload†	Min-K%	holdout10	legacy	−0.27	0.20	-0.476 ± 0.550	inconclusive	0.19	0.02
GPT-2	Client GA, upload†	Min-K%	holdout10	legacy_pooled	0.10	0.12	-0.016 ± 0.190	inconclusive	0.07	0.06
GPT-2	Client GA, upload†	Ref.-calib.	primary	legacy	−0.16	0.06	-0.213 ± 0.708	inconclusive	0.19	0.08
GPT-2	Client GA, upload†	Ref.-calib.	primary	legacy_pooled	0.02	−0.01	0.029 ± 0.110	inconclusive	0.08	0.06
GPT-2	Client GA, upload†	Ref.-calib.	holdout10	legacy	−0.25	−0.01	-0.239 ± 0.701	inconclusive	0.24	0.07
GPT-2	Client GA, upload†	Ref.-calib.	holdout10	legacy_pooled	−0.05	−0.08	0.033 ± 0.102	inconclusive	0.08	0.06
GPT-2	Client GA, global	Loss thr.	primary	legacy	−0.00	0.00	-0.006 ± 0.005	typical higher	0.31	0.22
GPT-2	Client GA, global	Loss thr.	primary	legacy_pooled	0.00	0.00	-0.001 ± 0.012	inconclusive	0.11	0.10
GPT-2	Client GA, global	Loss thr.	holdout10	legacy	−0.00	0.01	-0.011 ± 0.008	typical higher	0.35	0.25
GPT-2	Client GA, global	Loss thr.	holdout10	legacy_pooled	0.01	0.01	-0.005 ± 0.014	inconclusive	0.07	0.17
GPT-2	Client GA, global	Min-K%	primary	legacy	0.00	0.01	-0.004 ± 0.010	inconclusive	0.23	0.18
GPT-2	Client GA, global	Min-K%	primary	legacy_pooled	0.01	0.01	-0.003 ± 0.015	inconclusive	0.09	0.07
GPT-2	Client GA, global	Min-K%	holdout10	legacy	0.01	0.02	-0.007 ± 0.014	inconclusive	0.24	0.21
GPT-2	Client GA, global	Min-K%	holdout10	legacy_pooled	0.02	0.02	0.000 ± 0.024	inconclusive	0.10	0.13
GPT-2	Client GA, global	Ref.-calib.	primary	legacy	0.02	−0.00	0.022 ± 0.148	inconclusive	0.06	0.10
GPT-2	Client GA, global	Ref.-calib.	primary	legacy_pooled	0.00	0.00	-0.000 ± 0.048	inconclusive	0.09	0.04
GPT-2	Client GA, global	Ref.-calib.	holdout10	legacy	0.01	−0.01	0.019 ± 0.145	inconclusive	0.22	0.07
GPT-2	Client GA, global	Ref.-calib.	holdout10	legacy_pooled	−0.01	−0.00	-0.005 ± 0.052	inconclusive	0.07	0.13
GPT-2	Client GA, upload†	Loss thr.	primary	legacy	−0.31	0.12	-0.433 ± 0.411	typical higher	0.08	0.10
GPT-2	Client GA, upload†	Loss thr.	primary	legacy_pooled	0.04	0.02	0.013 ± 0.178	inconclusive	0.06	0.08
GPT-2	Client GA, upload†	Loss thr.	holdout10	legacy	−0.22	0.29	-0.513 ± 0.379	typical higher	0.16	0.04
GPT-2	Client GA, upload†	Loss thr.	holdout10	legacy_pooled	0.15	0.23	-0.079 ± 0.060	typical higher	0.09	0.05
GPT-2	Client GA, upload†	Min-K%	primary	legacy	−0.24	0.06	-0.291 ± 0.450	inconclusive	0.11	0.14
GPT-2	Client GA, upload†	Min-K%	primary	legacy_pooled	−0.01	−0.01	0.002 ± 0.134	inconclusive	0.10	0.10
GPT-2	Client GA, upload†	Min-K%	holdout10	legacy	−0.11	0.24	-0.349 ± 0.445	inconclusive	0.19	0.01
GPT-2	Client GA, upload†	Min-K%	holdout10	legacy_pooled	0.18	0.17	0.005 ± 0.097	inconclusive	0.06	0.06
GPT-2	Client GA, upload†	Ref.-calib.	primary	legacy	−0.13	0.04	-0.170 ± 0.370	inconclusive	0.09	0.07
GPT-2	Client GA, upload†	Ref.-calib.	primary	legacy_pooled	0.01	−0.02	0.029 ± 0.117	inconclusive	0.06	0.07
GPT-2	Client GA, upload†	Ref.-calib.	holdout10	legacy	−0.13	0.06	-0.184 ± 0.368	inconclusive	0.15	0.10
GPT-2	Client GA, upload†	Ref.-calib.	holdout10	legacy_pooled	0.04	−0.01	0.048 ± 0.051	inconclusive	0.11	0.10
GPT-2	Client GA, global	Loss thr.	primary	legacy	−0.02	−0.02	-0.005 ± 0.016	inconclusive	0.29	0.24
GPT-2	Client GA, global	Loss thr.	primary	legacy_pooled	−0.04	−0.03	-0.017 ± 0.019	inconclusive	0.14	0.13
GPT-2	Client GA, global	Loss thr.	holdout10	legacy	−0.02	−0.01	-0.004 ± 0.016	inconclusive	0.33	0.27
GPT-2	Client GA, global	Loss thr.	holdout10	legacy_pooled	−0.03	−0.01	-0.018 ± 0.024	inconclusive	0.09	0.20
GPT-2	Client GA, global	Min-K%	primary	legacy	−0.01	−0.01	-0.006 ± 0.022	inconclusive	0.21	0.20
GPT-2	Client GA, global	Min-K%	primary	legacy_pooled	−0.02	−0.01	-0.005 ± 0.033	inconclusive	0.11	0.10

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Table S17 (continued)

Model	Method	Attack	Non-members	Definition	Gap out.	Gap typ.	Difference	Verdict	Adv. out.	Adv. typ.
GPT-2	Client GA, global	Min-K%	holdout10	legacy	−0.01	−0.00	-0.006 ± 0.023	inconclusive	0.22	0.24
GPT-2	Client GA, global	Min-K%	holdout10	legacy_pooled	−0.01	−0.01	-0.000 ± 0.029	inconclusive	0.12	0.16
GPT-2	Client GA, global	Ref.-calib.	primary	legacy	−0.04	−0.03	-0.009 ± 0.066	inconclusive	0.03	0.14
GPT-2	Client GA, global	Ref.-calib.	primary	legacy_pooled	−0.04	−0.03	-0.008 ± 0.016	inconclusive	0.12	0.09
GPT-2	Client GA, global	Ref.-calib.	holdout10	legacy	−0.04	−0.04	-0.000 ± 0.050	inconclusive	0.19	0.01
GPT-2	Client GA, global	Ref.-calib.	holdout10	legacy_pooled	−0.04	−0.04	-0.006 ± 0.042	inconclusive	0.03	0.07
GPT-2	Client GA, upload†	Loss thr.	primary	legacy	−0.37	0.08	-0.452 ± 0.551	inconclusive	0.15	0.14
GPT-2	Client GA, upload†	Loss thr.	primary	legacy_pooled	0.01	0.00	0.013 ± 0.230	inconclusive	0.12	0.12
GPT-2	Client GA, upload†	Loss thr.	holdout10	legacy	−0.32	0.20	-0.525 ± 0.506	typical higher	0.17	0.06
GPT-2	Client GA, upload†	Loss thr.	holdout10	legacy_pooled	0.05	0.15	-0.092 ± 0.056	typical higher	0.08	0.07
GPT-2	Client GA, upload†	Min-K%	primary	legacy	−0.32	0.04	-0.357 ± 0.462	inconclusive	0.14	0.14
GPT-2	Client GA, upload†	Min-K%	primary	legacy_pooled	−0.04	−0.04	-0.005 ± 0.155	inconclusive	0.13	0.12
GPT-2	Client GA, upload†	Min-K%	holdout10	legacy	−0.23	0.17	-0.398 ± 0.458	inconclusive	0.14	0.06
GPT-2	Client GA, upload†	Min-K%	holdout10	legacy_pooled	0.08	0.10	-0.019 ± 0.161	inconclusive	0.05	0.05
GPT-2	Client GA, upload†	Ref.-calib.	primary	legacy	−0.12	−0.02	-0.092 ± 0.540	inconclusive	0.18	0.10
GPT-2	Client GA, upload†	Ref.-calib.	primary	legacy_pooled	−0.02	−0.06	0.038 ± 0.129	inconclusive	0.11	0.10
GPT-2	Client GA, upload†	Ref.-calib.	holdout10	legacy	−0.21	−0.10	-0.108 ± 0.533	inconclusive	0.19	0.02
GPT-2	Client GA, upload†	Ref.-calib.	holdout10	legacy_pooled	−0.12	−0.13	0.013 ± 0.035	inconclusive	0.07	0.06
GPT-2	Client GA, global	Loss thr.	primary	legacy	0.02	0.05	-0.025 ± 0.011	typical higher	0.35	0.17
GPT-2	Client GA, global	Loss thr.	primary	legacy_pooled	0.07	0.07	0.002 ± 0.018	inconclusive	0.06	0.05
GPT-2	Client GA, global	Loss thr.	holdout10	legacy	0.03	0.05	-0.023 ± 0.004	typical higher	0.37	0.21
GPT-2	Client GA, global	Loss thr.	holdout10	legacy_pooled	0.09	0.07	0.020 ± 0.012	outliers higher	0.01	0.12
GPT-2	Client GA, global	Min-K%	primary	legacy	0.04	0.05	-0.008 ± 0.003	typical higher	0.28	0.12
GPT-2	Client GA, global	Min-K%	primary	legacy_pooled	0.06	0.06	-0.007 ± 0.016	inconclusive	0.04	0.02
GPT-2	Client GA, global	Min-K%	holdout10	legacy	0.05	0.05	-0.000 ± 0.005	inconclusive	0.27	0.18
GPT-2	Client GA, global	Min-K%	holdout10	legacy_pooled	0.06	0.06	-0.001 ± 0.015	inconclusive	0.05	0.09
GPT-2	Client GA, global	Ref.-calib.	primary	legacy	0.09	0.12	-0.035 ± 0.009	typical higher	0.10	0.04
GPT-2	Client GA, global	Ref.-calib.	primary	legacy_pooled	0.08	0.12	-0.041 ± 0.025	typical higher	0.01	0.08
GPT-2	Client GA, global	Ref.-calib.	holdout10	legacy	0.07	0.12	-0.044 ± 0.020	typical higher	0.26	0.22
GPT-2	Client GA, global	Ref.-calib.	holdout10	legacy_pooled	0.08	0.12	-0.039 ± 0.011	typical higher	0.18	0.25
DistilGPT-2	Client GA, upload†	Loss thr.	primary	legacy	−0.44	0.16	-0.595 ± 0.165	typical higher	0.14	0.05
DistilGPT-2	Client GA, upload†	Loss thr.	primary	legacy_pooled	0.02	0.01	0.012 ± 0.080	inconclusive	0.08	0.07
DistilGPT-2	Client GA, upload†	Loss thr.	holdout10	legacy	−0.45	0.24	-0.690 ± 0.164	typical higher	0.15	0.02
DistilGPT-2	Client GA, upload†	Loss thr.	holdout10	legacy_pooled	0.01	0.14	-0.137 ± 0.025	typical higher	0.05	0.05
DistilGPT-2	Client GA, upload†	Min-K%	primary	legacy	−0.38	0.11	-0.488 ± 0.182	typical higher	0.16	0.08
DistilGPT-2	Client GA, upload†	Min-K%	primary	legacy_pooled	−0.04	−0.02	-0.023 ± 0.025	inconclusive	0.12	0.09
DistilGPT-2	Client GA, upload†	Min-K%	holdout10	legacy	−0.32	0.21	-0.533 ± 0.191	typical higher	0.16	0.00
DistilGPT-2	Client GA, upload†	Min-K%	holdout10	legacy_pooled	0.09	0.10	-0.009 ± 0.044	inconclusive	0.03	0.05
DistilGPT-2	Client GA, upload†	Ref.-calib.	primary	legacy	−0.24	0.08	-0.321 ± 0.142	typical higher	0.18	0.05
DistilGPT-2	Client GA, upload†	Ref.-calib.	primary	legacy_pooled	−0.01	−0.02	0.015 ± 0.089	inconclusive	0.07	0.05
DistilGPT-2	Client GA, upload†	Ref.-calib.	holdout10	legacy	−0.34	−0.01	-0.326 ± 0.159	typical higher	0.17	0.07
DistilGPT-2	Client GA, upload†	Ref.-calib.	holdout10	legacy_pooled	−0.10	−0.10	0.004 ± 0.032	inconclusive	0.04	0.04

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Table S17 (continued)

Model	Method	Attack	Non-members	Definition	Gap out.	Gap typ.	Difference	Verdict	Adv. out.	Adv. typ.
DistilGPT-2	Client GA, global	Loss thr.	primary	legacy	−0.02	−0.01	-0.012 ± 0.017	inconclusive	0.29	0.22
DistilGPT-2	Client GA, global	Loss thr.	primary	legacy_pooled	−0.01	−0.02	0.005 ± 0.013	inconclusive	0.12	0.10
DistilGPT-2	Client GA, global	Loss thr.	holdout10	legacy	−0.03	−0.01	-0.022 ± 0.029	inconclusive	0.33	0.26
DistilGPT-2	Client GA, global	Loss thr.	holdout10	legacy_pooled	−0.02	−0.01	-0.012 ± 0.032	inconclusive	0.06	0.18
DistilGPT-2	Client GA, global	Min-K%	primary	legacy	−0.03	−0.00	-0.024 ± 0.034	inconclusive	0.21	0.18
DistilGPT-2	Client GA, global	Min-K%	primary	legacy_pooled	−0.01	−0.01	-0.001 ± 0.023	inconclusive	0.09	0.08
DistilGPT-2	Client GA, global	Min-K%	holdout10	legacy	−0.02	0.01	-0.029 ± 0.049	inconclusive	0.22	0.20
DistilGPT-2	Client GA, global	Min-K%	holdout10	legacy_pooled	−0.00	0.01	-0.012 ± 0.029	inconclusive	0.10	0.12
DistilGPT-2	Client GA, global	Ref.-calib.	primary	legacy	−0.01	−0.04	0.025 ± 0.034	inconclusive	0.05	0.12
DistilGPT-2	Client GA, global	Ref.-calib.	primary	legacy_pooled	−0.03	−0.03	0.003 ± 0.012	inconclusive	0.09	0.06
DistilGPT-2	Client GA, global	Ref.-calib.	holdout10	legacy	−0.03	−0.06	0.028 ± 0.042	inconclusive	0.18	0.04
DistilGPT-2	Client GA, global	Ref.-calib.	holdout10	legacy_pooled	−0.05	−0.05	0.001 ± 0.032	inconclusive	0.04	0.08
GPT-2	Client GA, upload†	Loss thr.	primary	legacy	−0.43	0.14	-0.576 ± 0.158	typical higher	0.12	0.07
GPT-2	Client GA, upload†	Loss thr.	primary	legacy_pooled	0.05	0.00	0.052 ± 0.127	inconclusive	0.07	0.10
GPT-2	Client GA, upload†	Loss thr.	holdout10	legacy	−0.43	0.24	-0.669 ± 0.149	typical higher	0.13	0.04
GPT-2	Client GA, upload†	Loss thr.	holdout10	legacy_pooled	0.06	0.15	-0.088 ± 0.077	typical higher	0.07	0.05
GPT-2	Client GA, upload†	Min-K%	primary	legacy	−0.36	0.09	-0.448 ± 0.191	typical higher	0.15	0.10
GPT-2	Client GA, upload†	Min-K%	primary	legacy_pooled	0.00	−0.03	0.033 ± 0.055	inconclusive	0.10	0.11
GPT-2	Client GA, upload†	Min-K%	holdout10	legacy	−0.30	0.21	-0.503 ± 0.191	typical higher	0.16	0.03
GPT-2	Client GA, upload†	Min-K%	holdout10	legacy_pooled	0.11	0.10	0.014 ± 0.092	inconclusive	0.06	0.08
GPT-2	Client GA, upload†	Ref.-calib.	primary	legacy	−0.22	0.06	-0.281 ± 0.144	typical higher	0.18	0.05
GPT-2	Client GA, upload†	Ref.-calib.	primary	legacy_pooled	0.03	−0.04	0.076 ± 0.094	inconclusive	0.06	0.08
GPT-2	Client GA, upload†	Ref.-calib.	holdout10	legacy	−0.32	−0.02	-0.303 ± 0.151	typical higher	0.15	0.06
GPT-2	Client GA, upload†	Ref.-calib.	holdout10	legacy_pooled	−0.06	−0.11	0.046 ± 0.062	inconclusive	0.06	0.04
GPT-2	Client GA, global	Loss thr.	primary	legacy	−0.00	0.01	-0.012 ± 0.005	typical higher	0.31	0.20
GPT-2	Client GA, global	Loss thr.	primary	legacy_pooled	0.00	0.01	-0.007 ± 0.019	inconclusive	0.12	0.08
GPT-2	Client GA, global	Loss thr.	holdout10	legacy	−0.00	0.01	-0.017 ± 0.008	typical higher	0.35	0.24
GPT-2	Client GA, global	Loss thr.	holdout10	legacy_pooled	0.01	0.02	-0.013 ± 0.018	inconclusive	0.06	0.17
GPT-2	Client GA, global	Min-K%	primary	legacy	0.00	0.01	-0.010 ± 0.013	inconclusive	0.23	0.17
GPT-2	Client GA, global	Min-K%	primary	legacy_pooled	0.01	0.02	-0.009 ± 0.012	inconclusive	0.10	0.06
GPT-2	Client GA, global	Min-K%	holdout10	legacy	0.01	0.02	-0.007 ± 0.019	inconclusive	0.24	0.21
GPT-2	Client GA, global	Min-K%	holdout10	legacy_pooled	0.02	0.02	-0.003 ± 0.008	inconclusive	0.10	0.13
GPT-2	Client GA, global	Ref.-calib.	primary	legacy	0.02	0.01	0.014 ± 0.037	inconclusive	0.06	0.08
GPT-2	Client GA, global	Ref.-calib.	primary	legacy_pooled	0.00	0.01	-0.010 ± 0.017	inconclusive	0.09	0.03
GPT-2	Client GA, global	Ref.-calib.	holdout10	legacy	0.01	−0.00	0.014 ± 0.039	inconclusive	0.22	0.07
GPT-2	Client GA, global	Ref.-calib.	holdout10	legacy_pooled	−0.01	0.00	-0.009 ± 0.013	inconclusive	0.07	0.13
GPT-2-medium	Client GA, upload†	Loss thr.	primary	legacy	−0.36	0.19	-0.551 ± 0.250	typical higher	0.15	0.05
GPT-2-medium	Client GA, upload†	Loss thr.	primary	legacy_pooled	0.13	0.08	0.055 ± 0.085	inconclusive	0.04	0.04
GPT-2-medium	Client GA, upload†	Loss thr.	holdout10	legacy	−0.37	0.28	-0.649 ± 0.268	typical higher	0.18	0.02
GPT-2-medium	Client GA, upload†	Loss thr.	holdout10	legacy_pooled	0.11	0.21	-0.101 ± 0.053	typical higher	0.06	0.04
GPT-2-medium	Client GA, upload†	Min-K%	primary	legacy	−0.27	0.14	-0.412 ± 0.249	typical higher	0.14	0.08
GPT-2-medium	Client GA, upload†	Min-K%	primary	legacy_pooled	0.07	0.05	0.020 ± 0.076	inconclusive	0.09	0.06

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Table S17 (continued)

Model	Method	Attack	Non-members	Definition	Gap out.	Gap typ.	Difference	Verdict	Adv. out.	Adv. typ.
GPT-2-medium	Client GA, upload†	Min-K%	holdout10	legacy	−0.21	0.27	-0.478 ± 0.265	typical higher	0.18	0.01
GPT-2-medium	Client GA, upload†	Min-K%	holdout10	legacy_pooled	0.11	0.21	-0.099 ± 0.053	typical higher	0.05	0.04
GPT-2-medium	Client GA, upload†	Ref.-calib.	primary	legacy	−0.14	0.09	-0.222 ± 0.260	inconclusive	0.16	0.04
GPT-2-medium	Client GA, upload†	Ref.-calib.	primary	legacy_pooled	0.06	0.01	0.049 ± 0.052	inconclusive	0.03	0.03
GPT-2-medium	Client GA, upload†	Ref.-calib.	holdout10	legacy	−0.25	−0.01	-0.247 ± 0.286	inconclusive	0.19	0.06
GPT-2-medium	Client GA, upload†	Ref.-calib.	holdout10	legacy_pooled	−0.10	−0.05	-0.047 ± 0.047	typical higher	0.06	0.04
GPT-2-medium	Client GA, global	Loss thr.	primary	legacy	0.03	0.05	-0.025 ± 0.009	typical higher	0.33	0.17
GPT-2-medium	Client GA, global	Loss thr.	primary	legacy_pooled	0.07	0.07	-0.006 ± 0.011	inconclusive	0.10	0.05
GPT-2-medium	Client GA, global	Loss thr.	holdout10	legacy	0.03	0.06	-0.029 ± 0.011	typical higher	0.36	0.22
GPT-2-medium	Client GA, global	Loss thr.	holdout10	legacy_pooled	0.07	0.08	-0.004 ± 0.012	inconclusive	0.03	0.14
GPT-2-medium	Client GA, global	Min-K%	primary	legacy	0.04	0.05	-0.011 ± 0.009	typical higher	0.25	0.16
GPT-2-medium	Client GA, global	Min-K%	primary	legacy_pooled	0.06	0.07	-0.009 ± 0.016	inconclusive	0.10	0.05
GPT-2-medium	Client GA, global	Min-K%	holdout10	legacy	0.04	0.06	-0.011 ± 0.013	inconclusive	0.27	0.21
GPT-2-medium	Client GA, global	Min-K%	holdout10	legacy_pooled	0.06	0.07	-0.009 ± 0.016	inconclusive	0.06	0.13
GPT-2-medium	Client GA, global	Ref.-calib.	primary	legacy	0.06	0.11	-0.045 ± 0.016	typical higher	0.10	0.03
GPT-2-medium	Client GA, global	Ref.-calib.	primary	legacy_pooled	0.06	0.11	-0.047 ± 0.015	typical higher	0.03	0.07
GPT-2-medium	Client GA, global	Ref.-calib.	holdout10	legacy	0.05	0.11	-0.055 ± 0.016	typical higher	0.25	0.18
GPT-2-medium	Client GA, global	Ref.-calib.	holdout10	legacy_pooled	0.06	0.11	-0.048 ± 0.015	typical higher	0.19	0.20
Pythia-1B (LoRA)	GA†	Loss thr.	primary	legacy	−0.31	0.12	-0.430 ± 0.242	typical higher	0.17	0.12
Pythia-1B (LoRA)	GA†	Loss thr.	primary	legacy_pooled	0.13	0.08	0.047 ± 0.033	outliers higher	0.10	0.09
Pythia-1B (LoRA)	GA†	Loss thr.	holdout10	legacy	−0.28	0.23	-0.509 ± 0.240	typical higher	0.24	0.11
Pythia-1B (LoRA)	GA†	Loss thr.	holdout10	legacy_pooled	0.16	0.22	-0.065 ± 0.091	inconclusive	0.17	0.13
Pythia-1B (LoRA)	GA†	Min-K%	primary	legacy	−0.27	0.15	-0.418 ± 0.073	typical higher	0.11	0.08
Pythia-1B (LoRA)	GA†	Min-K%	primary	legacy_pooled	0.13	0.08	0.054 ± 0.077	inconclusive	0.09	0.08
Pythia-1B (LoRA)	GA†	Min-K%	holdout10	legacy	−0.27	0.17	-0.444 ± 0.079	typical higher	0.14	0.09
Pythia-1B (LoRA)	GA†	Min-K%	holdout10	legacy_pooled	0.12	0.11	0.010 ± 0.125	inconclusive	0.16	0.09
Pythia-1B (LoRA)	GA†	Ref.-calib.	primary	legacy	−0.10	0.02	-0.116 ± 0.249	inconclusive	0.17	0.11
Pythia-1B (LoRA)	GA†	Ref.-calib.	primary	legacy_pooled	0.01	−0.01	0.018 ± 0.035	inconclusive	0.09	0.09
Pythia-1B (LoRA)	GA†	Ref.-calib.	holdout10	legacy	−0.19	−0.04	-0.155 ± 0.238	inconclusive	0.24	0.17
Pythia-1B (LoRA)	GA†	Ref.-calib.	holdout10	legacy_pooled	−0.13	−0.04	-0.088 ± 0.080	typical higher	0.16	0.18
Pythia-1B (LoRA)	GD†	Loss thr.	primary	legacy	−0.70	−0.25	-0.458 ± 0.034	typical higher	0.40	0.46
Pythia-1B (LoRA)	GD†	Loss thr.	primary	legacy_pooled	−0.20	−0.31	0.112 ± 0.054	outliers higher	0.42	0.46
Pythia-1B (LoRA)	GD†	Loss thr.	holdout10	legacy	−0.68	−0.17	-0.510 ± 0.033	typical higher	0.36	0.43
Pythia-1B (LoRA)	GD†	Loss thr.	holdout10	legacy_pooled	−0.18	−0.21	0.028 ± 0.024	outliers higher	0.37	0.43
Pythia-1B (LoRA)	GD†	Min-K%	primary	legacy	−0.62	−0.26	-0.367 ± 0.039	typical higher	0.41	0.46
Pythia-1B (LoRA)	GD†	Min-K%	primary	legacy_pooled	−0.24	−0.32	0.078 ± 0.061	outliers higher	0.42	0.46
Pythia-1B (LoRA)	GD†	Min-K%	holdout10	legacy	−0.57	−0.18	-0.392 ± 0.038	typical higher	0.37	0.43
Pythia-1B (LoRA)	GD†	Min-K%	holdout10	legacy_pooled	−0.19	−0.23	0.040 ± 0.037	outliers higher	0.38	0.43
Pythia-1B (LoRA)	GD†	Ref.-calib.	primary	legacy	−0.45	−0.39	-0.060 ± 0.034	typical higher	0.41	0.46
Pythia-1B (LoRA)	GD†	Ref.-calib.	primary	legacy_pooled	−0.33	−0.42	0.090 ± 0.024	outliers higher	0.42	0.46
Pythia-1B (LoRA)	GD†	Ref.-calib.	holdout10	legacy	−0.57	−0.51	-0.064 ± 0.042	typical higher	0.37	0.42
Pythia-1B (LoRA)	GD†	Ref.-calib.	holdout10	legacy_pooled	−0.50	−0.53	0.031 ± 0.030	outliers higher	0.37	0.43

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Table S17 (continued)

Model	Method	Attack	Non-members	Definition	Gap out.	Gap typ.	Difference	Verdict	Adv. out.	Adv. typ.
Pythia-1B (LoRA)	NPO	Loss thr.	primary	legacy	−0.75	−0.25	-0.493 ± 0.017	typical higher	0.45	0.47
Pythia-1B (LoRA)	NPO	Loss thr.	primary	legacy_pooled	−0.24	−0.32	0.080 ± 0.042	outliers higher	0.46	0.47
Pythia-1B (LoRA)	NPO	Loss thr.	holdout10	legacy	−0.73	−0.18	-0.550 ± 0.031	typical higher	0.41	0.44
Pythia-1B (LoRA)	NPO	Loss thr.	holdout10	legacy_pooled	−0.23	−0.22	-0.006 ± 0.053	inconclusive	0.42	0.44
Pythia-1B (LoRA)	NPO	Min-K%	primary	legacy	−0.67	−0.26	-0.406 ± 0.024	typical higher	0.46	0.47
Pythia-1B (LoRA)	NPO	Min-K%	primary	legacy_pooled	−0.29	−0.33	0.044 ± 0.043	outliers higher	0.46	0.47
Pythia-1B (LoRA)	NPO	Min-K%	holdout10	legacy	−0.62	−0.19	-0.433 ± 0.036	typical higher	0.43	0.44
Pythia-1B (LoRA)	NPO	Min-K%	holdout10	legacy_pooled	−0.24	−0.24	0.008 ± 0.064	inconclusive	0.42	0.45
Pythia-1B (LoRA)	NPO	Ref.-calib.	primary	legacy	−0.50	−0.39	-0.109 ± 0.034	typical higher	0.46	0.46
Pythia-1B (LoRA)	NPO	Ref.-calib.	primary	legacy_pooled	−0.37	−0.43	0.059 ± 0.023	outliers higher	0.46	0.47
Pythia-1B (LoRA)	NPO	Ref.-calib.	holdout10	legacy	−0.63	−0.51	-0.120 ± 0.063	typical higher	0.43	0.43
Pythia-1B (LoRA)	NPO	Ref.-calib.	holdout10	legacy_pooled	−0.54	−0.54	-0.007 ± 0.073	inconclusive	0.41	0.43

Mean ± half-width of the 95% t-interval over seeds (up to 5); a single value means one seed. Verdict: outliers higher (CI lower bound > 0), typical higher (CI upper bound < 0) or inconclusive. †Model collapsed: retain loss rose more than threefold in every seed. GA: all four central models; GD: Pythia-1B (LoRA); client-GA upload: all six federated runs (intermediate model before recovery; the released global model is not collapsed).

3 Low false-positive rates, version-change attacks and control

Table S18: TPR at 0.1% FPR with scores pooled over seeds, its per-seed mean, and TPR at 1% FPR within the base-difficulty strata (outlier forget vs outlier non-members), pooled the same way.

Model	Run	Method	Attack	Non-members	Seeds	Negatives	TPR@0.1% pooled	TPR at zero false positives (400 non-members)	TPR@1% outlier	TPR@1% typical
DistilGPT-2	central	Retrain	Loss thr.	primary	5	2000	0.000	0.001 ± 0.003	0.000	0.013
DistilGPT-2	central	Retrain	Loss thr.	holdout10	5	2000	0.000	0.000 ± 0.000	0.017	0.029
DistilGPT-2	central	Retrain	Min-K%	primary	5	2000	0.000	0.001 ± 0.001	0.000	0.008
DistilGPT-2	central	Retrain	Min-K%	holdout10	5	2000	0.000	0.000 ± 0.000	0.006	0.023
DistilGPT-2	central	GA†	Loss thr.	primary	5	2000	0.000	0.003 ± 0.003	0.006	0.004
DistilGPT-2	central	GA†	Loss thr.	holdout10	5	2000	0.002	0.025 ± 0.022	0.006	0.009
DistilGPT-2	central	GA†	Min-K%	primary	5	2000	0.000	0.002 ± 0.001	0.009	0.004
DistilGPT-2	central	GA†	Min-K%	holdout10	5	2000	0.003	0.001 ± 0.003	0.005	0.014
DistilGPT-2	central	GD	Loss thr.	primary	5	2000	0.001	0.000 ± 0.000	0.000	0.004
DistilGPT-2	central	GD	Loss thr.	holdout10	5	2000	0.002	0.002 ± 0.003	0.016	0.009
DistilGPT-2	central	GD	Min-K%	primary	5	2000	0.001	0.001 ± 0.001	0.000	0.004
DistilGPT-2	central	GD	Min-K%	holdout10	5	2000	0.001	0.002 ± 0.002	0.007	0.006
DistilGPT-2	central	NPO	Loss thr.	primary	5	2000	0.000	0.001 ± 0.001	0.000	0.003
DistilGPT-2	central	NPO	Loss thr.	holdout10	5	2000	0.000	0.000 ± 0.000	0.005	0.003
DistilGPT-2	central	NPO	Min-K%	primary	5	2000	0.000	0.000 ± 0.000	0.000	0.000

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Table S18 (continued)

Model	Run	Method	Attack	Non-members	Seeds	Negatives	TPR@0.1% pooled	TPR at zero false positives (400 non-members)	TPR@1% outlier	TPR@1% typical
DistilGPT-2	central	NPO	Min-K%	holdout10	5	2000	0.000	0.000 ± 0.000	0.006	0.000
GPT-2	central	Retrain	Loss thr.	primary	5	2000	0.004	0.004 ± 0.004	0.000	0.020
GPT-2	central	Retrain	Loss thr.	holdout10	5	2000	0.000	0.000 ± 0.000	0.028	0.033
GPT-2	central	Retrain	Min-K%	primary	5	2000	0.002	0.003 ± 0.003	0.018	0.010
GPT-2	central	Retrain	Min-K%	holdout10	5	2000	0.000	0.000 ± 0.000	0.001	0.032
GPT-2	central	GA†	Loss thr.	primary	5	2000	0.001	0.003 ± 0.003	0.003	0.004
GPT-2	central	GA†	Loss thr.	holdout10	5	2000	0.002	0.015 ± 0.021	0.028	0.018
GPT-2	central	GA†	Min-K%	primary	5	2000	0.001	0.001 ± 0.002	0.003	0.006
GPT-2	central	GA†	Min-K%	holdout10	5	2000	0.004	0.007 ± 0.013	0.016	0.028
GPT-2	central	GD	Loss thr.	primary	5	2000	0.002	0.002 ± 0.004	0.000	0.007
GPT-2	central	GD	Loss thr.	holdout10	5	2000	0.001	0.002 ± 0.003	0.014	0.013
GPT-2	central	GD	Min-K%	primary	5	2000	0.002	0.001 ± 0.003	0.003	0.006
GPT-2	central	GD	Min-K%	holdout10	5	2000	0.000	0.001 ± 0.003	0.015	0.009
GPT-2	central	NPO	Loss thr.	primary	5	2000	0.002	0.001 ± 0.001	0.000	0.009
GPT-2	central	NPO	Loss thr.	holdout10	5	2000	0.000	0.001 ± 0.003	0.004	0.009
GPT-2	central	NPO	Min-K%	primary	5	2000	0.001	0.001 ± 0.001	0.002	0.004
GPT-2	central	NPO	Min-K%	holdout10	5	2000	0.000	0.000 ± 0.000	0.011	0.004
GPT-2-medium	central	Retrain	Loss thr.	primary	5	2000	0.004	0.003 ± 0.003	0.000	0.017
GPT-2-medium	central	Retrain	Loss thr.	holdout10	5	2000	0.004	0.004 ± 0.003	0.018	0.029
GPT-2-medium	central	Retrain	Min-K%	primary	5	2000	0.004	0.003 ± 0.003	0.000	0.014
GPT-2-medium	central	Retrain	Min-K%	holdout10	5	2000	0.004	0.004 ± 0.003	0.000	0.030
GPT-2-medium	central	GA†	Loss thr.	primary	5	2000	0.001	0.001 ± 0.001	0.015	0.010
GPT-2-medium	central	GA†	Loss thr.	holdout10	5	2000	0.003	0.009 ± 0.012	0.040	0.032
GPT-2-medium	central	GA†	Min-K%	primary	5	2000	0.000	0.001 ± 0.002	0.012	0.008
GPT-2-medium	central	GA†	Min-K%	holdout10	5	2000	0.000	0.002 ± 0.004	0.030	0.012
GPT-2-medium	central	GD	Loss thr.	primary	5	2000	0.000	0.000 ± 0.000	0.000	0.000
GPT-2-medium	central	GD	Loss thr.	holdout10	5	2000	0.000	0.000 ± 0.000	0.000	0.000
GPT-2-medium	central	GD	Min-K%	primary	5	2000	0.000	0.001 ± 0.001	0.000	0.000
GPT-2-medium	central	GD	Min-K%	holdout10	5	2000	0.000	0.000 ± 0.000	0.000	0.000
GPT-2-medium	central	NPO	Loss thr.	primary	5	2000	0.000	0.000 ± 0.000	0.000	0.000
GPT-2-medium	central	NPO	Loss thr.	holdout10	5	2000	0.000	0.000 ± 0.000	0.000	0.000
GPT-2-medium	central	NPO	Min-K%	primary	5	2000	0.000	0.000 ± 0.000	0.000	0.000
GPT-2-medium	central	NPO	Min-K%	holdout10	5	2000	0.000	0.000 ± 0.000	0.000	0.000
GPT-2	by_author_K5	Retrain	Loss thr.	primary	3	1200	0.003	0.003 ± 0.004	0.000	0.011
GPT-2	by_author_K5	Retrain	Loss thr.	holdout10	3	1200	0.000	0.000 ± 0.000	0.031	0.030
GPT-2	by_author_K5	Retrain	Min-K%	primary	3	1200	0.003	0.003 ± 0.000	0.008	0.007
GPT-2	by_author_K5	Retrain	Min-K%	holdout10	3	1200	0.000	0.000 ± 0.000	0.000	0.026
GPT-2	by_author_K5	Client GA, upload†	Loss thr.	primary	3	1200	0.000	0.001 ± 0.004	0.000	0.002
GPT-2	by_author_K5	Client GA, upload†	Loss thr.	holdout10	3	1200	0.003	0.014 ± 0.045	0.002	0.014
GPT-2	by_author_K5	Client GA, upload†	Min-K%	primary	3	1200	0.000	0.000 ± 0.000	0.003	0.002

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Table S18 (continued)

Model	Run	Method	Attack	Non-members	Seeds	Negatives	TPR@0.1% pooled	TPR at zero false positives (400 non-members)	TPR@1% outlier	TPR@1% typical
GPT-2	by_author_K5	Client GA, upload†	Min-K%	holdout10	3	1200	0.003	0.003 ± 0.011	0.010	0.013
GPT-2	by_author_K5	Client GA, global	Loss thr.	primary	3	1200	0.002	0.002 ± 0.004	0.000	0.015
GPT-2	by_author_K5	Client GA, global	Loss thr.	holdout10	3	1200	0.000	0.000 ± 0.000	0.035	0.032
GPT-2	by_author_K5	Client GA, global	Min-K%	primary	3	1200	0.001	0.001 ± 0.004	0.015	0.011
GPT-2	by_author_K5	Client GA, global	Min-K%	holdout10	3	1200	0.000	0.000 ± 0.000	0.008	0.025
GPT-2	iid_K3	Retrain	Loss thr.	primary	3	1200	0.003	0.003 ± 0.004	0.000	0.021
GPT-2	iid_K3	Retrain	Loss thr.	holdout10	3	1200	0.000	0.000 ± 0.000	0.033	0.033
GPT-2	iid_K3	Retrain	Min-K%	primary	3	1200	0.003	0.003 ± 0.004	0.008	0.011
GPT-2	iid_K3	Retrain	Min-K%	holdout10	3	1200	0.000	0.000 ± 0.000	0.002	0.032
GPT-2	iid_K3	Client GA, upload†	Loss thr.	primary	3	1200	0.002	0.001 ± 0.004	0.000	0.005
GPT-2	iid_K3	Client GA, upload†	Loss thr.	holdout10	3	1200	0.001	0.005 ± 0.006	0.043	0.006
GPT-2	iid_K3	Client GA, upload†	Min-K%	primary	3	1200	0.000	0.000 ± 0.000	0.003	0.005
GPT-2	iid_K3	Client GA, upload†	Min-K%	holdout10	3	1200	0.002	0.002 ± 0.004	0.022	0.028
GPT-2	iid_K3	Client GA, global	Loss thr.	primary	3	1200	0.002	0.002 ± 0.004	0.000	0.012
GPT-2	iid_K3	Client GA, global	Loss thr.	holdout10	3	1200	0.000	0.001 ± 0.004	0.033	0.029
GPT-2	iid_K3	Client GA, global	Min-K%	primary	3	1200	0.003	0.003 ± 0.006	0.008	0.014
GPT-2	iid_K3	Client GA, global	Min-K%	holdout10	3	1200	0.000	0.001 ± 0.004	0.008	0.026
GPT-2	iid_K10	Retrain	Loss thr.	primary	3	1200	0.005	0.003 ± 0.007	0.000	0.014
GPT-2	iid_K10	Retrain	Loss thr.	holdout10	3	1200	0.000	0.000 ± 0.000	0.024	0.026
GPT-2	iid_K10	Retrain	Min-K%	primary	3	1200	0.005	0.003 ± 0.007	0.003	0.011
GPT-2	iid_K10	Retrain	Min-K%	holdout10	3	1200	0.000	0.000 ± 0.000	0.000	0.023
GPT-2	iid_K10	Client GA, upload†	Loss thr.	primary	3	1200	0.000	0.000 ± 0.000	0.000	0.001
GPT-2	iid_K10	Client GA, upload†	Loss thr.	holdout10	3	1200	0.000	0.001 ± 0.004	0.008	0.009
GPT-2	iid_K10	Client GA, upload†	Min-K%	primary	3	1200	0.000	0.000 ± 0.000	0.000	0.000
GPT-2	iid_K10	Client GA, upload†	Min-K%	holdout10	3	1200	0.000	0.000 ± 0.000	0.008	0.007
GPT-2	iid_K10	Client GA, global	Loss thr.	primary	3	1200	0.005	0.004 ± 0.004	0.000	0.021
GPT-2	iid_K10	Client GA, global	Loss thr.	holdout10	3	1200	0.000	0.000 ± 0.000	0.043	0.030
GPT-2	iid_K10	Client GA, global	Min-K%	primary	3	1200	0.005	0.003 ± 0.007	0.003	0.016
GPT-2	iid_K10	Client GA, global	Min-K%	holdout10	3	1200	0.000	0.000 ± 0.000	0.006	0.033
DistilGPT-2	iid_K5	Retrain	Loss thr.	primary	5	2000	0.000	0.001 ± 0.003	0.000	0.009
DistilGPT-2	iid_K5	Retrain	Loss thr.	holdout10	5	2000	0.000	0.000 ± 0.000	0.012	0.021
DistilGPT-2	iid_K5	Retrain	Min-K%	primary	5	2000	0.000	0.001 ± 0.001	0.000	0.004
DistilGPT-2	iid_K5	Retrain	Min-K%	holdout10	5	2000	0.000	0.000 ± 0.000	0.006	0.015
DistilGPT-2	iid_K5	Client GA, upload†	Loss thr.	primary	5	2000	0.002	0.004 ± 0.006	0.006	0.005
DistilGPT-2	iid_K5	Client GA, upload†	Loss thr.	holdout10	5	2000	0.002	0.007 ± 0.005	0.004	0.012
DistilGPT-2	iid_K5	Client GA, upload†	Min-K%	primary	5	2000	0.002	0.001 ± 0.001	0.003	0.010
DistilGPT-2	iid_K5	Client GA, upload†	Min-K%	holdout10	5	2000	0.004	0.004 ± 0.005	0.007	0.013
DistilGPT-2	iid_K5	Client GA, global	Loss thr.	primary	5	2000	0.000	0.003 ± 0.003	0.002	0.007
DistilGPT-2	iid_K5	Client GA, global	Loss thr.	holdout10	5	2000	0.001	0.001 ± 0.002	0.012	0.016
DistilGPT-2	iid_K5	Client GA, global	Min-K%	primary	5	2000	0.000	0.001 ± 0.003	0.005	0.005

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Table S18 (continued)

Model	Run	Method	Attack	Non-members	Seeds	Negatives	TPR@0.1% pooled	TPR at zero false positives (400 non-members)	TPR@1% outlier	TPR@1% typical
DistilGPT-2	iid_K5	Client GA, global	Min-K%	holdout10	5	2000	0.000	0.000 ± 0.000	0.008	0.009
GPT-2	iid_K5	Retrain	Loss thr.	primary	5	2000	0.002	0.003 ± 0.003	0.000	0.012
GPT-2	iid_K5	Retrain	Loss thr.	holdout10	5	2000	0.000	0.000 ± 0.000	0.029	0.030
GPT-2	iid_K5	Retrain	Min-K%	primary	5	2000	0.001	0.002 ± 0.003	0.014	0.010
GPT-2	iid_K5	Retrain	Min-K%	holdout10	5	2000	0.000	0.000 ± 0.000	0.000	0.026
GPT-2	iid_K5	Client GA, upload†	Loss thr.	primary	5	2000	0.001	0.002 ± 0.002	0.003	0.006
GPT-2	iid_K5	Client GA, upload†	Loss thr.	holdout10	5	2000	0.002	0.004 ± 0.004	0.028	0.014
GPT-2	iid_K5	Client GA, upload†	Min-K%	primary	5	2000	0.001	0.000 ± 0.000	0.003	0.005
GPT-2	iid_K5	Client GA, upload†	Min-K%	holdout10	5	2000	0.005	0.003 ± 0.004	0.018	0.026
GPT-2	iid_K5	Client GA, global	Loss thr.	primary	5	2000	0.002	0.001 ± 0.002	0.000	0.011
GPT-2	iid_K5	Client GA, global	Loss thr.	holdout10	5	2000	0.000	0.000 ± 0.000	0.027	0.028
GPT-2	iid_K5	Client GA, global	Min-K%	primary	5	2000	0.002	0.001 ± 0.002	0.018	0.010
GPT-2	iid_K5	Client GA, global	Min-K%	holdout10	5	2000	0.000	0.000 ± 0.000	0.004	0.019
GPT-2-medium	iid_K5	Retrain	Loss thr.	primary	5	2000	0.003	0.004 ± 0.003	0.000	0.015
GPT-2-medium	iid_K5	Retrain	Loss thr.	holdout10	5	2000	0.002	0.002 ± 0.001	0.024	0.030
GPT-2-medium	iid_K5	Retrain	Min-K%	primary	5	2000	0.003	0.003 ± 0.003	0.006	0.015
GPT-2-medium	iid_K5	Retrain	Min-K%	holdout10	5	2000	0.002	0.002 ± 0.001	0.000	0.027
GPT-2-medium	iid_K5	Client GA, upload†	Loss thr.	primary	5	2000	0.002	0.001 ± 0.002	0.021	0.010
GPT-2-medium	iid_K5	Client GA, upload†	Loss thr.	holdout10	5	2000	0.001	0.004 ± 0.006	0.037	0.027
GPT-2-medium	iid_K5	Client GA, upload†	Min-K%	primary	5	2000	0.000	0.000 ± 0.000	0.009	0.002
GPT-2-medium	iid_K5	Client GA, upload†	Min-K%	holdout10	5	2000	0.001	0.003 ± 0.004	0.037	0.017
GPT-2-medium	iid_K5	Client GA, global	Loss thr.	primary	5	2000	0.003	0.003 ± 0.001	0.000	0.015
GPT-2-medium	iid_K5	Client GA, global	Loss thr.	holdout10	5	2000	0.003	0.003 ± 0.004	0.030	0.033
GPT-2-medium	iid_K5	Client GA, global	Min-K%	primary	5	2000	0.003	0.003 ± 0.001	0.004	0.014
GPT-2-medium	iid_K5	Client GA, global	Min-K%	holdout10	5	2000	0.003	0.003 ± 0.004	0.001	0.026
Pythia-1B (LoRA)	central	Retrain	Loss thr.	primary	5	2000	0.000	0.003 ± 0.004	0.000	0.019
Pythia-1B (LoRA)	central	Retrain	Loss thr.	holdout10	5	2000	0.000	0.000 ± 0.000	0.036	0.034
Pythia-1B (LoRA)	central	Retrain	Min-K%	primary	5	2000	0.000	0.002 ± 0.006	0.000	0.013
Pythia-1B (LoRA)	central	Retrain	Min-K%	holdout10	5	2000	0.000	0.000 ± 0.000	0.012	0.032
Pythia-1B (LoRA)	central	GA†	Loss thr.	primary	5	2000	0.000	0.001 ± 0.002	0.003	0.001
Pythia-1B (LoRA)	central	GA†	Loss thr.	holdout10	5	2000	0.000	0.009 ± 0.012	0.004	0.006
Pythia-1B (LoRA)	central	GA†	Min-K%	primary	5	2000	0.000	0.001 ± 0.001	0.005	0.001
Pythia-1B (LoRA)	central	GA†	Min-K%	holdout10	5	2000	0.000	0.010 ± 0.013	0.004	0.004
Pythia-1B (LoRA)	central	GD†	Loss thr.	primary	5	2000	0.000	0.000 ± 0.000	0.000	0.000
Pythia-1B (LoRA)	central	GD†	Loss thr.	holdout10	5	2000	0.000	0.000 ± 0.000	0.000	0.000
Pythia-1B (LoRA)	central	GD†	Min-K%	primary	5	2000	0.000	0.000 ± 0.000	0.000	0.000
Pythia-1B (LoRA)	central	GD†	Min-K%	holdout10	5	2000	0.000	0.001 ± 0.001	0.000	0.000
Pythia-1B (LoRA)	central	NPO	Loss thr.	primary	5	2000	0.000	0.000 ± 0.000	0.000	0.000
Pythia-1B (LoRA)	central	NPO	Loss thr.	holdout10	5	2000	0.000	0.000 ± 0.000	0.000	0.000
Pythia-1B (LoRA)	central	NPO	Min-K%	primary	5	2000	0.000	0.000 ± 0.000	0.000	0.000

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Table S18 (continued)

Model	Run	Method	Attack	Non-members	Seeds	Negatives	TPR@0.1% pooled	TPR at zero false positives (400 non-members)	TPR@1% outlier	TPR@1% typical
Pythia-1B (LoRA)	central	NPO	Min-K%	holdout10	5	2000	0.000	0.000 ± 0.000	0.000	0.000

Pooling puts runs with different models on one threshold; with 2,000 negatives, 0.1% FPR corresponds to 2 false positives. TPR at zero false positives (400 non-members): the per-seed TPR@0.1%, which allows no false positive at that size, as mean and t-interval over seeds (no pooling). †Model collapsed: retain loss rose more than threefold in every seed. GA: all four central models; GD: Pythia-1B (LoRA); client-GA upload: all six federated runs (intermediate model before recovery; the released global model is not collapsed).

Table S19: H4/H5: update-signal attacks. Score change (black-box, pre − post) and gradient alignment with $\Delta\theta$ (white-box), and their gaps to retrain (same seed).

						Score change		Grad. alignment	
Model	Run	Method	View	Attack	NM	AUC	Gap	AUC	Gap
DistilGPT-2	central	Retrain	release	Loss thr.	primary	0.99 ± 0.01	0.00 ± 0.00	0.98 ± 0.01	0.00 ± 0.00
DistilGPT-2	central	Retrain	release	Loss thr.	holdout10	0.99 ± 0.01	0.00 ± 0.00	0.98 ± 0.01	0.00 ± 0.00
DistilGPT-2	central	Retrain	release	Min-K%	primary	0.98 ± 0.01	0.00 ± 0.00	0.98 ± 0.01	0.00 ± 0.00
DistilGPT-2	central	Retrain	release	Min-K%	holdout10	0.97 ± 0.01	0.00 ± 0.00	0.98 ± 0.01	0.00 ± 0.00
DistilGPT-2	central	GA†	release	Loss thr.	primary	0.55 ± 0.03	-0.44 ± 0.03	0.55 ± 0.05	-0.43 ± 0.05
DistilGPT-2	central	GA†	release	Loss thr.	holdout10	0.47 ± 0.11	-0.52 ± 0.12	0.46 ± 0.08	-0.51 ± 0.09
DistilGPT-2	central	GA†	release	Min-K%	primary	0.60 ± 0.03	-0.38 ± 0.03	0.55 ± 0.05	-0.43 ± 0.05
DistilGPT-2	central	GA†	release	Min-K%	holdout10	0.54 ± 0.06	-0.44 ± 0.06	0.46 ± 0.08	-0.51 ± 0.09
DistilGPT-2	central	GD	release	Loss thr.	primary	0.78 ± 0.05	-0.21 ± 0.05	0.75 ± 0.04	-0.23 ± 0.05
DistilGPT-2	central	GD	release	Loss thr.	holdout10	0.62 ± 0.02	-0.37 ± 0.02	0.62 ± 0.03	-0.36 ± 0.04
DistilGPT-2	central	GD	release	Min-K%	primary	0.77 ± 0.04	-0.20 ± 0.05	0.75 ± 0.04	-0.23 ± 0.05
DistilGPT-2	central	GD	release	Min-K%	holdout10	0.60 ± 0.02	-0.37 ± 0.03	0.62 ± 0.03	-0.36 ± 0.04
DistilGPT-2	central	NPO	release	Loss thr.	primary	0.80 ± 0.03	-0.19 ± 0.04	0.75 ± 0.04	-0.23 ± 0.05
DistilGPT-2	central	NPO	release	Loss thr.	holdout10	0.72 ± 0.06	-0.27 ± 0.06	0.66 ± 0.04	-0.31 ± 0.05
DistilGPT-2	central	NPO	release	Min-K%	primary	0.78 ± 0.04	-0.20 ± 0.04	0.75 ± 0.04	-0.23 ± 0.05
DistilGPT-2	central	NPO	release	Min-K%	holdout10	0.66 ± 0.04	-0.31 ± 0.04	0.66 ± 0.04	-0.31 ± 0.05
GPT-2	central	Retrain	release	Loss thr.	primary	0.99 ± 0.00	0.00 ± 0.00	0.98 ± 0.01	0.00 ± 0.00
GPT-2	central	Retrain	release	Loss thr.	holdout10	0.99 ± 0.00	0.00 ± 0.00	0.98 ± 0.01	0.00 ± 0.00
GPT-2	central	Retrain	release	Min-K%	primary	0.98 ± 0.00	0.00 ± 0.00	0.98 ± 0.01	0.00 ± 0.00
GPT-2	central	Retrain	release	Min-K%	holdout10	0.98 ± 0.00	0.00 ± 0.00	0.98 ± 0.01	0.00 ± 0.00
GPT-2	central	GA†	release	Loss thr.	primary	0.57 ± 0.04	-0.42 ± 0.04	0.59 ± 0.04	-0.39 ± 0.04
GPT-2	central	GA†	release	Loss thr.	holdout10	0.48 ± 0.10	-0.51 ± 0.10	0.51 ± 0.07	-0.47 ± 0.07
GPT-2	central	GA†	release	Min-K%	primary	0.62 ± 0.04	-0.36 ± 0.04	0.59 ± 0.04	-0.39 ± 0.04
GPT-2	central	GA†	release	Min-K%	holdout10	0.53 ± 0.03	-0.45 ± 0.04	0.51 ± 0.07	-0.47 ± 0.07
GPT-2	central	GD	release	Loss thr.	primary	0.81 ± 0.05	-0.18 ± 0.05	0.81 ± 0.04	-0.17 ± 0.05
GPT-2	central	GD	release	Loss thr.	holdout10	0.66 ± 0.06	-0.33 ± 0.06	0.69 ± 0.05	-0.29 ± 0.06
GPT-2	central	GD	release	Min-K%	primary	0.82 ± 0.06	-0.16 ± 0.06	0.81 ± 0.04	-0.17 ± 0.05
GPT-2	central	GD	release	Min-K%	holdout10	0.66 ± 0.06	-0.32 ± 0.06	0.69 ± 0.05	-0.29 ± 0.06
GPT-2	central	NPO	release	Loss thr.	primary	0.83 ± 0.08	-0.16 ± 0.08	0.79 ± 0.03	-0.19 ± 0.04

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Table S19 (continued)

Model	Run	Method	View	Attack	NM	Score change		Grad. alignment	
						AUC	Gap	AUC	Gap
GPT-2	central	NPO	release	Loss thr.	holdout10	0.75 ± 0.08	-0.24 ± 0.08	0.71 ± 0.05	-0.26 ± 0.05
GPT-2	central	NPO	release	Min-K%	primary	0.82 ± 0.09	-0.16 ± 0.09	0.79 ± 0.03	-0.19 ± 0.04
GPT-2	central	NPO	release	Min-K%	holdout10	0.74 ± 0.09	-0.24 ± 0.09	0.71 ± 0.05	-0.26 ± 0.05
GPT-2-medium	central	Retrain	release	Loss thr.	primary	0.99 ± 0.00	0.00 ± 0.00	0.99 ± 0.00	0.00 ± 0.00
GPT-2-medium	central	Retrain	release	Loss thr.	holdout10	0.99 ± 0.00	0.00 ± 0.00	0.99 ± 0.00	0.00 ± 0.00
GPT-2-medium	central	Retrain	release	Min-K%	primary	0.99 ± 0.00	0.00 ± 0.00	0.99 ± 0.00	0.00 ± 0.00
GPT-2-medium	central	Retrain	release	Min-K%	holdout10	0.99 ± 0.00	0.00 ± 0.00	0.99 ± 0.00	0.00 ± 0.00
GPT-2-medium	central	GA†	release	Loss thr.	primary	0.54 ± 0.04	-0.45 ± 0.03	0.55 ± 0.03	-0.44 ± 0.03
GPT-2-medium	central	GA†	release	Loss thr.	holdout10	0.49 ± 0.12	-0.51 ± 0.12	0.48 ± 0.10	-0.51 ± 0.10
GPT-2-medium	central	GA†	release	Min-K%	primary	0.63 ± 0.04	-0.37 ± 0.05	0.55 ± 0.03	-0.44 ± 0.03
GPT-2-medium	central	GA†	release	Min-K%	holdout10	0.58 ± 0.13	-0.42 ± 0.13	0.48 ± 0.10	-0.51 ± 0.10
GPT-2-medium	central	GD	release	Loss thr.	primary	0.97 ± 0.04	-0.02 ± 0.03	0.76 ± 0.04	-0.23 ± 0.04
GPT-2-medium	central	GD	release	Loss thr.	holdout10	0.95 ± 0.05	-0.04 ± 0.05	0.65 ± 0.04	-0.34 ± 0.04
GPT-2-medium	central	GD	release	Min-K%	primary	0.97 ± 0.03	-0.02 ± 0.03	0.76 ± 0.04	-0.23 ± 0.04
GPT-2-medium	central	GD	release	Min-K%	holdout10	0.95 ± 0.05	-0.05 ± 0.05	0.65 ± 0.04	-0.34 ± 0.04
GPT-2-medium	central	NPO	release	Loss thr.	primary	0.93 ± 0.08	-0.06 ± 0.08	0.82 ± 0.04	-0.17 ± 0.04
GPT-2-medium	central	NPO	release	Loss thr.	holdout10	0.90 ± 0.10	-0.09 ± 0.10	0.78 ± 0.05	-0.22 ± 0.05
GPT-2-medium	central	NPO	release	Min-K%	primary	0.93 ± 0.08	-0.06 ± 0.08	0.82 ± 0.04	-0.17 ± 0.04
GPT-2-medium	central	NPO	release	Min-K%	holdout10	0.90 ± 0.10	-0.10 ± 0.10	0.78 ± 0.05	-0.22 ± 0.05
GPT-2	by_author_K5	Retrain	any client	Loss thr.	primary	0.97 ± 0.01	0.00 ± 0.00	0.97 ± 0.01	0.00 ± 0.00
GPT-2	by_author_K5	Retrain	any client	Loss thr.	holdout10	0.97 ± 0.02	0.00 ± 0.00	0.96 ± 0.02	0.00 ± 0.00
GPT-2	by_author_K5	Retrain	any client	Min-K%	primary	0.94 ± 0.03	0.00 ± 0.00	0.97 ± 0.01	0.00 ± 0.00
GPT-2	by_author_K5	Retrain	any client	Min-K%	holdout10	0.94 ± 0.04	0.00 ± 0.00	0.96 ± 0.02	0.00 ± 0.00
GPT-2	by_author_K5	Client GA, upload†	server*	Loss thr.	primary	0.57 ± 0.07	-0.40 ± 0.09	0.64 ± 0.05	-0.34 ± 0.05
GPT-2	by_author_K5	Client GA, upload†	server*	Loss thr.	holdout10	0.49 ± 0.28	-0.47 ± 0.28	0.55 ± 0.14	-0.42 ± 0.15
GPT-2	by_author_K5	Client GA, upload†	server*	Min-K%	primary	0.60 ± 0.11	-0.34 ± 0.14	0.64 ± 0.05	-0.34 ± 0.05
GPT-2	by_author_K5	Client GA, upload†	server*	Min-K%	holdout10	0.50 ± 0.16	-0.43 ± 0.19	0.55 ± 0.14	-0.42 ± 0.15
GPT-2	by_author_K5	Client GA, global	any client	Loss thr.	primary	0.80 ± 0.20	-0.17 ± 0.19	0.75 ± 0.08	-0.22 ± 0.07
GPT-2	by_author_K5	Client GA, global	any client	Loss thr.	holdout10	0.78 ± 0.21	-0.19 ± 0.20	0.68 ± 0.05	-0.28 ± 0.05
GPT-2	by_author_K5	Client GA, global	any client	Min-K%	primary	0.75 ± 0.22	-0.19 ± 0.19	0.75 ± 0.08	-0.22 ± 0.07
GPT-2	by_author_K5	Client GA, global	any client	Min-K%	holdout10	0.72 ± 0.20	-0.22 ± 0.16	0.68 ± 0.05	-0.28 ± 0.05
GPT-2	iid_K3	Retrain	any client	Loss thr.	primary	0.92 ± 0.02	0.00 ± 0.00	0.92 ± 0.01	0.00 ± 0.00
GPT-2	iid_K3	Retrain	any client	Loss thr.	holdout10	0.93 ± 0.01	0.00 ± 0.00	0.92 ± 0.02	0.00 ± 0.00
GPT-2	iid_K3	Retrain	any client	Min-K%	primary	0.87 ± 0.02	0.00 ± 0.00	0.92 ± 0.01	0.00 ± 0.00
GPT-2	iid_K3	Retrain	any client	Min-K%	holdout10	0.88 ± 0.02	0.00 ± 0.00	0.92 ± 0.02	0.00 ± 0.00
GPT-2	iid_K3	Client GA, upload†	server*	Loss thr.	primary	0.57 ± 0.01	-0.35 ± 0.01	0.63 ± 0.07	-0.29 ± 0.08
GPT-2	iid_K3	Client GA, upload†	server*	Loss thr.	holdout10	0.43 ± 0.16	-0.50 ± 0.16	0.54 ± 0.14	-0.38 ± 0.15
GPT-2	iid_K3	Client GA, upload†	server*	Min-K%	primary	0.58 ± 0.08	-0.29 ± 0.07	0.63 ± 0.07	-0.29 ± 0.08
GPT-2	iid_K3	Client GA, upload†	server*	Min-K%	holdout10	0.44 ± 0.14	-0.43 ± 0.14	0.54 ± 0.14	-0.38 ± 0.15
GPT-2	iid_K3	Client GA, global	any client	Loss thr.	primary	0.87 ± 0.15	-0.05 ± 0.15	0.79 ± 0.12	-0.13 ± 0.13

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Table S19 (continued)

						Score change		Grad. alignment	
Model	**Run**	**Method**	**View**	**Attack**	**NM**	**AUC**	**Gap**	**AUC**	**Gap**
GPT-2	iid_K3	Client GA, global	any client	Loss thr.	holdout10	0.85 ± 0.16	-0.08 ± 0.17	0.72 ± 0.11	-0.20 ± 0.12
GPT-2	iid_K3	Client GA, global	any client	Min-K%	primary	0.80 ± 0.19	-0.07 ± 0.21	0.79 ± 0.12	-0.13 ± 0.13
GPT-2	iid_K3	Client GA, global	any client	Min-K%	holdout10	0.79 ± 0.16	-0.08 ± 0.16	0.72 ± 0.11	-0.20 ± 0.12
GPT-2	iid_K10	Retrain	any client	Loss thr.	primary	0.99 ± 0.01	0.00 ± 0.00	0.99 ± 0.01	0.00 ± 0.00
GPT-2	iid_K10	Retrain	any client	Loss thr.	holdout10	0.99 ± 0.00	0.00 ± 0.00	0.99 ± 0.00	0.00 ± 0.00
GPT-2	iid_K10	Retrain	any client	Min-K%	primary	0.98 ± 0.01	0.00 ± 0.00	0.99 ± 0.01	0.00 ± 0.00
GPT-2	iid_K10	Retrain	any client	Min-K%	holdout10	0.98 ± 0.00	0.00 ± 0.00	0.99 ± 0.00	0.00 ± 0.00
GPT-2	iid_K10	Client GA, upload†	server*	Loss thr.	primary	0.61 ± 0.10	-0.38 ± 0.10	0.56 ± 0.05	-0.44 ± 0.06
GPT-2	iid_K10	Client GA, upload†	server*	Loss thr.	holdout10	0.53 ± 0.22	-0.47 ± 0.22	0.50 ± 0.11	-0.49 ± 0.12
GPT-2	iid_K10	Client GA, upload†	server*	Min-K%	primary	0.61 ± 0.08	-0.37 ± 0.07	0.56 ± 0.05	-0.44 ± 0.06
GPT-2	iid_K10	Client GA, upload†	server*	Min-K%	holdout10	0.52 ± 0.12	-0.46 ± 0.12	0.50 ± 0.11	-0.49 ± 0.12
GPT-2	iid_K10	Client GA, global	any client	Loss thr.	primary	0.66 ± 0.06	-0.34 ± 0.07	0.65 ± 0.06	-0.34 ± 0.06
GPT-2	iid_K10	Client GA, global	any client	Loss thr.	holdout10	0.60 ± 0.02	-0.39 ± 0.02	0.60 ± 0.02	-0.39 ± 0.02
GPT-2	iid_K10	Client GA, global	any client	Min-K%	primary	0.62 ± 0.08	-0.36 ± 0.08	0.65 ± 0.06	-0.34 ± 0.06
GPT-2	iid_K10	Client GA, global	any client	Min-K%	holdout10	0.59 ± 0.03	-0.39 ± 0.03	0.60 ± 0.02	-0.39 ± 0.02
DistilGPT-2	iid_K5	Retrain	any client	Loss thr.	primary	0.96 ± 0.01	0.00 ± 0.00	0.96 ± 0.01	0.00 ± 0.00
DistilGPT-2	iid_K5	Retrain	any client	Loss thr.	holdout10	0.96 ± 0.01	0.00 ± 0.00	0.95 ± 0.01	0.00 ± 0.00
DistilGPT-2	iid_K5	Retrain	any client	Min-K%	primary	0.92 ± 0.01	0.00 ± 0.00	0.96 ± 0.01	0.00 ± 0.00
DistilGPT-2	iid_K5	Retrain	any client	Min-K%	holdout10	0.91 ± 0.01	0.00 ± 0.00	0.95 ± 0.01	0.00 ± 0.00
DistilGPT-2	iid_K5	Client GA, upload†	server*	Loss thr.	primary	0.56 ± 0.05	-0.40 ± 0.05	0.62 ± 0.02	-0.34 ± 0.02
DistilGPT-2	iid_K5	Client GA, upload†	server*	Loss thr.	holdout10	0.52 ± 0.07	-0.44 ± 0.07	0.60 ± 0.04	-0.36 ± 0.05
DistilGPT-2	iid_K5	Client GA, upload†	server*	Min-K%	primary	0.57 ± 0.05	-0.35 ± 0.06	0.62 ± 0.02	-0.34 ± 0.02
DistilGPT-2	iid_K5	Client GA, upload†	server*	Min-K%	holdout10	0.50 ± 0.05	-0.41 ± 0.04	0.60 ± 0.04	-0.36 ± 0.05
DistilGPT-2	iid_K5	Client GA, global	any client	Loss thr.	primary	0.74 ± 0.08	-0.22 ± 0.08	0.67 ± 0.04	-0.29 ± 0.05
DistilGPT-2	iid_K5	Client GA, global	any client	Loss thr.	holdout10	0.72 ± 0.06	-0.23 ± 0.06	0.65 ± 0.03	-0.30 ± 0.04
DistilGPT-2	iid_K5	Client GA, global	any client	Min-K%	primary	0.69 ± 0.10	-0.23 ± 0.10	0.67 ± 0.04	-0.29 ± 0.05
DistilGPT-2	iid_K5	Client GA, global	any client	Min-K%	holdout10	0.65 ± 0.08	-0.26 ± 0.08	0.65 ± 0.03	-0.30 ± 0.04
GPT-2	iid_K5	Retrain	any client	Loss thr.	primary	0.97 ± 0.01	0.00 ± 0.00	0.97 ± 0.01	0.00 ± 0.00
GPT-2	iid_K5	Retrain	any client	Loss thr.	holdout10	0.97 ± 0.01	0.00 ± 0.00	0.96 ± 0.01	0.00 ± 0.00
GPT-2	iid_K5	Retrain	any client	Min-K%	primary	0.94 ± 0.00	0.00 ± 0.00	0.97 ± 0.01	0.00 ± 0.00
GPT-2	iid_K5	Retrain	any client	Min-K%	holdout10	0.93 ± 0.01	0.00 ± 0.00	0.96 ± 0.01	0.00 ± 0.00
GPT-2	iid_K5	Client GA, upload†	server*	Loss thr.	primary	0.58 ± 0.04	-0.39 ± 0.04	0.64 ± 0.02	-0.33 ± 0.02
GPT-2	iid_K5	Client GA, upload†	server*	Loss thr.	holdout10	0.52 ± 0.07	-0.44 ± 0.07	0.58 ± 0.03	-0.38 ± 0.03
GPT-2	iid_K5	Client GA, upload†	server*	Min-K%	primary	0.59 ± 0.03	-0.35 ± 0.03	0.64 ± 0.02	-0.33 ± 0.02
GPT-2	iid_K5	Client GA, upload†	server*	Min-K%	holdout10	0.51 ± 0.08	-0.43 ± 0.07	0.58 ± 0.03	-0.38 ± 0.03
GPT-2	iid_K5	Client GA, global	any client	Loss thr.	primary	0.77 ± 0.11	-0.20 ± 0.11	0.74 ± 0.07	-0.23 ± 0.07
GPT-2	iid_K5	Client GA, global	any client	Loss thr.	holdout10	0.76 ± 0.09	-0.21 ± 0.09	0.70 ± 0.04	-0.26 ± 0.04
GPT-2	iid_K5	Client GA, global	any client	Min-K%	primary	0.72 ± 0.09	-0.22 ± 0.09	0.74 ± 0.07	-0.23 ± 0.07
GPT-2	iid_K5	Client GA, global	any client	Min-K%	holdout10	0.70 ± 0.08	-0.23 ± 0.08	0.70 ± 0.04	-0.26 ± 0.04
GPT-2-medium	iid_K5	Retrain	any client	Loss thr.	primary	0.98 ± 0.00	0.00 ± 0.00	0.98 ± 0.00	0.00 ± 0.00

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Table S19 (continued)

Model	Run	Method	View	Attack	NM	Score change AUC	Score change Gap	Grad. alignment AUC	Grad. alignment Gap
GPT-2-medium	iid_K5	Retrain	any client	Loss thr.	holdout10	0.98 ± 0.00	0.00 ± 0.00	0.98 ± 0.00	0.00 ± 0.00
GPT-2-medium	iid_K5	Retrain	any client	Min-K%	primary	0.97 ± 0.00	0.00 ± 0.00	0.98 ± 0.00	0.00 ± 0.00
GPT-2-medium	iid_K5	Retrain	any client	Min-K%	holdout10	0.97 ± 0.00	0.00 ± 0.00	0.98 ± 0.00	0.00 ± 0.00
GPT-2-medium	iid_K5	Client GA, upload†	server*	Loss thr.	primary	0.54 ± 0.03	-0.44 ± 0.03	0.64 ± 0.02	-0.35 ± 0.02
GPT-2-medium	iid_K5	Client GA, upload†	server*	Loss thr.	holdout10	0.50 ± 0.08	-0.48 ± 0.08	0.59 ± 0.05	-0.40 ± 0.05
GPT-2-medium	iid_K5	Client GA, upload†	server*	Min-K%	primary	0.56 ± 0.06	-0.41 ± 0.06	0.64 ± 0.02	-0.35 ± 0.02
GPT-2-medium	iid_K5	Client GA, upload†	server*	Min-K%	holdout10	0.48 ± 0.07	-0.50 ± 0.07	0.59 ± 0.05	-0.40 ± 0.05
GPT-2-medium	iid_K5	Client GA, global	any client	Loss thr.	primary	0.84 ± 0.06	-0.14 ± 0.06	0.80 ± 0.05	-0.18 ± 0.05
GPT-2-medium	iid_K5	Client GA, global	any client	Loss thr.	holdout10	0.84 ± 0.06	-0.15 ± 0.06	0.76 ± 0.04	-0.22 ± 0.03
GPT-2-medium	iid_K5	Client GA, global	any client	Min-K%	primary	0.79 ± 0.05	-0.18 ± 0.05	0.80 ± 0.05	-0.18 ± 0.05
GPT-2-medium	iid_K5	Client GA, global	any client	Min-K%	holdout10	0.78 ± 0.06	-0.19 ± 0.06	0.76 ± 0.04	-0.22 ± 0.03
Pythia-1B (LoRA)	central	Retrain	release	Loss thr.	primary	0.98 ± 0.01	0.00 ± 0.00	0.97 ± 0.01	0.00 ± 0.00
Pythia-1B (LoRA)	central	Retrain	release	Loss thr.	holdout10	0.98 ± 0.00	0.00 ± 0.00	0.98 ± 0.01	0.00 ± 0.00
Pythia-1B (LoRA)	central	Retrain	release	Min-K%	primary	0.97 ± 0.01	0.00 ± 0.00	0.97 ± 0.01	0.00 ± 0.00
Pythia-1B (LoRA)	central	Retrain	release	Min-K%	holdout10	0.97 ± 0.01	0.00 ± 0.00	0.98 ± 0.01	0.00 ± 0.00
Pythia-1B (LoRA)	central	GA†	release	Loss thr.	primary	0.59 ± 0.16	-0.39 ± 0.16	0.51 ± 0.04	-0.47 ± 0.04
Pythia-1B (LoRA)	central	GA†	release	Loss thr.	holdout10	0.51 ± 0.24	-0.47 ± 0.24	0.43 ± 0.12	-0.54 ± 0.12
Pythia-1B (LoRA)	central	GA†	release	Min-K%	primary	0.61 ± 0.15	-0.35 ± 0.14	0.51 ± 0.04	-0.47 ± 0.04
Pythia-1B (LoRA)	central	GA†	release	Min-K%	holdout10	0.61 ± 0.17	-0.36 ± 0.17	0.43 ± 0.12	-0.54 ± 0.12
Pythia-1B (LoRA)	central	GD†	release	Loss thr.	primary	0.95 ± 0.04	-0.03 ± 0.04	0.66 ± 0.04	-0.32 ± 0.04
Pythia-1B (LoRA)	central	GD†	release	Loss thr.	holdout10	0.91 ± 0.06	-0.07 ± 0.06	0.58 ± 0.02	-0.39 ± 0.03
Pythia-1B (LoRA)	central	GD†	release	Min-K%	primary	0.95 ± 0.04	-0.01 ± 0.04	0.66 ± 0.04	-0.32 ± 0.04
Pythia-1B (LoRA)	central	GD†	release	Min-K%	holdout10	0.91 ± 0.06	-0.05 ± 0.06	0.58 ± 0.02	-0.39 ± 0.03
Pythia-1B (LoRA)	central	NPO	release	Loss thr.	primary	0.97 ± 0.03	-0.01 ± 0.03	0.71 ± 0.04	-0.27 ± 0.04
Pythia-1B (LoRA)	central	NPO	release	Loss thr.	holdout10	0.94 ± 0.05	-0.04 ± 0.05	0.65 ± 0.04	-0.32 ± 0.05
Pythia-1B (LoRA)	central	NPO	release	Min-K%	primary	0.97 ± 0.03	0.00 ± 0.03	0.71 ± 0.04	-0.27 ± 0.04
Pythia-1B (LoRA)	central	NPO	release	Min-K%	holdout10	0.94 ± 0.04	-0.03 ± 0.05	0.65 ± 0.04	-0.32 ± 0.05

Mean ± half-width of the 95% t-interval over seeds (up to 5); a single value means one seed. NM: non-member set. View: who observes the change from θ_{pre} (release: the two model versions; *the server, and all clients at the first recovery broadcast). †Model collapsed: retain loss rose more than threefold in every seed. GA: all four central models; GD: Pythia-1B (LoRA); client-GA upload: all six federated runs (intermediate model before recovery; the released global model is not collapsed).

Table S20: Random-direction control of the white-box attack: AUC over all (method, seed) draws, against the SD of the AUC for independent records.

Runs	Model	Non-members	Draws	Mean	SD	Range	Null SD	Outside 2σ	Grad. align.
central	DistilGPT-2	primary	20	0.494	0.021	0.456–0.531	0.020	0.10	0.757
central	DistilGPT-2	holdout10	20	0.481	0.053	0.381–0.583	0.020	0.50	0.681
central	GPT-2	primary	20	0.491	0.020	0.454–0.534	0.020	0.05	0.791
central	GPT-2	holdout10	20	0.478	0.045	0.399–0.559	0.020	0.45	0.722
central	GPT-2-medium	primary	20	0.505	0.027	0.471–0.591	0.020	0.05	0.782

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Table S20 (continued)

Runs	Model	Non-members	Draws	Mean	SD	Range	Null SD	Outside 2σ	Grad. align.
central	GPT-2-medium	holdout10	20	0.492	0.037	0.424–0.571	0.020	0.35	0.726
federated	GPT-2	primary	42	0.502	0.034	0.448–0.570	0.020	0.29	0.773
federated	GPT-2	holdout10	42	0.493	0.037	0.438–0.554	0.020	0.38	0.730
federated	DistilGPT-2	primary	15	0.499	0.032	0.449–0.547	0.020	0.20	0.752
federated	DistilGPT-2	holdout10	15	0.492	0.031	0.431–0.531	0.020	0.20	0.732
federated	GPT-2-medium	primary	15	0.485	0.027	0.433–0.529	0.020	0.13	0.809
federated	GPT-2-medium	holdout10	15	0.497	0.039	0.444–0.566	0.020	0.40	0.778
central	Pythia-1B (LoRA)	primary	20	0.484	0.025	0.432–0.529	0.020	0.15	0.710
central	Pythia-1B (LoRA)	holdout10	20	0.488	0.031	0.444–0.546	0.020	0.30	0.661

4 Costs, TOFU metrics and settings

Table S21: H6: cost of client-level gradient ascent with recovery against federated retraining, per setting (ratios paired by seed).

			Wall time (min)			Peak memory		Communication (GiB)		
Model	Sharding	K	GA	Retrain	Ratio	Alloc.	Reserved	GA	Retrain	Ratio
GPT-2	by_author	5	1.8 ± 0.0	5.3 ± 0.1	0.33 ± 0.01	1.00 ± 0.00	1.01 ± 0.03	8.34	29.67	0.28
GPT-2	iid	3	1.8 ± 0.0	5.3 ± 0.0	0.33 ± 0.00	1.00 ± 0.00	1.02 ± 0.08	4.64	14.83	0.31
GPT-2	iid	10	1.8 ± 0.0	5.3 ± 0.0	0.33 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	17.62	66.75	0.26
DistilGPT-2	iid	5	1.1 ± 0.0	3.5 ± 0.1	0.33 ± 0.00	1.00 ± 0.00	1.00 ± 0.01	5.49	19.53	0.28
GPT-2	iid	5	1.8 ± 0.0	5.3 ± 0.1	0.33 ± 0.00	1.00 ± 0.00	1.01 ± 0.02	8.34	29.67	0.28
GPT-2-medium	iid	5	4.8 ± 0.1	14.5 ± 0.2	0.33 ± 0.00	1.00 ± 0.01	1.01 ± 0.02	23.79	84.60	0.28

Mean ± half-width of the 95% t-interval over seeds (up to 5); a single value means one seed. GA: client gradient ascent plus recovery rounds. Peak memory as the ratio GA / retrain (allocated and reserved). Time and peak GPU memory are measured; communication is analytic (fp32 parameters).

Table S22: TOFU forget quality (KS test against the retrained model of the same seed) and model utility.

Model	Run	Method	Forget quality	Model utility	MU excl. 0
DistilGPT-2	central	Retrain	1.000 ± 0.000	0.020 ± 0.055	0.099
DistilGPT-2	central	GA†	0.000 ± 0.000	0.000 ± 0.000	–
DistilGPT-2	central	GD	0.085 ± 0.064	0.023 ± 0.030	0.028 ± 0.038
DistilGPT-2	central	NPO	0.051 ± 0.048	0.039 ± 0.038	0.039 ± 0.038
GPT-2	central	Retrain	1.000 ± 0.000	0.082 ± 0.052	0.082 ± 0.052
GPT-2	central	GA†	0.000 ± 0.000	0.000 ± 0.000	–
GPT-2	central	GD	0.001 ± 0.001	0.108 ± 0.034	0.108 ± 0.034
GPT-2	central	NPO	0.004 ± 0.007	0.085 ± 0.038	0.085 ± 0.038

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Table S22 (continued)

Model	Run	Method	Forget quality	Model utility	MU excl. 0
GPT-2-medium	central	Retrain	1.000 ± 0.000	0.260 ± 0.018	0.260 ± 0.018
GPT-2-medium	central	GA†	0.000 ± 0.000	0.000 ± 0.000	–
GPT-2-medium	central	GD	0.000 ± 0.000	0.026 ± 0.028	0.026 ± 0.028
GPT-2-medium	central	NPO	0.000 ± 0.001	0.193 ± 0.069	0.193 ± 0.069
GPT-2	by_author_K5	Retrain	1.000 ± 0.000	0.108 ± 0.054	0.108 ± 0.054
GPT-2	by_author_K5	Client GA, upload†	–	–	–
GPT-2	by_author_K5	Client GA, global	0.725 ± 1.170	0.082 ± 0.009	0.082 ± 0.009
GPT-2	iid_K3	Retrain	1.000 ± 0.000	0.141 ± 0.053	0.141 ± 0.053
GPT-2	iid_K3	Client GA, upload†	–	–	–
GPT-2	iid_K3	Client GA, global	0.954 ± 0.196	0.044 ± 0.116	0.066 ± 0.345
GPT-2	iid_K10	Retrain	1.000 ± 0.000	0.103 ± 0.075	0.103 ± 0.075
GPT-2	iid_K10	Client GA, upload†	–	–	–
GPT-2	iid_K10	Client GA, global	0.527 ± 0.279	0.122 ± 0.035	0.122 ± 0.035
DistilGPT-2	iid_K5	Retrain	1.000 ± 0.000	0.000 ± 0.000	–
DistilGPT-2	iid_K5	Client GA, upload†	–	–	–
DistilGPT-2	iid_K5	Client GA, global	0.780 ± 0.381	0.016 ± 0.035	0.041 ± 0.306
GPT-2	iid_K5	Retrain	1.000 ± 0.000	0.104 ± 0.034	0.104 ± 0.034
GPT-2	iid_K5	Client GA, upload†	–	–	–
GPT-2	iid_K5	Client GA, global	0.706 ± 0.509	0.087 ± 0.029	0.087 ± 0.029
GPT-2-medium	iid_K5	Retrain	1.000 ± 0.000	0.267 ± 0.020	0.267 ± 0.020
GPT-2-medium	iid_K5	Client GA, upload†	–	–	–
GPT-2-medium	iid_K5	Client GA, global	0.745 ± 0.236	0.257 ± 0.028	0.257 ± 0.028
Pythia-1B (LoRA)	central	Retrain	1.000 ± 0.000	0.332 ± 0.005	0.332 ± 0.005
Pythia-1B (LoRA)	central	GA†	0.000 ± 0.000	0.000 ± 0.000	–
Pythia-1B (LoRA)	central	GD†	0.001 ± 0.002	0.006 ± 0.015	0.015 ± 0.162
Pythia-1B (LoRA)	central	NPO	0.000 ± 0.001	0.106 ± 0.069	0.106 ± 0.069

Mean ± half-width of the 95% t-interval over seeds (up to 5); a single value means one seed. Rows with model utility 0: 37 of 128. MU excl. 0: model utility without those rows (sensitivity check). Retrain rows with model utility 0 are not collapsed: one of the nine harmonic-mean components is 0, ROUGE on real authors (DistilGPT-2, central: 4/5 seeds; DistilGPT-2, iid_K5: 5/5 seeds). †Model collapsed: retain loss rose more than threefold in every seed. GA: all four central models; GD: Pythia-1B (LoRA); client-GA upload: all six federated runs (intermediate model before recovery; the released global model is not collapsed).

Table S23: Settings of each campaign, from the resolved configs.

Setting	central	fed_ext	fed_iid	lora
models	distilgpt2, gpt2, gpt2-medium	gpt2	distilgpt2, gpt2, gpt2-medium	EleutherAI/pythia-1b
seeds	0, 1, 2, 3, 4	0, 1, 2	0, 1, 2, 3, 4	0, 1, 2, 3, 4
model.precision	"fp16_amp"	"fp16_amp"	"fp16_amp"	"fp16_amp"
model.lora	null	null	null	–
data.max_length	128	128	128	128
data.max_samples	null	null	null	null

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Table S23 (continued)

Setting	central	fed_ext	fed_iid	lora
data.outlier_fraction	0.3	0.3	0.3	0.3
data.max_new_tokens	128	128	128	128
unlearn.finetune_epochs	3	3	3	3
unlearn.finetune_lr	2e-05	2e-05	2e-05	0.0001
unlearn.train_batch_size	16	16	16	16
unlearn.epochs	2	2	2	2
unlearn.learning_rate	1e-05	1e-05	1e-05	5e-05
unlearn.batch_size	4	4	4	4
unlearn.eval_batch_size	64	64	64	64
unlearn.retain_weight	1.0	1.0	1.0	1.0
unlearn.beta	0.1	0.1	0.1	0.1
attack.min_k	0.2	0.2	0.2	0.2
federated.num_clients	–	10 / 3 / 5	5	–
federated.sharding	–	”by_author” / ”iid”	”iid”	–
federated.rounds	–	8	8	–
federated.local_epochs	–	1	1	–
federated.client_unlearn_epochs	–	2	2	–
federated.rho_factor	–	2.0	2.0	–
federated.recovery_rounds	–	2	2	–
model.lora.alpha	–	–	–	32
model.lora.base_dtype	–	–	–	”float16”
model.lora.dropout	–	–	–	0.05
model.lora.gradient_checkpointing	–	–	–	true
model.lora.r	–	–	–	16
model.lora.target_modules.gpt_neox	–	–	–	[”query_key_value”, ”dense”]
model.lora.target_modules.llama	–	–	–	[”q_proj”, ”k_proj”, ”v_proj”, ”o_proj”]

Table S24: Phase times of seed 0 run alone (pilot, LoRA pilot) and with a second process on the other GPU (campaign).

Model	Run	Method	Phase	Alone (s)	Campaign (s)	Ratio	Same GPU
GPT-2	central	Retrain	base_scoring	11.1	10.3	0.92	yes
GPT-2	central	Retrain	finetune	142.3	134.2	0.94	yes
GPT-2	central	Retrain	pre_eval	131.7	115.2	0.87	yes
GPT-2	central	Retrain	method	125.5	118.8	0.95	yes
GPT-2	central	Retrain	post_eval	123.6	108.3	0.88	yes
GPT-2	central	Retrain	alignment	73.3	72.5	0.99	yes
GPT-2	central	GA†	base_scoring	11.1	10.3	0.92	yes
GPT-2	central	GA†	finetune	142.3	134.2	0.94	yes
GPT-2	central	GA†	pre_eval	131.7	115.2	0.87	yes
GPT-2	central	GA†	method	22.2	21.7	0.98	yes

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Table S24 (continued)

Model	Run	Method	Phase	Alone (s)	Campaign (s)	Ratio	Same GPU
GPT-2	central	GA†	post_eval	144.0	127.6	0.89	yes
GPT-2	central	GA†	alignment	74.9	71.0	0.95	yes
GPT-2	central	GD	base_scoring	11.1	10.3	0.92	yes
GPT-2	central	GD	finetune	142.3	134.2	0.94	yes
GPT-2	central	GD	pre_eval	131.7	115.2	0.87	yes
GPT-2	central	GD	method	32.0	31.0	0.97	yes
GPT-2	central	GD	post_eval	113.8	100.4	0.88	yes
GPT-2	central	GD	alignment	74.8	71.4	0.95	yes
GPT-2	central	NPO	base_scoring	11.1	10.3	0.92	yes
GPT-2	central	NPO	finetune	142.3	134.2	0.94	yes
GPT-2	central	NPO	pre_eval	131.7	115.2	0.87	yes
GPT-2	central	NPO	method	40.2	38.9	0.97	yes
GPT-2	central	NPO	post_eval	143.7	127.0	0.88	yes
GPT-2	central	NPO	alignment	74.7	71.0	0.95	yes
GPT-2	iid_K5	Retrain	base_scoring	11.0	10.4	0.95	yes
GPT-2	iid_K5	Retrain	pre_eval	132.5	123.7	0.93	yes
GPT-2	iid_K5	Retrain	method	332.6	319.9	0.96	yes
GPT-2	iid_K5	Retrain	post_eval	124.7	116.5	0.93	yes
GPT-2	iid_K5	Retrain	alignment	75.1	69.9	0.93	yes
GPT-2	iid_K5	Retrain	fedavg	376.4	361.8	0.96	yes
GPT-2	iid_K5	Client GA, upload†	base_scoring	11.0	10.4	0.95	yes
GPT-2	iid_K5	Client GA, upload†	pre_eval	132.5	123.7	0.93	yes
GPT-2	iid_K5	Client GA, upload†	method	28.3	27.7	0.98	yes
GPT-2	iid_K5	Client GA, upload†	post_eval	42.2	39.3	0.93	yes
GPT-2	iid_K5	Client GA, upload†	alignment	73.9	71.6	0.97	yes
GPT-2	iid_K5	Client GA, upload†	fedavg	376.4	361.8	0.96	yes
GPT-2	iid_K5	Client GA, upload†	client_unlearn	28.3	27.7	0.98	yes
GPT-2	iid_K5	Client GA, global	base_scoring	11.0	10.4	0.95	yes
GPT-2	iid_K5	Client GA, global	pre_eval	132.5	123.7	0.93	yes
GPT-2	iid_K5	Client GA, global	method	110.6	106.8	0.97	yes
GPT-2	iid_K5	Client GA, global	post_eval	131.3	122.6	0.93	yes
GPT-2	iid_K5	Client GA, global	alignment	74.5	70.7	0.95	yes
GPT-2	iid_K5	Client GA, global	fedavg	376.4	361.8	0.96	yes
GPT-2	iid_K5	Client GA, global	client_unlearn	28.3	27.7	0.98	yes
GPT-2	iid_K5	Client GA, global	recovery	82.3	79.1	0.96	yes
Pythia-1B (LoRA)	central	Retrain	base_scoring	15.4	13.9	0.90	yes
Pythia-1B (LoRA)	central	Retrain	finetune	339.3	332.0	0.98	yes
Pythia-1B (LoRA)	central	Retrain	pre_eval	189.1	180.9	0.96	yes
Pythia-1B (LoRA)	central	Retrain	method	295.5	282.6	0.96	yes
Pythia-1B (LoRA)	central	Retrain	post_eval	175.8	167.8	0.95	yes
Pythia-1B (LoRA)	central	Retrain	alignment	147.7	140.1	0.95	yes
Pythia-1B (LoRA)	central	GA†	base_scoring	15.4	13.9	0.90	yes

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Table S24 (continued)

Model	Run	Method	Phase	Alone (s)	Campaign (s)	Ratio	Same GPU
Pythia-1B (LoRA)	central	GA†	finetune	339.3	332.0	0.98	yes
Pythia-1B (LoRA)	central	GA†	pre_eval	189.1	180.9	0.96	yes
Pythia-1B (LoRA)	central	GA†	method	28.0	27.3	0.98	yes
Pythia-1B (LoRA)	central	GA†	post_eval	215.7	206.7	0.96	yes
Pythia-1B (LoRA)	central	GA†	alignment	147.7	140.5	0.95	yes
Pythia-1B (LoRA)	central	GD†	base_scoring	15.4	13.9	0.90	yes
Pythia-1B (LoRA)	central	GD†	finetune	339.3	332.0	0.98	yes
Pythia-1B (LoRA)	central	GD†	pre_eval	189.1	180.9	0.96	yes
Pythia-1B (LoRA)	central	GD†	method	53.9	53.0	0.98	yes
Pythia-1B (LoRA)	central	GD†	post_eval	217.0	207.1	0.95	yes
Pythia-1B (LoRA)	central	GD†	alignment	147.9	140.5	0.95	yes
Pythia-1B (LoRA)	central	NPO	base_scoring	15.4	13.9	0.90	yes
Pythia-1B (LoRA)	central	NPO	finetune	339.3	332.0	0.98	yes
Pythia-1B (LoRA)	central	NPO	pre_eval	189.1	180.9	0.96	yes
Pythia-1B (LoRA)	central	NPO	method	74.5	73.5	0.99	yes
Pythia-1B (LoRA)	central	NPO	post_eval	216.2	207.7	0.96	yes
Pythia-1B (LoRA)	central	NPO	alignment	147.9	140.3	0.95	yes

†Model collapsed: retain loss rose more than threefold in 5/5 seeds. GA: both central models; GD: Pythia-1B (LoRA); client-GA upload: GPT-2 iid_K5 (intermediate model before recovery; the released global model is not collapsed).